

BINET KAMAT

TEST OF

INTELLIGENCE

1ST -A

TEST FOR AGE - 3

BKT

1. POINTING TO PARTS OF THE BODY (3 out of 4) [Nose, Eyes, Mouth, Hair]
2. NAMING FAMILIAR OBJECTS (3 out of 5) [Key, Coins, Penknife, Watch, Pencil]
3. REPEATING TWO DIGITS (1out of 3: one every half second) (4-2;3-7;6-4;7-2)
C - 5B
4. ENUMERATION OF OBJECTS IN A picture (at least 2 objects in any one picture.)
[(A). Railway station (B) Reception (C) Motor accident.]
5. REPEATING 6 TO 7 SYLLABLES (1 out of 3; read 1 every half, second)
(a) I have a little dog
(b) The dog runs after the cat.
(c) In summer the sun is hot.
6. COMPARISON OF LINES (3 out of 3; or 5 out of 6.)

ALTERNATIVE ITEMS

- 1 Giving sex
- 2 Giving proper names (what is your name?)

1ST - B

REPEATING DIGITS

DIGITS	AGE	ITEM, NO	
2	3	3	(1/3) {4-2; 3-7; 6-4; 7-2.}
3	4	1	(1/3) {6-4-1 ; 3-5-2 ; 8-3-7}
4	6	1	(1/3) { 4 - 7 - 3 - 9 ; 2 - 8 - 5 - 4 ; 7 - 2 - 6 - 1 }
5 *	8	2	(1/3) {3 - 1 - 7 - 5 - 9; 4 - 2 - 3 - 8 - 5; 9 - 8 - 1 - 7 - 6 .}
6 *	12	2	(1/3) ALTERNATIVES { 3 - 7 - 4 - 8 - 5 - 9; 5 - 2 - 1 - 7 - 4 - 6 ; 4 - 7 - 1 - 5 - 8 - 2 }
7 *	16	6	(1/2) { 2 - 1 - 8 - 3 - 4 - 7 - 9; 9 - 7 - 2 - 8 - 4 - 6 - 5 }
8 *	22	2	(1/3) { 7 - 2 - 5 - 3 - 4 - 8 - 9 - 6; 4 - 9 - 8 - 5 - 3 - 7 - 6 - 2; 8 - 3 - 7 - 9 - 5 - 4 - 2 - 6 }

NOTE * Read 1 per sec., others read 1 every half - second.

2ND - A

TEST FOR AGE - 4

c = 3b

1. REPEATING 3 DIGITS (1 out of 3 . read 1 every half - second) [6 - 4 - 1; 3 - 5 - 2; 8 - 3 - 7]
2. DISCRIMINATION OF FORMS (Kuhlman , 7 out of 10)
3. COMPREHENSION, FIRST DEGREE (2 out of 3).

- (a). What must you do when you are sleepy?
- (b). What ought you to do when you are cold?
- (c). What ought you to do when you are hungry?

c = 6b

4. REPEATING 12 TO 13 SYLLABLES (1 out of 3 absolutely correct or 2 with one error each)
 - (a). The boy's name is John. He is a very good boy.
 - (b). When the train comes you will hear the whistle blow.
 - (c). We are going to have a good time in the country.

c = 7b

5. COUNTING FOUR PAISA (coins).
6. COPYING A SQUARE (1 out of 3).

ALTERNATIVE ITEM

1. COMPARISON OF TWO WEIGHT (3/3, 4/5)

2ND - B

COMPREHENSION

AGE - 4 ITEM NO. 3 FIRST DEGREE (2 OUT OF 3) AFTER 20 SEC REPEAT AGAIN.

- (a) What must you do when are sleepy?
- (b) What ought you to do when you are cold?
- (c). What ought you to do when you are hungry?

AGE - 6 ITEM NO.2 SECOND DEGREE (2 out of 3. time ½ min.) after 5 to 10 seconds repeat again).
what is the thing to do -

- (a) If it is raining when you start for school?
- (b) If you find that your house is on fire?
- (c) If you are going some where & miss the train? (or car or bus)

AGE = 8 ITEM NO.3 THIRD DEGREE (2 OUT OF 3; AFTER 5 - 10 SEC READ AGAIN & THEN ALLOW 30 SECONDS MORE).

- (a) What's the thing for you to do when you have broken something which belongs to someone else?
- (b) What is the thing for you to do when you are on your way to school & notice that you are in danger of being late?

AGE = 12 ITEM NO. ALTERNATIVES 3 FOURTH DEGREE (2 OUT OF 3 ; TIME ½ MINUTE REPEAT AFTER 5 SEC)

- (a) What ought you to say when someone asks your opinion about a person you don't know very well?
- (b) What ought you to do before beginning something important?
- (c) Why should we judge a person more by his actions than by his words?

3RD -A

TEST FOR AGE - 5

1. AESTHETIC COMPARISON (no errors). Present the card.
c = 7b
2. DEFINITIONS IN TERMS OF USE (4 out of 6. wait for 1 min each for response)
EXAMPLE : WHAT IS A CHAIR {Chair, Spoon, Doll, Pencil, Blanket}
3. THREE COMMISSIONS (all 3 correct in proper sequences) example, put the key on table close the door, bring the toy from the chair).
4. DISTINGUISHING RIGHT AND LEFT (3 out of 3 or 4 out of 5).
(a) Show me your right hand (b) Show me your left ear (c) Show me your right eye. (d) Show me your left-hand (e) show me your right ear. (f) Show me your left eye.
- c = 7b 5. Naming four coins (3 out of 4 one at a time) {50 p, Rs -1, Rs - 2, Rs - 5 }.
- c = 7b 6. COUNTING 13 PAPER STRIPS OR COINS (1 out of 2 without errors).

ALTERNATIVES ITEMS:

- 1 FORE NOON AND AFTER NOON
Is it morning or afternoon? (in the morning)
Is it afternoon or morning? (in the afternoon)
Is it evening or morning? (in the evening)
2. GIVING FAMILY NAME (Full name)
3. GIVING AGE.

3RD -B

ENUMERATION OF OBJECTS IN PICTURE

AGE - 3. ITEM NO 4 (Atleast 3 objects is any one picture).

- (a) Railway station
- (b) Reception
- (c) Motor accident {list of person, object in detail}.

DISCRIMINATION OF PICTURES

AGE - 6. ITEM NO 5 (2 out of 3 picture described time limit of 2 min for each).

Ex: what is this picture about? { action, characteristics }

INTERPRETATION OF PICTURE

AGE 12 ITEM NO 6 (3 out of 4 time limit 2 min for each picture).

Ex: what is this picture about? Or explain the picture?

- a) Railway station
- b) Reception
- c) Motor accident
- d) Domestic scene.

{Beyond what is visible: emotion or situation}.

4TH - A

TEST FOR AGE - 6

c=3b

1. REPEATING 4 DIGITS (1 out of 3. read 1 every half - second).{ 4 - 7 - 3 - 9; 2 - 8 - 5 - 4 : 7 - 2 - 6 - 1 }

c=4b

2.COMPREHENSION, SECOND DEGREE. (2 out of 3, time allowed ½ min). (After 5 to 10 seconds repeat again).

WHAT IS THE THING TO DO:-

(a) If it is raining when you start for school? (b) If you find that your house is on fire? (c) If you are going some where and miss the train or bus or car?

3.divided card (2 out of 3 trials . 1 min each)

4.GIVING NUMBER OF FINGER (no errors) -- one hand, two hands and then both hands

C = 5B

5.DESCRPTION OF PICTURES (2 out of 3 or 2 min each)—Railway station, Reception at home, Motor Accident.

6.MISSING FEATURES (3 out of 4. time ½ min each).

ALTERNATIVES ITEM

A) NAMING COLOURS (no errors). Point to the colours in the order Red, Yellow, Blue & Green.

4TH - B

REPEATING SYLLABLES

AGE	ITEM. NO		SYLLABLES
3	5	6-7 (1 OUT OF 3)	(a) i have a little dog. (b). The dog runs after the cat.(c). In summer the sun is hot
4	4	12 - 13 (1 out of 3 absolutely correct or 2 with 1 error each)	(A) The boys name is john. He is a very good boy. (B) When the train comes you will hear the whistle blow (C) We are going to have a good time is the country.
7	1	16 - 18 (1 out of 3 absolutely correct or 2 with 1 error each)	(A) We will go out for a long walk. Please give me my pretty straw hat. (B) We are having a fine time .we found a little mouse in the trap. (C) Walter had a fine time on his holiday. He went fishing every day.
10	2	20 - 22 (1 out of 3 absolutely correct or 2 with 1 error each)	(A) The apple tree makes a cool pleasant shade on the ground where the children are playing. (B) It is nearly half past 1 o' clock : (C) The house is very quiet & the cat has gone to sleep. (d) In summer the days are very warm & fine; in winter it snows and I am cold.
19	5	30 (1 out of 2 absolutely correct)	(a) Rama likes very much to go to his grand mother because she tells him funny stories. (b) Yesterday i saw a pretty little dog in the street. It had curly brown hair.

5TH - A

TEST FOR AGE - 7

c-6b

1. REPEATING 16 - 18 SYLLABLES (1 out of 3 absolutely correct or 2 with 1 error each).
 (a) We will go out for a long walk. Please give me my pretty straw hat.
 (b) We are having a fine time .we found a little mouse in the trap.
 (c) Walter had a fine time on his holiday. He went fishing every day.

2. COPYING A DIAMOND (2 out of 3).

c=9b

3. REPEATING 3 DIGITS REVERSED (1 out of 3 read 1 per second) {2 - 8 - 3; 4 - 2 - 7; 9 - 5 - 8}
 4. NAMING DAYS OF WEEK (order correct, 15 sec, 2 out of 3 correct).
 (a). What day comes before Tuesday?
 (b) What day comes before Thursday?
 (c) What day comes before Friday?

5. COUNTING BACKWARDS 20 TO 1 (40 seconds. 1 errors allowed).

c=9b

6. GIVING DIFFERENCES FROM MEMORY (2 out of 3 ; 1 min each) (a) Fly - Butterfly (b) Stone - Potato (c) Wood - Glass

ALTERNATIVE ITEM

1. GIVING DAY OF WEEK & DATE OF THE MONTH. (Both correct. the second within 3 days of the actual date).

- (a) What day of the week is today? (b) What day of the month is today?

5TH - B

COMPARISON OF WEIGHTS

AGE	ITEM. NO.	COMPARISON OF 2 WEIGHTS (3 OUT OF 3 OR 4 OUT OF 5).
4	ALTERNATIVE-1	{3 - 15; 15 - 3 ; 3 - 15; 15 - 3; 3 - 15 }.
10	1	ARRANGING OF 5 WEIGHTS (2 out of 3).
COUNTING PAPER STRIPS OR SIMILAR COINS		
4	5.	COUNTING 4 PAPER STRIPS OR COINS (Counting & tallying = total).
5	6	COUNTING 13 PAPER STRIPS OR COINS (1 out of 2 without error).
NAMING THE VALUE OF COINS		
5	5	NAMING 4 COINS (3 out of 4) { 50 Np, Rs 1, Rs 2, Rs 5 , (Give one at a time)
8	1	FINDING VALUES OF COINS (15 sec) + total?
8	5	Coins and total? naming 6 coins (no error) { 50 Np, Rs 1, Rs 2, Rs 5, Rs 10, Rs 20.}

6TH - A

TEST FOR AGE - 8

c-7b

1. FINDING VALUES OF COINS (15 sec) + total.

Coins and total)?

c-3b

2. REPEATING 5 DIGITS (1 out of 3, read 1 per second).

{3-1-7-5-9; 4-2-3-8-5; 9-8-1-7-6}.

c-4b

3. COMPREHENSION, THIRD DEGREE (2 out of 3, read once, after 5-10 sec read second time & allow 30 seconds more)

WHAT IS THE THING FOR YOU TO DO: -

- (a) When you have broken something which belongs to some one else?
- (b) When you are on your way to school and notice that you are in danger of being late?
- (c) What would you do if a playmate hits you without meaning to do so?

c-8b

4. DEFINITION - SUPERIOR USE (2 out of 4, 1 min each). { Balloon, Tiger, Soldier, Football}

c-7b

5. NAMING 6 COINS (No error) { 50 Np, Rs 1, Rs 2, Rs 5, Rs 10, Rs 20.}

c-8b

6. READING & REPORT: (2 facts; 10 errors; 2 min)

ALTERNATIVES ITEMS:

(1) TYING A SLIPKNOT (model kept before the child: (2-min) (2). Ball & Field (Inferior plan) circle of 6 cms.

6TH - B

DEFINITION OF WORDS

AGE	ITEM. NO	DEFINITION IN TERMS OF USE (4 OUT OF 6. WAIT FOR 1 MIN FOR REPLY)
4	2	{Chair, Horse, Spoon, Doll, Pencil, Blanket}. (Action / Functional use)
8	4	DEFINITION - SUPERIOR USE (2 out of 4, 1 min each) {Balloon, Tiger, Soldier, Police, boll, / Football} (description of Shape, Size, Colour, action, characteristics etc.,)
12	3	DEFINING ABSTRACT WORDS (3 out of 5, 1 min to start definition). {Pitty, Revenge, Charity, Envy, Injustice}
READ AND REPORT		
8	6	(2 facts, 10 errors, 2 min)
9	5	(6 FACTS, 5 ERROR, 1 MIN)
10	6	(8 FACTS, 2 ERRORS, 40 SECONDS)

7TH - A

TEST FOR AGE - 9

c-9b

1. REPEATING 4 DIGITS REVERSED (1 out of 3, read 1 per second)
 { 6-5-2-8; 4-9-3-7; 3-6-2-9 }.
2. MAKING CHANGE (2 out of 3 correct. 15 sec each. read only once again. if required)
 (a) Rs 4 worth: Rs 10 given. How much do you get back?
 (b) Rs 12 worth : 15 given
 (c) Rs 11 worth, Rs 20 given. How much should we get back?

c-9b

3. GIVEN SIMILARITIES - 2 THINGS (2 out of 4. 1 min each).
 (a) Mango - Banana (b) Iron - Silver (c) Steamship Tanga (d) wood - coal.
4. using 3 words is a sentence (2 out of 3. 1 min each repeat once . if required)
 (a) Boy, Ball, River (b) Work, money, men. (c) Trees, Rivers, Lakes.

c-8b

5. READING AND REPORT : (6 facts 5 errors : 1 min)

c-10 b

6. FREE ASSOCIATION - 35 WORDS IN 3 MIN. (record every ½ min score separately). c=10b

ALTERNATIVE ITEMS

- (1) VOCABULARY - 20 WORDS.

7TH - B

REPEATING DIGIST REVERSED

AGE ITEM NO. DIGITS

7	3	3	{ 2-8-3; 4-2-7; 9-5-8 } (1 out of 3)
9	1	4	{ 6-5-2-8; 4-9-3-7; 3-6-2-9 } (1 out of 3)
12	4	5	{ 3-1-8-7-9; 6-9-4-8-2; 5-2-9-6-1 } (1 out of 3)
16	4	6	{ 4-7-1-9-5-2; 5-8-3-2-4; 7-5-2-6-3-8 } (1 out of 3)
AGE	ITEM NO	DIFFERENCES (2 OUT OF 3; 1 MIN EACH)	
7	6	(a) Fly - Butter fly (b) stone- potato (c) wood - glass	
16	3	(a) President -& PM or Governor & CM (2 difference)	
9	3	SIMILARITIES - 2 THINGS (2 out of 4) (a) Mango - Banana (b) Iron - Silver (c) Bus - Auto or steamship / Tanga (d) wooden and charcoal	
14	5	SIMILARITIES - 3 THINGS (3 out of 5, 1 min each) (a) Wool, cotton, silk, (b) snake - cow, sparrow (c) book, teacher, newspaper (d) Scissor, Paisa, Piece of wire (e) Rose, Potato, Tree.	

8TH - A

TEST FOR AGE - 10

C-7B

1. ARRANGING 5 WEIGHTS (2 correct out of 3, trails) { 15, 12, 9, 6, 3 }.
2. REPEATING 20 - 22 SYLLABLES (1 out of 3 or 2 with 1 error each).

- (a) The apple tree makes a cool pleasant shade on the ground where the children are playing.
- (b) It is nearly half - past 10' clock, the house is very quiet and the cat has gone to sleep.
- (a) In summer the days are very warm & fine in winter it snows and i am cold.

3. NAMING THE MONTHS (20 seconds : 1 error in naming , 2 checks out of 3 correct)
Name all the months of the year, what month comes before April, before July, before November.
4. DRAWING DESIGNS FROM MEMORY (1 correct & 1- half correct, expose 10seconds).
5. FINDING RHYMES (3 rhymes for each word , 1 min for each, 2 out of 3 correct).

C-8b

6. READING AND REPORT (8 facts : 2 errors: 40 seconds).

c= 10 b ALTERNATIVES ITEMS

(1)VOCABULARY - 25 words (2) GIVING MONTH & YEAR (both correct).

8TH - B

VOCABULARY

FREE ASSOCIATION

AGE	ITEM NO	NO OF WORDS	AGE	ITEM NO	WORDS	TIME
9	ALTERNATIVE - 1	20	9	6	35	In 3min (Record every ½ sec score separately).
10	ALTERNATIVE - 1	25	16	ALTERNATIVE - 2	60	60 in 3min
12	ALTERNATIVE - 1	30				
14	ALTERNATIVE - 1	40				
16	ALTERNATIVE - 1	45				

9TH - A

TEST FOR AGE - 12

C-12B

1. DETECTING ABSURDITIES (3 out of 5 - ½ min each . after 5 seconds read a second time).
2. CONSTRUCTION PUZZLE (Healy 2 Fernald. 3 times in 5 minutes).

C-8B

3. DEFINING ABSTRACT WORDS (3 out of 5. 1 min each to start the definition)
{Pity, revenge, charity, envy, injustice}.

C-9B

4. REPEATING 5 DIGITS REVERSED (1 out of 3. read 1 per second).
{ 3 - 1 - 8 - 7 - 9; 6 - 9 - 4 - 8 - 2; 5 - 2 - 9 - 6 - 1 }
5. INTERPRETATION OF FABLES (4 marks, 2 correct or the equivalent).
in half credits, each fable correctly interpreted carrying 2 marks. time limit 2 min after final query is put

- (a) Hercules and the Wagoner (b) "The Milkmaid and her plans" (c) the fox and the crow
(b) "The farmer and the stork". (e) "The miller, his son, and the donkey".

C-5b

6. INTERPRETATION OF PICTURES (3 out of 4. time limit 2 min for each picture). {Railway station, reception, motor accident, and domestic scene}.

C-10

ALTERNATIVE ITEMS

- (1) VOCABULARY - 30 WORDS

C-3b

- (1) REPEATING 6 DIGITS, (3 - 7 - 4 - 8 - 5 - 9; 5 - 2 - 1 - 7 - 4 - 6; 4 - 7 - 1 - 5 - 8 - 2).
- (2) COMPREHENSION 4TH DEGREE QUESTIONS? (card no 4 B)

9TH - B

AGE - 12 ITEM NO - 1 DETECTING ABSURDITIES (3 OUT OF 5. ½ MIN FOR EACH)

- (a) 'A man said; " I know a road from my house to the town, which is down hill all the way to the town & down hill all the way back home'.
- (b) 'An engine driver said that the more carriages he had on his train, the faster he could go'.
- (c) 'Yesterday the police found the body of a girl cut into eighteen pieces they believe that she killed herself'.
- (d) There was a railway accident yesterday but it was not very serious. Only 48 people were killed.
- (E) 'A bicycle rider being thrown from his bicycle in an accident, struck his head against a stone and was instantly killed. They picked him up and carried him to the hospital and they do not think he will get well again'.

10TH - A

TEST FOR AGE - 14

1. INDUCTION TEST - FINDING A RULE. (Get rule by 6th folding, unfold & show the paper to the child after he gives his answer to each cutting). {1 - 2, 2 - 2, 3 - 4, 5 - 16, 6 - 32}.
2. DISSECTED SENTENCES (2 out of 3, 1 min each).
3. ARITHMETICAL REASONING (2 out of 3, 1 min each).
 - (a) If 2 pebbles cost Rs 5, how many pebbles can you buy for 50?
 - (b) If a man's salary is Rs 20 a week and he spends Rs 14 a week, how long will it take him to save Rs 300?
 - (c) At Rs 15 a yard, how much will 7 feet of cloth cost?
4. PROBLEM OF ENCLOSED BOXES (3 out of 4. ½ min for each problem).
One large box containing.
 - (a) 2 smaller, (1 inside of each).
 - (b) 2 smaller, 2 inside of each
 - (c) 3 smaller, 3 inside of each
 - (d) 4 smaller, 4 inside of each.

ANSWER {5,7,13,21}.

10TH - B

C-9B

5. GIVING SIMILARITIES - 3 THINGS (3 out of 5, 1 min each).
 - (a) Wool, cotton, silk.
 - (b) Snake, cow, sparrow.
 - (c) Book, teacher, newspaper.
 - (d) scissors, coin, piece of wire
 - (e) Rose, potato, tree.

6. BALL - FIELD, SUPERIOR PLAN.

ALTERNATIVE ITEMS:

C-10B

1. VOCABULARY 40 WORDS FROM VERNACULAR LIST.

11TH - A

TEST FOR AGE - 16

1. INTERPRETATION OF FABLES (8 MARKS)

(A) Hercules and the Wagoner (b) "the milkmaid and her plans" (c) the fox and the crow (d) the farmer and the stork (e) the miller, his son, and the donkey.

2. REVERSING HANDS OF CLOCK (2 OUT OF 3, ERROR -MUST NOT EXCEED 3 MINUTES, TIME LIMIT 1/2 MIN FOR EACH).

(A) " Suppose it is 6.22 'o' clock, that is twenty - two minutes. After 6 can you see in your mind where the large hand would be and where the small hand would be"?

(b) "Now suppose that the two hands were to change, so that the large hand takes the place of the small hand, and the small hand takes the place of the large hand. What time would it then be? similarity with 11:10., 2:46,

ANSWER - {4:30 TO 4:35; 1:53 TO 1:58; 9:10 TO 9:15}.

C=9B.

3. GIVING DIFFERENCE BETWEEN PRESIDENT AND PM OR GOVERNOR & CM.
(2 out of 4 in any forms)

C=9B

4. REPEATING 6 DIGITS REVERSED (1 out of 3 read 1 per second)

{4-7-1-9-5-2; 5-8-3-2-9-4; 7-5-2-6-3-8}.

11TH - B

5. PROBLEM QUESTIONS (2 out of 3, query on (a) & (b) read a second time after second. time limit 1 min to begin reply).

(a) 'A man who was walking in the wood of outside this town stopped suddenly, very much frightened, and then ran to the nearest policeman, saying that he had just seen hanging from the limb of a tree a ... a what?

(b) 'My neighbour has been having strange visitors. First a doctor came to his house, then a lawyer, then a clergyman (or priest) what to do you think happened there?

(c) 'A villager who had came to town for the first time in his life saw a man riding along a street. As the man rode by the villager said: "this man is lazy; he walks sitting down! What was that man riding on that caused the villager to say". He walks sitting down".

C=3B

6. REPEATING 7 DIGITS (1 out of 2 read 1 per sec) {2-1-8-3-4-7-9; 9-7-2-8-4-6-5}.

ALTERNATIVE ITEMS:

C=10B (1) VOCABULARY 45 WORDS (2) FREE ASSOCIATION - 60 WORDS IN 3 MIN.

12TH

TEST FOR AGE - 19

1. USING A CODE (vernacular code to be used, 2 errors, 6 min)
2. INGENUITY TEST (2 out of 3, 5 min each).
 - (a). A mother sent her boy to the river to get 7 pints of water. She gave him a 3 – pint vessel and a 5 – pint vessel. Show how the boy measures 7- pints? Begin with 5 – pint vessel?
 - (b). 5 and 7 – pint vessels to get 8. Begin with 5 – pint vessel.
 - (c). 4 and 9 – pint vessels to get 7. Begin with 4 – pint vessel.
3. DIFFERENCES IN ABSTRACT TERMS. (3 out of 4)
 - (a) Laziness and idleness.
 - (b) Pride and vanity.
 - (c) Poverty and misery.
 - (d) Dishonour and disrepute.
4. BINET'S PAPER CUTTING TEST.(same before XIV,l, drawn on piece of paper) 2 min.
c-6b
5. REPEATING 30 SYLLABLES (1 out of 2)
6. REVERSING TRIANGLE IN IMAGINATION (2 min)
ALTERNATIVE 1.
VOCABULARY 55 WORDS.

13TH

TEST FOR AGE - 22 (VERY SUPERIOR ADULT)

1. COMPREHENSION OF PHYSICAL RELATION. (2 out of 3)
 - (a). Problem regarding the path of cannon ball,
 - (b). Problem regarding the weight of a fish in water,
 - (c). Difficulty of hitting a distant mark.
2. REPEATING EIGHT DIGITS (1 OUT OF 3)
7 – 2 – 5 – 3 – 4 – 8 – 9 – 6; 4 – 9 – 8 – 5 – 3 – 7 – 6 – 2; 8 – 3 – 7 – 9 – 5 – 4 – 2 – 6;
3. REPEATING THOUGHT OF PASSAGE HEARD.
“Many opinions have been given on the value of life. Some call it good, others call it bad. it would be more nearly correct to say that it is mediocre. For on the hand our happiness is never as great as we should like, and on the other hand our misfortunes are never as great as our enemies would wish for us. it is this mediocrity of life which prevents it from being radically unjust”.
4. REVERSING TRIANGLE IN IMAGINATION (BINET'S FORM) (time limit 2 minutes to complete to diagram).
5. REPEATING SEVEN DIGITS REVERSED (1 out of 3)
{4 – 1 – 6 – 2 – 5 – 9 – 3; 3 – 8 – 2 – 6 – 4 – 7 – 5; 9 – 4 – 5 – 2 – 8 – 3 – 7}
6. FREE ASSOCIATION OF WORDS (80 words in 3 minutes)
ALTERNATIVE ITEM 1. VOCABULARY 60 WORDS.

14TH

(A) HERCULES AND THE WAGONER'.

'A man was driving along a country road, when the wheels suddenly sank in a deep rut. The man did nothing but look at the wagon and call loudly to Hercules to come and help him. Hercules came up, looked at the man, and said: " put your shoulder to the wheel, my man, and whip up your oxen." then he went away and left the driver.

(B) 'THE MILKMAID AND HER PLANS'.

' A milkmaid was carrying her pail of milk on her head, and was thinking to herself: the money for this milk will buy four hens; the hens will lay at least 100 eggs; the eggs will produce at least 75 chicks; and with the money which the chicks will bring i can buy a new dress to wear instead of the ragged one i have on. at this moment she looked down at herself, trying to think how she would look in her new dress, but as she did so, the pail of milk slipped from her head and dashed upon the ground. Thus all her imaginary schemes perished in a moment.

15TH

(C) THE FOX AND THE CROW'.

' A crow, having stolen a bit of meat, perched in a tree and held it in her beak. a fox, seeing her, wished to secure the meat, and spoke to the crow: " how handsome you are and i have heard that the beauty of your voice is equal to that of your form and feathers. Will you not sing for me, so that i may judge whether this is true? The crow was so pleased that she opened her mouth to sing and dropped the meat which the fox immediately ate'.

(D) 'THE FORMER AND THE STORK'.

'A farmer set some traps to catch cranes, which had been eating his seed. With them he caught a stork. The stork which had not really been stealing, begged the farmer to spare his life, saying that he was a bird of excellent character, that he was not at all like the cranes and that the farmer should have pity on him. But the farmer said, "I have caught you with these robbers the cranes, and you have got to die with them."

(E) THE MILLER, HIS SON, AND THE DONKEY.

'A miller and his son were driving their donkey to a neighbouring town to sell him. They had not gone far when a child saw them and cried out, "what fools those fellows are to be trudging along on foot when one of them might be riding!" the old man, hearing this, made his son get on the donkey, while he himself walked. Soon they came upon some men "Look". Said one of them, "see that lazy boy riding while his old father has to walk," on hearing this the miller made his son get off, and he climbed upon the donkey himself. Further on they met a company of women, who shouted out, "why you lazy old fellow to ride along so comfortably while your poor boy there can hardly keep pace by the side of you! And so the good-natured miller took his boy up behind him and both of them rode. As they came to the town a citizen said to them, "why, you cruel fellows! You two are better able to carry the poor little donkey than he is to carry you." "Very well," said the miller, "we will try." so both of them jumped to the ground, got some ropes, tied the donkey's legs to a pole, and tried to carry him. But as they crossed the bridge the donkey became frightened, kicked loose and fell into the stream.'

Domain- wise classification of items adapted to BKT*

(*Lezak, M.D.(1983), Neuropsychological Assessment(2nd Ed.) Oxford University Press, P 288-291).

Domains ↕	LANGUAGE	MEMORY		REASONING			CONCEPTUAL THINKING	VISUO MOTOR	SOCIAL INTELLIGENCE
		Meaningful	Non Meaningful	Non Verbal	Verbal	Numerical			
22	6, A1	3	2,5	1,4					
19	A1	5		6	1	2	3	4	
16	A1,2		4,6	2	5		1,3		
14	2,A1			6		1,3,4	5		
12	3,A1		4,A2	2	1		5		6,A3
10	5,A1	2,6	4	1					3,A2
9	4,6,A1	5	1			2	3		
8	4	6	1,2	A2		1		A1	3,5
7		1	3	3,6		5	6	2	4,A1
6	A1		1			4			2,5
5	2	3		2,A1		6			1,5,5,A1,2,3
4		4	1	6		5		6	3
3	1,2	5	3						4,A1,2
Age ↑		Meaningful	Non Meaningful	Non Verbal	Verbal	Numerical	CONCEPTUAL THINKING		
LANGUAGE		MEMORY		REASONING			CONCEPTUAL THINKING		

BINET-KAMAT TEST OF INTELLIGENCE
(RECORDING SHEET)

Name of Client: Gender: Date of Birth: / /20.... Age: Yrs Months School: Class: Socio-economic Status: Language(s) Spoken:	Date of Testing : / / 20.... Client Registration No: Name & Address of the Parent/Guardian
--	--

Item No	AGE LEVELS										
	III	IV	V	VI	VII	VIII	IX	X	XII	XIV	XVI
1.											
2.											
3.											
4.											
5.											
6.											
Alt #1											
Alt #2											
Alt #3											

Basal Age (BA): (in months)

Terminal Age (TA): (in months)

Mental Age (MA): (in months)

IQ: { MA/TA X 100 }

Behavioural Observations:

Diagnostic Impression }
& Recommendation(s) }

(Name of the Psychologist/Tester)
Reg. No:

Signature }
& Date }

BINET KAMAT

TEST OF

INTELLIGENCE

CHAPTER TEN

TESTS FOR YEAR III

III. 1. *Pointing to parts of the body (3 out of 4)*

TEXT AND PROCEDURE

'Show me your nose.'

'Put your finger on your nose.'

Same with eyes mouth and hand.

If two or three repetitions fail to bring a response, overcome timidly by pointing to chin or ear and questioned.

'Is this your nose?'

Answer is by winking, opening the mouth, etc., should be counted as satisfactory.

Remarks: Three correct responses out of four are required. This test is much easier than the next one - 'naming objects'. At this stage the association between the name of an object and the object itself is made, but the association between this pair and the muscular movements of the vocal organization is not yet made. This illustrates beautifully the way in which our psychological organization is slowly built up into more and more complex wholes.

1. In what follows, the text of the tests and the procedure in detail are given in English. The tests and test procedure in the two Indian languages, Kannada and Marathi, are published separately by Purohit and Purohit, Poona. The English version is given only for English readers and for comparison between this revision and previous revisions. It should, however, never be used with Indian children. With them the Indian language version which is their mother-tongue and the school language should only be used.

2. This test is passed by nearly all the children who are just 2 years old. Being the first of the third year it should naturally be passed by nearly all children who have completed 2 years and have not yet completed 3 years.

Really at this beginning of the scale we require smaller fractions of age than one year. It would be proper, for example, to take children between 2 years and 2 years 2 months and require 50 per cent of these to pass this test. This would be the most scientific way of fixing this lower end. Similarly for III, 2 we should take children between 2 years 2 months and 2 years 4 months and so on more complex wholes. Further there is the question whether the child explores and studies its own body first or that of a doll or another person - its mother, brother, sister or servant. Probably explores those limbs of its own which are visible to it, then those of a second person, then compares and generalizes in its own way and thus finally attains the concept of a nose, a mouth and a hand and so on.

III. 2. *Naming Familiar Objects. (3 out of 5)*

TEXT AND PROCEDURE

Use (1) a key; (2) a paisa; (3) a closed penknife; (4) a watch; and (5) an ordinary lead pencil (taking care to use things in common use).

'What is this?' or

'Tell me what this is.'

Remarks. This test is harder than the previous one as is evident from the percentage of children who pass it. The previous test requires the association of auditory images of the names of things and visual images. This test requires over and above this the association of kinesthetic and muscular movement images with the previous group. Hence it is more advantaged. It is interesting to note that the most common objects from the everyday experience of little children of this age selected by Binet hold good in such a different environment as the Indian. In scoring this test baby talk should be allowed. Children of this age pronounce the words in a strange way, or they use abbreviated forms of the names. It is doubtful, however, if onomatopoeic words should be allowed. Thus they call a clock or a watch a 'tick-tick'. This may mean that onomatopoeic words come earlier than that the proper names. On the contrary some children indulge in such talk and their parents encourage it out of endearment even after their vocabulary is sufficiently enlarged. If such is the case the test should be scored plus; otherwise minus.

III. 3. Repeating Two Digits (1 pot of 3. Order correct. Read 1 every half second)

TEXT AND PROCEDURE

Listen: Say 4-2' (Practice Series)
Now say: 3-7; 6-4; 7-2.

Remarks. With little children the series should be given rather rapidly. Terman prescribes an interval of one second between two digits. The interval is found to be too long with very little children, who begin to utter a digit before the next digit comes. Hence one has to give the digits rather rapidly, about one every half second. This test is in Binet's original test; but is omitted by Terman probably because he finds it too easy for year III. Burt has it in year III. With little children below the 1-year level it is very difficult to secure their attention. Hence the trial series and even the first series may be spoiled and may have to be repeated more than once, before attention is secured and silence broken. Then give the second and third series uttering them only once. There is very great divergence in the location of the repetition of 'digit' tests by the various workers. This difference is due mostly to want of uniformity in the procedure; some give the digits every half-second, other every second. Some will have done with the whole hierarch of digits, one after the other. Others will give them in their appropriate places in the several age groups. In particular, Burt's location of the test appears strange. His result must be due to the fact that he gives the whole hierarchy at once. On the one hand this means that, especially with the later tests, there has been much recent practice in the repetition of digits; on the other hand it might lead to fatigue towards the end. On the whole it appears that this procedure makes the tests more easy.

In the present revision the tests are meant to be given only in their appropriate places in the several age-groups. The location of the several series by different workers is given below:

	Binet	Terman	Burt	Present revision
2 digits	III	-----	III	III
3 digits	IV	III	IV	IV
4 digits	--	VI	V	VI
5 digits	VIII	VII	VI	VIII
6 digits	---	X(1 of 2)	VIII	XII
7 digits	XV	XIV(1 of 2)	XI	XVI(1 of 2)
8 digits	--	XIX	--	XXII

It will be seen the location of the test in the present revision agrees best with Binet's.

III. 4. Enumeration of Objects in a Picture. (At least two objects in a one Picture. At least two objects in any one picture are satisfactory for this test.)

TEXT AND PROCEDURE

- A) Railway Station. 1
B) Reception. 2
C) Motor accident. 3
Say,

'Now I am going to show you a pretty picture. Tell me what you see in this picture.'

'Look at the picture and tell me everything you can see in it.'

'Show me the -----'. (Only one such question is permissible)

'That is fine; now tell me everything that you see in the picture.'

'And what else?'

Remarks. Present these pictures before the child in the order shown above. In order to pass this test the child must name at least *two* objects in any *one* picture *spontaneously*. This test is more difficult than naming actual objects. This additional difficulty here appears to be the transformation of pictures of two dimensions to actual objects of three dimensions. Of course, the pictures cast on the retina in any case are of two dimensions and a skillful artist draws his pictures in perspective, but however skillful the artist the reality of the actual things is wanting in these pictures. Thus the front of the motor-car in picture No. 3 above is perceived by several children as a writing stage though the same children would never make the mistake with the actual motor car. This stage in the mental development of children is called by Binet 'the identification stage'. Terman called it 'the enumeration stage'. Then comes the stage of description about the sixth year of age and finally the stage of interpretation about the twelfth year. What is required for description is the grasping of the properties or actions of individual things; for interpretation what is required is the grasping of the relation of all the things or object and finding only one meaning for the whole picture. This test requires the highest type of conceptual process requiring analysis and synthesis. These pictures are selected from among a large number and are found to give the best results.

III. 5. Repeating six to seven syllables. (1 out of 3)

TEXT AND PROCEDURE

'Can you say, "mamma"'

'Now say, "nice kitty,"'

So much is for practice or to overcome shyness. Then continue asking the child to say,

a. 'I have a little dog.'

b. 'The dog runs after the cat.'

c. 'In summer the sun is hot.'

Remarks. For a plus in this test any one of the three sentences must be correctly repeated after only one reading. There should be no transposition or omission of any of the words. Indistinct talk owing to baby pronunciation should be ignored; for example, many children of this age cannot pronounce as 'r' or a 'k'. In Kannada or Marathi script each letter written is a complete syllables, but in the Roman script each syllable usually requires two or more letters! So although the number of written letters is less in the Indian scripts, ability to repeat speech depends only on sound, that is, only on the number of syllables. In this test the syllables to be repeated are syllables having sense and not non-sense syllables the number of syllables repeated would be much less. If the syllables are sense syllables somehow the associations of the meaning of words come to help of children. Besides familiar speech sounds require a number of muscular and vocal co-ordination in which the child must have already had practice.

III. 6. Comparison of Lines. (3 out of 3: or 5 out of 6)

TEXT AND PROCEDURE

Put the card before the child with the lines in a horizontal position. Say, 'Here are two lines. This is one and this is another. Look closely and tell me which one is longer. Put your finger on the longest one.'

Reverse the position of lines and say,

'Now show me the longest.'

Turn the card, make a third presentation and repeat the same formula.

Remarks: The greatest difficulty with children of the third and fourth years is to secure their attention. The children are very fidgety; after securing a child's attention, although you may place the card before him, he casts a casual glance at it and turns it over and looks at other side. This is only one instance of the great *curiosity* of children in looking at things from different sides and of their pleasure in *doing* things and turning and turning them and analyzing them. In order to fix attention better and make the formula more acceptable we have added the words- 'This is one and this is another.' The original formula, 'See these lines,' is found to be too bald and unsuited to children of this age. They never wait to observe the lines very closely but turn over the cards. At this age we have to repeat the formula again and again until attention is secured. So the original formula will have to be repeated several times until this object is attended. With the new formula attention is secured more quickly and hence it is better to use the new formula but it does not seem to make the test easier in any way. Terman used the superlative form of 'long' also in the alternative formula. But in the Indian languages there are no single words for superlative and roundabout way of signifying the superlative are of no use. Hence one formula is enough. The child must point definitely to one of the two lines. If the child points to the wrong line never make a suggestion by look of displeasure or by any kind of sound; accept any kind of answer with equal coolness and encouragement. According to Binet, success in this test depends more on comprehension of direction than on actual discrimination of lengths.

III. Alternative I. Giving Sex.

TEXT AND PROCEDURE

To a boy: 'Are you a boy or girl?'

To a girl: 'Are you a girl or a boy?'

If there be no answer, say,

'Are you a girl?' (If a boy), 'Are you a boy?' (If a girl).

'Well what are you? Are you a boy or a girl?' (or vice versa)

Remarks. Note that with boy we end the questions with 'a girl' and with girls we end it with 'a boy'. The reason is that children have a tendency to repeat the later alternative without thinking. This is form of automatism. The word 'little' in Terman's formula of 'a little boy' is omitted as it does not in any way improve the comprehension of the formula; one of the contrary there is a tendency of contrast 'little' with 'big' as if the word 'little' were emphasized. Real sex consciousness is not required in answering this question; for children seem to make distinctions between boys and girls mostly by their dresses and costumes and by the way they are address in terms of pronouns. In giving the formula the words 'boy' or 'girl' should be clearly emphasized.

III. Alternative 2. Giving Proper Name.

TEXT AND PROCEDURE

'What is your name?'

How do people call you?'

Remarks. This is a new test. Binet's test is to give the family name. It is not usual however with Indian children to be called by their family name nor does the family name of their parents occur so often in daily household talk. Thus this test if it is to be retained at all goes to the fifth year. As the family name test was found to be unsuitable for this age, the proper name was tried and was found to give fairly good results. The test has, therefore, been put down as an alternative test for this year, as all the other proper tests of this year are better than this one.

Note on Location of Tests of Year III. In strict mathematical conformity the tests of this year must be arranged in order of difficulty in such a way that test 1 should be passed by about 50 percent of children between 2 years and 2 years 2 months; test 2 should be passed by about 50 percent of children between 2 years 2 months and 2 years 4 months; test 3 by 50 percent of children between 2 years 4 months and 2 years 6 months and so on. The fractions of mental age to be determined for this year are very small and very significant and hence the scale is required to be more sensitive. The following table gives the percentage of passes for several groups of children considered from this point of view. It should be noted, however, that the number of children in each such group is very small and hence too much importance should not be attached to the discrepancies that are found.

In the table given below, III, 1, III, 2 etc., are the first, second and so on tests of year III. The number of children examined for each age group and each test is shown in brackets. Thus there were 9 children examined of chronological age 2 years 0 month to 2 years 2 months. All these passed the first test of year III; hence the percentage of passes for this group is 100 and the number of children is shown in brackets, as (9). Similarly 4 children were examined in the group 2 years 6 months to 2 years 8 months chronological age. Of these 3 passed the test III, 6; hence the percentage of passing this test is 75 and the number of children examined is shown in brackets below this as (4).

Percentage of passes with number of children examined.

Age group Of children Yr. M.		III, 1	III, 2	III, 3	III, 4	III, 5	III, 6	III, Alt. 1	III, Alt. 2
2	0 to	100%	33%	33%	22%	13%	0%	0%	25%
2	2	(9)	(9)	(9)	(9)	(8)	(4)	(8)	(4)
2	2 to	100%	50%	68%	39%	25%	25%	13%	66%
2	4	(8)	(8)	(6)	(8)	(8)	(4)	(8)	(3)
2	4 to	100%	25%	33%	25%	0%	50%	0%	50%
2	6	(4)	(4)	(3)	(4)	(3)	(2)	(4)	(2)
2	6 to	100%	75%	100%	50%	75%	75%	50%	66%
2	8	(4)	(4)	(4)	(4)	(4)	(4)	(4)	(3)
2	8 to	100%	66%	100%	33%	66%	66%	33%	100%
2	10	(3)	(3)	(3)	(3)	(3)	(3)	(3)	(2)
2	10 to	100%	100%	100%	84%	70%	70%	56%	75%
3	0	(7)	(7)	(6)	(7)	(7)	(7)	(7)	(4)

CHAPTER ELEVEN

TESTS FOR YEAR IV

IV. 1. Repeating Three Digits. (1 out of 3. Order correct. Read 1 every half-second)

TEXTS AND PROCEDURE

'Listen: say 7-3.' (Practice series)

'Now say: 6-4-1; 3-5-2; 8-3-7.

Remarks. See remarks under III, 3. If the examination of a child is to begin with this year does there in with this test. Begin with the next test. 'Discrimination of form: or the next after that. 'Comprehensive first

Shyness is overcome. If the child is self-without shows differences, coax him and say, 'I hence on can repeat them till that. "Very often the practice series will have to be repeated several times until silence is broken. Always observe the rule however, that only three attempts by a child are allowed and plus should be scored if any one attempt is correct after the series is given only once by the examiner.

IV. 2. Discrimination of Forms. (Kuhlman. 7 out of 10)

TEXT AND PROCEDURE

Use the double set of geometrical figures supplied in the packet of test material. One of these should be cut out so that the several geometrical figures are separated.

Place the uncut card before the child, put the cut out figure of a circle in the position marked with a X on the uncut card and say: ('Show me one like this'. At the same time pass the finger around the circumference on the cut out circle.)

'Do you see all these things? (Simultaneously run the finger over the various forms on the uncut card.)

'And do you see this one?'

'Now, find another one just like this.'

'Take the square next, then the triangle, and then the others in any order.
Correct first error, saying, 'No, find one just like this.'

Remarks. At this level of intelligence children just begin to discriminate between different shapes of figures. This discrimination comes as a result of constant comparison and contrast of figures. The child has to see squares and circles several times and constantly compare and contrast them with one another, so that distinctive images of a square and a circle are fixed in his mind and later on he can carry in his mind these images and compare them with the actual perception of a square that is placed before him. Then only can he see the square as a square. The child who has not carried out these processes in his mind sees no difference between a square and a circle. The same thing is implied in the test of copying square assigned to this same year, IV, 6. Many children draw a round figure like a circle in copying a square. This results from the failure in to grasp the difference in the shapes of two figures more than the failure practice of muscular co-ordination. * Similarly even with adults. The writer remembers how very difficult it was for him for a long time to distinguish between two brothers who looked alike. It was only after seeing the brothers several times and in different setting and observing the minute details of their respective features that he learnt to distinguish between them. Then he began to laugh at himself for having failed to distinguish between them when there were so many distinctive marks in their features. Similarly with children who learn to distinguish between simple figures- simple from our point of view- only after careful observation, comparison, and contrast. To distinguish between the several figures in this test it is not necessary that the child should know the names of the figures; nor is it necessary that

he should know how to draw them. It is enough for the child to be able to grasp the image of the figure shown and retain it and compare it with the figures on the card he has to go back several times to the figure shown, and carry the image as many times for comparison with the figures in the card.

*It is true that the image of a figure may be in part kinesthetic and that ability to grasp difference in shape may be bound up with practice in muscular co-ordination. But in the test of drawing a square it does not appear to the writer as though kinesthetic imagery plays any important part. There is room here for further experiment.

IV. 3. Comprehension. First Degree. (2 out of 3)

TEXT AND PROCEDURE

- a) 'What must you do when you are sleepy?'
- b) 'What ought you to do when you are cold?'
- c) 'What ought you to do when you are hungry?'

After waiting about 20 seconds for an answer the questions may be repeated more than once. It is most important at this age to secure attention and for this purpose it is frequently necessary that the questions should be repeated more than once.

Remarks. These are the simplest of the comprehension questions. They are again to be found in year VI, years VIII and year XII. It may be asked what makes the questions harder and harder. One answer is that the number of items woven into the situations are increasing, making the situations more and more complex and hence more and more difficult to imagine. Even in the question of this year, by making the children call up the image of the situation the questions are made more difficult than they would be if the children were made to meet the actual situations. In such tests the sub-tests have all to be of the same difficulty. There are so in this age, but they are not exactly so in the higher ages. In the latter case there is the danger of the score of the test being determined only by the most difficult of the sub-tests. On the whole these comprehension questions are very good in that they make children think in the abstract and such thinking is a quality only of superior intelligence. In repeating the questions the formulae may be slightly modified (as shown in brackets in the vernacular versions) because the children may not be equally familiar with the different kinds of phraseology. Questions (a) and (b) were suggested by Binet in 1990 but were not standardized. They were standardized for the first time in the Stanford revision, which also added the last one namely (c).

IV. 4. Repeating Twelve to Thirty Syllables. (1 out of 3 absolutely correct: or 2 with 1 error each).

TEXT AND PROCEDURE

- 'Listen, say this: "Where is Kitty?"' (For practice only)
'Now say this:
- a) "The boy's name is John. He is a very good boy".
 - b) "When the train comes you will hear the whistle blow".
 - c) "We are going to have a good time in the country."

Remarks. With the test of repeating syllables as also with that of repeating digits, only one reading is allowed. After securing the attention of the child the first sub-test should be given. If the child does not respond owing to timidity or crossness repeat the same formula. But this cannot be counted for success. Then the second and the third sub-test should be given once only. Defects in pronunciation should be ignored. Binet, Burt, and Bobertag standardized a sentence or sentences of 10 syllables and located the test in year V. But the statistics of the present study agree with Terman's in locating the test with 12 or 13 syllables in year IV.

IV. 5. Counting Four Paise (Coins).

TEXT AND PROCEDURE

Place the paise on the table before the child in a horizontal row.

'See these paise. Count them and tell me how many they are:

Count them with your finger this way.'

'One, - 'Now go ahead.'

'No, count them with your finger, this way.'

The test is passed only by counting tallies with the pointing.

Lastly ask, 'How many?'

Remarks. The most important thing in this test is that the child should count 'one, two, three, four; loudly, at the same time putting his finger on the paise, and that the pointing and counting should tally; and so schooling or learning to recite numbers has no effect on the ability to pass this test. It requires an intelligent interest in numbers and the child's spontaneous interest in them makes him handle things and count them. Quite a number of children who are paid great attention while being taught the numbers fail this test by not putting their finger on the paise to tally with their counting. They are slow in pointing with their finger while their lips move faster in reciting the numbers. The performance of this test requires higher mental ability than that required for mere mechanical counting of numbers. This test however does not require a full concept of the number four on the part of children. Probably what the children know at this age is that four is more than one and two and is a fairly large number. It is also very doubtful whether the child recognizes the whole group as four or calls the last paise as 'four'. But the reply to the last question 'How many' is generally 'four'. Do not score the test plus unless the answer to this last question is correct. The procedure of this test, which is after Terman, goes a little further than Binet. Who simply say, 'Count then aloud; count them with your finger.' Terman adds, 'Count them with your finger this way- one'. Binet locates the test in year V; but Burt who follows exactly the same procedure locates it in year IV.

IV. 6. Copying a square.. (1 out of 3 pencils)

TEXT AND PROCEDURE

Put the printed model of a square before the child and say; 'You see that (point to the Square).. I want you to make one just like it. Make it right here'. (Show where it is to be drawn.) Go ahead. I know you can do it nicely.

Don't run the finger around the four sides. Give three trials, saying catch time, 'Make it exactly like this'.

After the child has drawn all three, ask;

'Which one do you like the best?'

Remarks. The model used should be the one supplied in the packet of test material. If another is used care should be taken to see that it is of the right size that is about 4 centimeter on each side. It should also be drawn in thick black lines, preferably in Indian ink. The child should be asked to draw in pencil, as children of this age cannot handle a pen properly. Success in this test depends; it appears, more on the ability to distinguish between different kinds of forms than on the acquisition muscular co-ordination. (See remarks under IV, 2. P. 167). Children generally are found to cast a casual glance at the figure and finish off the drawing at once. Those who have not reached the mental level draw a round figure. The formation of the corner is the hardest thing in this as well as in the 'diamond' test. (VII, 2). The corner drawn by immature children in the 'diamond' test is more interesting.

Only one success out of three is enough for a pass in this test. The answers should be carefully compared with the model-scoring card before assessing. The square drawn should be recognizably a square but mathematical accuracy or anything near it is not required. Even with squares drawn by adults the perpendicular side will always be less than the horizontal side. This is one of the optical illusions. Note also

whether the child draws the square continuously or draws the opposite side together. Sometimes in the formal case one of the corners is a little rounded off. The first trial of the child is generally the best. This proves that the task is a heavy one from the point of view of the child, requiring very great effort and attention. During the second and third attempts the child, because of fatigue, cannot bring all his energy and effort to bear.

Binet requires the child to draw with pen and ink and further gives only one trial. This seems to make the test slightly more difficult. He locates it in year V. It is not however made as difficult as the unfamiliarity of the material would suggest. This proves that the more important factor for success in this test is sufficient mental development enabling discrimination between different kinds of forms.

IV, Alternative. Comparison of Two Weights. (3 out of 3; or 4 out of 5)

TEXT AND PROCEDURE

Use the two boxes of 3 grams and 15 gram in weight and present them during the trials in the following order, 3-15; 15-3; 3-15; 15-3; 3-15.

'You see these blocks. They look just alike, but one of them is heavy and one is light. Try them and tell me which one is heavier.'

If the child does not know how to proceed and simply lifts any one blindly and hands it over, say, 'No, that is not the way. You must take the boxes in your hands and try them, like this.'

Show the child the proper method of weighing the boxes in the hands by taking the boxes in your own hands one after the other. In the second trial, the weights are shuffled and their positions reversed. In the third trial, the positions are the same as in the first.

Materials. Stiff and strong cardboard pill-boxes about 4 centimeters in diameter make very good weight-boxes for this test as well as for X, 1. Five boxes of uniform size shape and color should be selected. They should be accurately weighted by stuffing them with lead foil and cotton until they weigh 3, 6, 9, 12 and 15 grams respectively. The lead should be exactly in the center of the boxes with cotton on both sides and should not be loose nor should it rattle. The lids should then be glued on permanently. The boxes should be marked underneath with distinguishing letters such as P, S, Y, C, H so that by silently turning them over the examiner can see which is which. The child, however, should not know that they are so marked. For the test of this year use only the first and the last, i.e. 3 grammes and 15 grammes.

Remarks. If the child has reached the mental age he generally hands over the right box. Terman accepts only 2 correct responses out of 3. This leaves a good deal to chance. Hence it is better to accept 3 out of 3 or 4 out of 5 since once the child understands what is required his responses are always correct. The difference in weights of these boxes is sufficiently large to be discriminated by children of this age. Backward children are always prone to hand over the box on the right (or left) hand side. This is an instance of stereotype to which backward children are generally prone. Success in this test seems to depend mainly on the ability to understand verbal instructions. The idea of making comparison and selecting one of two or many is rather difficult for children below this mental level. After the instructions are understood the requirement for reaching the goal are to be kept fixed in the mind until that goal is reached. This in itself requires a tremendous expenditure of energy in the form of sustained attention. This is like lifting a heavy weight and placing it on the table. Some may succeed in doing this while others may lift the weight three fourths of the height and then drop it through exhaustion of the strained muscles; others still may lift it to one-fourth of the height; and lastly there may be some who may not be able to lift the weight from the ground at all. The test appears to lie on the border of age IV and V, and inclines more on the side of the former. It is located in year IV in the present revision.

Burt presents the boxes of 3 grammes and 12 grammes and 6 grammes and 15 grammes alternately and gives three trials. All trials are required to be correct. He located the test in year V.

CHAPTER TWELVE

TESTS FOR YEAR V

V. 1. Aesthetic Comparison. (No errors)

TEXT AND PROCEDURE

Present the three pairs of pictures on the card before the child one at a time, hiding the other pairs from view by means of a piece of cardboard or sheet of paper. Say,

'Which of these two pictures is the prettiest?'

Remarks. The pictures for the test are the same as Binet's; but Indian costume is substituted for European. Thus the nature of the difficulty of the test is the same as that of the original. The test is found suitable for year V as was found by Terman. Binet located in the year VI. Burt located in the year IV. This test is very interesting test and children respond to it in a lively manner. It correlates highly with intelligence; which shows that aesthetic sense develops with intelligence. This throws doubt on the common supposition that artistic ability, which undoubtedly requires aesthetic sense, does not correlate with intelligence; that dull children often make good draftsman and artists. Similarly it is supposed that musical ability does not correlate with intelligence. The writer knows one or two backward children who are very good artists or singers. One boy was found in the course of testing to be backward in intelligence. His school work confirmed the same the same finding, but he had very good drawing and artistic ability. He could, for example, draw beautiful diagrams and pictures in his science paper, but his answer revealed that he had no clear grasp of scientific principles. Another case was that of a girl of about 8 or 9 years, who appeared very backward and could not talk, but could sing most beautifully. She could pick up any tune quickly and having heard it once or twice could sing it in proper notes without uttering the words. To the writer these stray instances appear only as exceptions. We are not likely to make a note of a hundred intelligent people who are good artists and singers but one or two dull or defective people who show high drawing or musical ability strike our imagination and we remember them and quote them as instances of popular beliefs, which by the way must have started in the same fashion. The real thing is that an intelligent person shows his intelligence in whatever direction his energy is directed. A few dull people finding that they cannot compete with their more cleverer fellows in their class studies or the ordinary business of everyday life may have somehow developed a liking for these subjects and may have been concentrating their attention on them and thus stand out in relief against their fellows; but this does not disprove the general rule.

V. 2. Definition in terms of use. (4 out of 6. Wait about one minute each time for response)

TEXT AND PROCEDURE

Use the words: chair, block (or hours), spoon, doll, pencil and blanket, Say, 'You have seen a chair. You know what chair is. Tell me, what is a chair?'

If the child does not give a definition but says, 'That is the chair' or 'A chair is a chair', etc., say, Yes, but tell me, what is a chair?'

If child is too shy or cross, coax him and repeat the same formula with first word several times. But once the child response never shows by gesture or nod or word that the definition is not correct. Accept every definition with approval. But if the definition is very bad do not give the impression that is the best definition, but just leave the child alone. It is not permissible, as some who are not properly initiated into the art do, to ask, 'What do you do with a spoon?'

The second word for definition is 'horse' or 'bullock'. If the child has seen horses and knows their common uses this word is to be preferred to 'bullock'. With 'bullock' the difficulty is that the child defines it very often as 'a cow'. Then it is necessary to ask the child further;

'A bullock is not a cow. Is it? Tell me what a bullock is'.

Remarks. Binet has included three test or definition in his scale. The first one, 'definition of familiar words in terms of use,' he locates in year VI; the second one, 'definition of familiar words in terms superior to use,' he locates in year IX; and the third one, 'definition of abstract words,' he locates in year XII. He has thus discovered a very ingenious way of classifying children's response according to their mental ages. The present test is a good test and there is no difficulty in determining whether the definition given by any child is superior to use. We shall discuss this question later. The objects chosen are all familiar object, so there is question of the child's not knowing the words. The test therefore examines only the child's power of definition or his power of expression. This power of expression is a sign of intelligence. Dull children know the nature of objects, but the development of abstract language necessary to give expression to their perceptual mass is not sufficiently advanced. A good deal depends also on the actual objects chosen. If the same words used in this year are used in year VIII also, for definitions superior to use, we could not say whether we would get suitable answers, or whether if we had used the words of the year VIII in year V also we would have got suitable answers for year V. It appears that some are more suitable for use definitions and others for class definitions. Terman uses different sets of words for the two years and they are standardized for those years only. We have followed Terman in this respect. It would have been better, however, if the same set of words had been used for the two ages, as was done by Binet and Burt; but the selection of the objects would require very great care. The selection of a different set of words has one great advantage, namely, that if a child starts giving use definitions, he usually continues to do so with the rest of the words. The testing can then go on with other kinds of test. At a later stage he is presented with new words, which give him the chance of attempting class definitions. Thus a second chance is given to child with words a little more difficult and he is obliged to do his best.

Failures are generally of the nature of silence or repetition of the same word or the reply, 'I don't know'.

V, 3. Three Commissions. (All three correct in proper sequence)

TEXT AND PROCEDURE

After getting up from the chair and moving with the child to the center of the room, say, 'Now, I want you to do something for me. Here is a key. I want you to put it on that chair over there, then I want you to shut (or open) that door, then bring me the box which you see over there (pointing in turn the to the object designated). Do you understand? Be sure to get it right. First put the key on the chair, then shut (open) the door, and then bring the box (again pointing). Go ahead.'

Remarks. In giving out the formula read out the instructions slowly and clearly. Stress the important and key words such as 'key', 'door', 'first', 'then', etc. While the child is carrying out the directions do not stare at him. A casual look of approval is all that is required. Never indicate or suggest the order of action by word or look or gesture, nor say any such thing as 'What is the first thing to do?' or 'What next?' The execution of the triple command requires the holding fast in the consciousness of the three commands in their proper order and the will to carry them out as directed by the memory. If the memory of their proper order slips, either the order of performing the commands is changed or one of the commands is forgotten or the child stops in the middle of the execution. The comprehension of the serial order of things must also be a great factor in the successful performance of this test. Binet located the test in year VII in his 1911 series; but it is found quite easy for year V.

V, 4. Distinguishing Right and Left. (3 out of 3; or 5 out of 6)

TEXT AND PROCEDURE

'Show me your right hand.'

'Show me your left ear.'

'Show me your right eye.'

If there be only one failure repeat the above formula substituting 'left' for 'right' and 'right' for 'left' in them.

Remarks. In reading the formula stress the key words 'right', 'left', 'hand', 'ear' etc. Don't suggest the answer by a glance or any other gesture. Nor should the examiner repeat the question by stressing 'right' or by simply saying 'right' if the child shows the left hand (as if the one that he showed were wrong). The child first shows the wrong limb and then corrects himself spontaneously, score him correct. Generally in such cases the rule is that the second answer should be taken into account for scoring. Our statistics show that the test is easy enough for year V. Binet locates it in year VII in his 1911 series. Terman puts it in year VI and Burt also in the same year. Binet and Burt use only 'right hand' and 'left ear' and require both to be correct.

There is a good deal of discussion on the question of space orientation with children. It has been observed and experimentally proved that children distinguish between up and down, that is vertical distinction in space, more readily than right and left. What is the cause of this? Several explanations are offered, any one or more of which may be responsible for this distinction.

(1) The most important cause for this appears to the writer to be that in vertical distinctions the earth as the fixed ground serves as an object of reference. This object of reference is so large and firm that you cannot go to the other side of it. Thus whatever is away from it is up and whatever is nearer to it is down. In the case of right and left there is no such fixed object of reference, although in considering the two hands there is of course the body to serve as such an object. But the difficulty is that the hands are symmetrically placed on either side of the body. The child has, therefore, distinguished between right and left by first having to find out the special functions of the right and the left hand: for instance the right hand is used for eating. The child does writing also with the right hand, but this comes at the later stage and many children have already learnt the distinction of right and left before they begin to write.

(2) Kinesthetic memory of sensation is another explanation given. It is said that children move objects up and down and the memory, involved in these muscular movements, is more distinct than the kinesthetic memory involved in distinguishing between objects to the right or left. In the latter case, it is suggested that no larger muscles of the body are involved in the distinction; at the most, it may be the movement of the eyes to the right or left; but on the other hand, it may be said that these right and left movements of the eyeballs are more important than the up and down movement of the same. Some psychologists give the name of *local signature* to these distinctions in space relations made by the movements of the eyeballs.

(3) The frequency with which a child makes distinction between *up* and *down*, and *right* and *left* offered as one of the explanations. It is alleged that distinctions between up and down occur the more often in a child's every day life than those between right and left.

(4) Finally, the distinction may be due to the *frequency*, which a child uses or hears the words in the language—up and down or right and left.

If you observe little children trying to learn right and left, and up and down distinctions it appears that they do so by the following steps:

(1) This slowly distinguishes between up and down by reference to the earth or ground. Anything away from it is *up*, anything close to it is *down*, or between right and left by associating certain actions of the right hand with the right hand and those of the left with the left.

(2) Secondly they associate the conventional verbal names 'up' or 'down', 'left' or 'right' with these directions as indicated in the first step. In this process, the distinctions between the horizontal directions are certainly harder than those between the vertical and hence appear to evolve more slowly.

The first period consists of about 6 hours or two *praharas*, the second of 3 hours or one *prahara* and the third of 3 hours or one *prahara*. Scientifically it is divided into 4 parts or *praharas* each of 3 hours. Hence questions are worded in two ways as shown above— one alternative being between morning and afternoon, and the other between morning and evening. These formulae work very well. In assessing the answer, afternoon [Mahayana (Kan). Dupar (Mar)] may be taken his midday meal though it may not be actually 12 O' clock, Time orientation hour by hour it not develops in children of this age.

One great objections raised against this test is that gives the choice between two alternative only and answering by mere guessing or answering at random would give 50% correct answer according to the law of chance. Binet included the test in his 1908 series in year VI, but omitted it in his 1911 revision. Burt puts it in year V. Our own statistics show that it is easy in for year V.

V. Alternative 2. Giving Family Name

TEXT AND PROCEDURE

'What is your name?'

'What is your Surname?'

If the child gives his proper name only, say,

'Yes, but what is your other name?'

'Mahadev what?'

'Is your name Mahadev Joshi?'

Remarks. This is generally located in III by European and American authorities. Indian children, however, are not accustomed to use their family name; neither they ever called by the family name nor to the usually hear people calling their parents by the same. The more common method of addressing people in the household is by their proper name. Hence the test is not suitable for children of India. If it is to be retained at all our statistics show that it should go up to year V, and so it is retained as an alternative test in year V. Objection can also be raised against this test that the ground that the more enlightened families in the habit of teaching their children the proper name and their family name, while in backward families even very much older children do not know their family name unless they have been sent to school and their names registered in full.

V. Alternative 3. Giving Age.

TEXT AND PROCEDURE

'How old are you?'

Remarks. This is another test that is not very suitable for Indian children, or for any children at all. In the first place the age as given by children may be either age last birthday or age next birthday. Both ways are correct; with Indian children it is more common for them to be distinguished by their age next birthday. Secondly, celebrating the birthday is not usual among Indian children. In illiterate families even adult do not know their age. Hence this test should be very sparingly used, and is retained here as an alternative test only for sake of comparison. Binet had it in his 1908 scale VI, but omitted it in his 1911 series. Many other investigators have also omitted it.

CHAPTER THIRTEEN

TESTS FOR YEAR VI

VI, 1. *Repeating Four Digits. (1 out of 3. Read one every half-second)*

TEXT AND PROCEDURE

Now, listen. I am going to say over numbers and after I am through, I want you to say exactly what I say. Listen closely and get them just right.'

'4-7-3-9; 2-8-5-4; 7-2-6-1.'

Success with a series that has been re-read should not be counted.

Remarks. See remarks for III, 3 and IV, 1. The digits should be read at the rate of one every half-second for the reason stated there. Though repeating digits appear a very simple test, actually it is far from being so. It puts a great strain on the mind and hence should not be given towards the end of the examination. It brings on fatigue very rapidly and should therefore be given as far as possible when the mind is fresh. Not should testing begin with this test as children are likely to fight shy of it.

VI, 2. *Comprehension. Second Degree. (2 out of 3. Time allowed minute)*

TEXT AND PROCEDURE

'What's the thing to do-

(a) If it is raining when you start for school?

(b) If you find that your house is on fire?

(c) If you are going somewhere and miss the train? (or train- car or motor-bus or ferry?)'

After 5 to 10 seconds repeat the question once more.

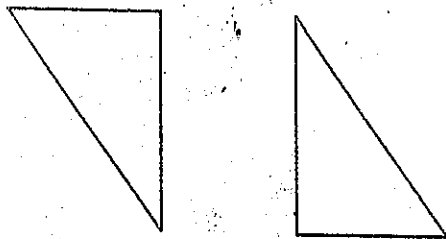
Remarks. See remarks under IV, 3. The question may be repeated a second time only. No more repetitions are allowed. Care should be taken, however, that attention is secured before the question is asked. Terman says, 'The three comprehension questions of this group were all suggested by Binet in 1905. Only one of them however, "What should you do if you were going to some place and missed your train?" was incorporated in the 1908 or 1911 series, and this was used in year X with seven others much harder.' The other two were standardized for the first time in the Stanford revision. In this test as also in some others Terman would allow the question to be repeated once or twice. As remarked elsewhere it makes a good deal of difference whether a question is repeated once, or twice, or thrice. We have therefore definitely laid down how many repetitions there should be in each case. In the present case the question may be repeated only once as directed above. The third question should be varied by using the conveyance most commonly in vogue at the place where the child is examined. Nowadays motor-buses ply between any two adjoining important towns several times during the day and two adjoining important towns several times during the night and hence this way of stating the question is very convenient. If the name of the adjoining town were put in perhaps the question would be a little easier.

VI, 3. *Divided Card. (2 trials out of 3 correct. 1 minute each. Or better, thrice in 3 minutes.)*

TEXT AND PROCEDURE

Select two stiff white cards each 5 cm. X 8 cm. Cut one of these diagonally into two triangles. Place the undivided card in front of the child with the longer side place right and left. Below this, *a little nearer the child*, place the triangles cut out of the second card in the following position:

Diagrams of two triangles is to be drawn



Then say,

'I want you to take these two pieces (touching the two triangles) and put them together so that they will look exactly like this (pointing to the uncut card).'

If the child sits silent without knowing what to do say;

'Turn these pieces round as you like and put them together so that they look exactly like this.'

After the child has finished, ask:

'Do they look like this?'

After each trial put the pieces in the position shown and repeat the short formula:

'Put these two pieces together so that they look exactly like this.'

If during the experiment the child turns over one of the two pieces, turn it over again and start again, as it is impossible to get the right construction with one of the pieces turned over upside down. The trial should be regarded as having begun from this new start.

Remarks. This test was scored both ways (i) by giving one minute for each performance and requiring only 2 correct performances out of 3, and (ii) by giving 3 minutes for 3 performances. The scoring was exactly the same in the both these ways. The latter method, that of giving altogether 3 minutes for 3 performances, has a distinct advantage in not disturbing the child when he is in a contemplative mood and trying to put the pieces right. This method is similar to that of Healy and Fernald's 'construction puzzle' test which gives a total of 5 minutes for 3 performances. (See XII, 2.) It is always better therefore to take the test in this way.

The test is similar to Ebbinghaus's 'completion' test. The child has first of all to form a clear idea of the goal to be reached, and secondly, under the influence of this idea to direct his efforts to the realization of this goal. Mentally backward children fail in his test either because they cannot form a clear idea of the goal to be reached or because they fail to hold it firmly in mind until it is reached. Lastly, the child must have the power of self-criticism, by which he compares his own performance with the model and finds out whether his performance is correct. Backward children are sometimes quite satisfied with their own performance though the performance may be quite ludicrous.

Burt, strictly following Binet and Simon's instructions, presents the triangles so that the hypotenuses are at right angles, but do not face towards each other. * He says. Terman's method of presentation is easier, but it does not appear to be so. On the contrary a child may by chance move the upper triangle in Burt's position slightly to the right and find that he accomplishes the feat. Burt's supplementary instructions, 'move them about and see if you can fit them together', agree with ours. This supplementary formula is put in so that we may be sure that a child tries his best and does not remain ideal for want of inability to understand the question as it is put in one form. Burt gives only one trial and gives a half-minute to it after Bobertag.

The test is located in year V by Binet and Terman, but in year VI by Burt and Bobertag. Our own statistics agree with the latter.

VI, 4. Giving Numbers of Fingers. (No error)

TEXT AND PROCEDURE

'How many fingers have you on one hand?'

'How many on the other hand?'

'How many on both hands together?'

Keep an eye on the child's fingers and if he is found to count them say,

'No, don't count. Tell me without counting.' Then repeat the question as before.

Remarks. In Indian nomenclature the thumbs are included in the fingers, so difficulty of excluding the thumbs never arises. This test is harder than 'counting 13 paise' (V, 6.). It requires on the part of the child a fuller concept of numbers. Unless this concept is formed the child cannot retain the memory of these numbers. The concept of numbers required for counting is decidedly lower than this concept. A certain interest in numbers has prompted the child to count his fingers probably several times and has made him remember this number. The concept of smaller numbers such as one, two, three or four seems to have been formed earlier and it would be interesting to find out at what age the child knows that he has one nose or two eyes.

*See, however, Kite's translation, p. 199. 'Put the two triangular pieces in front of the child in such a way that the two hypotenuses are not adjacent.'

Contrary to expectation the test is found to be a good test of intelligence as is shown by the regular rise in percentages of passes and the high correlation of the test with the scale as a whole. Coaching does not affect the test to any appreciable extent. Our statistics show that the test is easy enough for year VI. Burt assigns it to year VI also. Binet (1908) and Terman have assigned it to year VII.

VI, 5. *Description of Pictures.* (2 of the three pictures described or better Time limit 2 minutes each)

TEXT AND PROCEDURE

Use the pictures in the following order:

1. Railway station.
2. Reception at home
3. Motor accident

'What is the picture about?'

'What is this a picture of?'

'What is represented here?'

The questions may be repeated in one or the other form twice or thrice to encourage the child to answer so long as the time limit is not exceeded. Forms of questions that suggest particularly enumeration, or description, or interpretation should be avoided.

Remarks. It should be carefully noted as to what constitutes enumeration (required for age III), what description (required for age VI) and what interpretation (required for age XII). Burt says, 'replies giving a mere list of persons, objects or details' constitute enumeration; 'phrases indicating actions or characteristics' constitute description; and 'replies going beyond what is actually visible in the picture, and mentioning the situation or emotion it suggests' constitute interpretation. Sometimes replies make it difficult to decide whether they constitute enumeration or description. For example: 'Here is a man', or 'He has an umbrella in his hand'. These responses should be scored minus rather than plus for this age because they show only what is present in the picture and do not tell anything about the 'action or the characteristics' of the figure of the picture. The following answer would be plus: 'The man is going with an umbrella in his hand'; 'This man has caught hold of this man.'

The picture selected for this test are taken from *Manoranjan*, a high class Marathi magazines depicting Indian life. They are selected from among a large number after careful testing as to their suitability and are used by courtesy of the proprietor, *Manoranjan* Bombay. The pictures much contain sufficient number of familiar objects, animals, and men, easily recognizable by small children; secondly, they should also show certain

action which can be easily described; and lastly, they should contain only one theme which can be properly interpreted by carefully observing the different parts of the pictures and their relation one another. The pictures selected were found to satisfy very well all these conditions. Burt's or Terman's picture could not be used as the depict western life. Thus though the test is materially changed statistics of the present investigation so that they are almost of the same difficulty and serve the purpose of the three test—enumeration, description and interpretation—equally well.

This test shows very well the height of Binet's genius in evaluating children's answer. After studying these answers carefully he discerned clearly that about the age of three children can only recognize and name objects in pictures; about the age of six or seven children begin to recognize objects and their relation to one another so as to understand their mere outwardly visible actions; and about the age of twelve or fifteen their associative powers are so far developed that they can put together and combine all several parts of the picture in such a way as to find the inner purport of the picture. This ability manifests the highly synthetic power of the mind. The test is located in year VII by Binet, Terman and Bobertag and in year VI by Burt.

VI, 6. *Missing features. (3 out of 4. Time limit half minute).*

TEXT AND PROCEDURE

'There is something wrong with face. It is not all there. Part of it is left out. Look carefully and tell me, what part of the face is not there?'

If the child speaks of parts that are not drawn in the picture such as the body or the legs, say,
'No, I am talking about the face. Look again and tell me, what is left out of the face?'

In case of failure, say,
'See, the eye is gone.'

Then with the next picture, say,
'What is left out of this face?'

With picture (d) which is different from the others, say,
'What is left out of this picture?'

Remarks. As a rule children answer the questions promptly. The time limit is set down only as a guide in the case of children who sit silent so that after the lapse of the specified time the examiner may proceed to the next question. It is interesting to note that sometimes the children say, 'the other eye', or 'the other ear'.

This is one type of 'completion' test, in which the child is required to call up before his mind an image of an object that he has already seen and with which he is quite familiar and compare it with the present picture to find out the missing part. Other examples of such tests are filling in the gaps in sentences having meaning, making up a picture out of its parts, and rearranging a disarranged sentence.

The features of the pictures in this test are all truly Binet's but they are all Indianized by giving them Indian form of hair dressing, Indian ornaments and dress. The results are therefore strictly comparable with change inevitable. Still the picture has served the purpose in hand very well.

In Marathi, unfortunately there is only one word for mouth and face. So the formula of the question becomes a little unsuitable. But this can be avoided to some extent by using both the alternative forms of the word as shown in the Marathi test.

This procedure is found to answer the purpose very well.

VI, Alternative Naming Colours. (No errors)

TEXT AND PROCEDURE

Point to the colours in the order red, yellow, blue, green.

'What is the name of that colour?'

Sometimes the blue colour is named purple and with a good deal of plausibility. In this case, have ready some purple paper and ask the child what colour it is; and then ask him the name of the blue colour. In all other cases no second attempts are allowed.

Remarks. The test of naming colours is more difficult than the test of recognizing colours. As a rule, children recognize things first and then learn to associate the name with the things. Thus pointing to parts of the body is easier than naming these parts. In the case of colours the recognizing and discriminating between colours is itself a hard task quite unlike discriminating between form of familiar objects. Thus the test is hard enough for year VI, while the test of naming familiar objects' comes in year III. The conception of colour in the abstract, apart from the objects, is very difficult. Hence children take such a long time to recognize colors.

There is a good deal of divergence in the location of this test. Binet located it in year VII in his 1911 revision, Bobertag and Saffiotti in year VIII, while Terman and Burt locate it in year V. Our statistics put it in year VI.

CHAPTER FOURTEEN

TESTS FOR YEAR VII

VII, 1. Repeating Sixteen to Eighteen Syllables. (1 out of 3 absolutely correct; or 2 with one error each).

TEXT AND PROCEDURE

'Now, listen. I am going to say something and after I am through I want you to say it over just like I do. Understand? Listen carefully and be sure to say exactly what I say.'

After this, read these sentences slowly, emphatically and so as to convey the meaning clearly:

- (a) We will go out for a long walk. Please give me my pretty straw hat."
- (b) We are having a fine time. We found a little mouse in the trap."
- (c) Walter had a fine time on his holiday. He went fishing every day.'

After each response say, 'Did you get it right?'

Remarks. This test is one of repeating syllables having sense as distinct from nonsense syllables. The number of nonsense syllables that a child of the same age can give is much less. However, the syllables are to be given exactly in the same order and accurately. Mispronunciations due to tender age may be disregarded. Any transposition, omissions, or addition of syllables is therefore an error. Burt following Binet (1908) closely has only one sentence with 16 syllables and allows no error. Terman modifies this test by giving 3 chances with 16 to 18 syllables and requires one to be given correctly or two with one error each. The difficulty in both procedures seems to be very nearly the same. We have followed Terman. Our statistics show that the test is a little too difficult for year VI, but quite easy for year VII. We have located it in year VII. In the Indian languages the syllables are written in the form of letters. Hence we name the test as 'giving 16 to 18 letter' in the vernacular.

*In rendering these sentences into the Indian languages some changes are made in the meaning, but the requirements of the test are satisfied.

In all such memory tests the set of syllables or digits is to be read only once after securing the attention of the child. So responses obtained after second readings are invalid and should not be counted. Below the level of the sixth year of age, however, it is very difficult to secure the attention of the child. Hence re-reading was allowed with the first set, but this was not counted and later sets were given only once.

VII, 2. Copying a Diamond. (2 out of 3)

TEXT AND PROCEDURE

Place the model printed on the card before the child with the longer diagonal pointing directly towards him. If a copy is to be made by hand it should be done in Indian ink with sufficiently thick lines. The longer diagonal should be 8 centimeters and the shorter one 4 centimeters.

Give him pen, ink and paper and say,

'I want you to draw one exactly like this.'

Give three trials. As the child is drawing remind him with,

'Make it exactly like this one.' But do not pass the finger around the edge of the model.

After each drawing is finished ask the child encouragingly, 'Is it good?'

At the end ask,
'Which one is the best?'

Remarks. Score the diagram plus if it is recognizably a rhombus. The vertical diagonal must be longer than the horizontal. For guidance in scoring refer to the printed scoring card, which is made up of the actual responses of children. For a pass two of three drawing should be satisfactory. Until we actually observe children of this age making these drawings, the test appears to us to be too easy. But it is very interesting to observe children making the drawings. It is really a very heavy mental burden to children of this age. Some fail to make the angles, and instead of trying to make convex angles many of them try to make them concave or re-entrant. Some, not knowing how to make a bend at the angles, make circle and others irregular turns. Other again not being able to make the angles sufficiently bent, fail to make the last ends of the lines meet together and thus are either forced to join the gap by an additional line or to leave the gap open. The test, therefore, requires the child to form a mental image of the diagram as he would make it and then to guide his fingers under the influence of this projected image. A glance at the unsatisfactory drawings will make this clear. This is a very good test and ordinary training in schools seems to affect it but little. Binet (1911) and Burt locate it in year VI, Terman and Bobertag in year VII, and Saffioti in year VIII. Our own statistics warrant its inclusion in year VII.

VII, 3. Repeating Three Digits Reversed. (1 out of 3. Read one per second)

TEXT AND PROCEDURE

Listen carefully I am going to read some number now, but this time I want you to say them backwards. For example, if I should say 1,2,3 you should say 3,2,1. Do you understand?

'Read now, listen carefully, and be sure to say the numbers backwards.

"2-8-3; 4-2-7; 9-5-8."

If the child repeats the digits forward instead of backwards repeat the instruction emphasizing the word 'backwards'.

Remarks. This is the first test of 'repeating digits backwards'. At this age children are often found unable to comprehend what is meant by reverse order though it is made clear by illustration. It is very interesting to find that several children give the digits in the reverse order first arranging them in the descending order. For example, the above series are given back as 8-3-2, 7-4-2 and 9-8-5. However, in later years the test of repeating digits in the reverse order works as a better test than the one of repeating digits in the forward order, because the child has to fix the order of the digits firmly in his mind and then reverse them. This requires quite a feat of intelligence. It is not known how many children retain the memory by auditory images, how many by visual images and how many by motor images. Literate and intelligent pupils are sometimes seen to make finger moments tracing the digits in an effort to remember. This method certainly comes to the aid of their auditory images. It affords us also a great lesson in educational method, showing us that we should make use whatever possible of kinesthetic imagery along with auditory and visual, and that it is also very important to establish such kinesthetic imagery.

VII, 4. Naming Days of Week. (Order correct. 15 seconds. 2 out of 3 checks correct)

TEXT AND PROCEDURE

'You know the days of the week, do you not? Name the days of the week for me.'

If the child has not comprehended what is wanted and gives irrelevant answer, say again,

'No, that is not what I mean. I want you to name the days of week.'

After the child has finished, check up by asking,

'What day comes before Tuesday?'
'What day comes before Thursday?'
'What day comes before Friday?'

Remarks. This is one of the test of time orientation. It is very interesting to note how the sense of time develops in children. They begin to distinguish between different parts of the day about the fifth year (see V, alt. 1), days of the week and the month about the seventh year (see VII, 4 and VII, alt. 1), and months as parts of a year and the about the tenth year (see X, alt. 2).

The criticisms leveled against this test are that children give the names of days mechanically without having developed the necessary sense of time, and that the usual training in school or in enlightened homes makes the children learn the names of the days of the week mechanically, thus placing them in an advantageous position in comparison with other children of illiterate homes or poor surroundings. These criticisms are to a large extent removed by Terman's device of introducing checks. But it was found in the present study that children who could give the names of days of the week could also answer correctly the checks with very few exceptions. Keep a record of the time from the moment the child names the first day.

The test was located by Binet (1908) in year IX, but was dropped by him in his 1911 revision. Terman located it in year VII, Burt in year VI and Bobertag in year VIII, Burt strictly following Binet has no checks in his test, but requires the days to be given in 10 seconds.

VII, 5. Counting Backwards Twenty to One. (40 seconds. 1 error allowed.)

TEXT AND PROCEDURE

'You can count the numbers from 1 to 20, can you not?
Now count them for me.'

When the child has finished, say; 'Now count them backwards, like this, 20-19-18 and so on. Now go ahead.'

If the child says he cannot count backwards encourage him and say: 'I know you can count them backwards very well. Go on.'

Remarks. Binet wants the children to count down to 0. But it appears that this was not insisted upon, for children have no idea of zero and cannot count down to zero. All subsequent investigators rightly require counting from 20 to 1 only. Terman allows a maximum of 10 seconds for this test, and we have retained this time limit. Fifty percent of children of mental age between seven and eight years pass this test, of whom about 35 percent accomplish the performance within 20 seconds. In order to make the instructions quite clear without much worry it is better to follow the procedure of Dr. Simon and make the child count forwards first and then backwards. Backward children count a few numbers backwards and then go on counting forwards. If the child has been specifically taught in school to count backward, it is better not to give him this test.

To pass the test the child is required to be familiar with the numbers and with their concepts so that each succeeding number becomes familiar to him as greater by one than the previous number. Further the child has to keep constantly before his mind the object to be achieved, namely counting backwards and at the same time keep in mind the number that is uttered and imagine the number that is to come next. This requires a good deal of abstract number imagery. If the attentive attitude becomes slack at any moment the child immediately begins to count forwards as a result of the habit of counting forwards that has become automatic with him.

VII, 6. Giving Differences from Memory. (2 out of 3. 1 minute each)

TEXT AND PROCEDURE

'What is the difference between a fly and a butterfly?'

If the child is found not to understand this, proceed:

'You know flies, do you not? You have seen flies? And you know butterflies? Now, tell me the difference between a fly, and a butterfly?'

'In what way are these different?'

'Similarly, stone and potato; wood and glass. No supplementary questions are allowed as they may serve as leading or suggestive questions.

If the child says one is bigger or harder than the other, ask, 'Which is bigger (or harder)?'

Remarks This is a very good test of intelligence. It throws the child on his own resources and mental equipment. The child has first to call up the images of the objects and then compare and contrast them in the abstract. These are the highest types of conceptual processes. We acquire and accumulate all our knowledge by means of the dual associative power of noting differences and similarities. Our classification of matter and life into species and genera is based on this principle. An intelligent child of this mental age has therefore learnt to distinguish between common things by noting the differences. Of course the faculty of noting more minute differences develops with further acquisition of knowledge and the exercise of the mind in that direction. A botanist will find in two specimens minute differences which will be quite invisible to laymen. The differences between the common objects picked out for this test can be given by normal seven year old children.

To pass this test at this age any real difference noted, however trivial, is quite sufficient. Statistics show that the power of finding similarities between common things (IX, 3) is harder than the power of finding differences (VII, 6). In assessing the answers notice the tendency on the part of children to stereotypy. Starting for example, by saying that a butterfly is white and a fly is black the child continues to give the same difference in the case of other pairs. Though the differential characteristics may be true of all the pairs such answer should be scored minus. Weak minds after finding out one kind of difference area exhausted and proceed without thinking to apply the same difference to all the others.

Binet (1911) and Bobertag locate the test in year VIII. Our statistics agree with those of Terman and Burt in locating it in year VII.

VII, Alternative. Giving Day of Week and Day of Month. (Both correct, the second within 3 days of the actual date).

TEXT AND PROCEDURE

(a) What day of the week is today?"

(b) What day of the month is it?"

In the latter case, if the boy says he does not know, say, 'Give it roughly.'

Remarks. Before using Binet's test of 'giving day and date' it was considered whether it would be advisable to score the responses to the question on the Indian method of giving the names of months and days. But a little experience of testing with children showed that they knew the European names of months and the date better than the Indian. The Indian months are lunar months and counting of the days of the months stops at

the end of the bright half and the counting is done again in the dark half of the month, so that there are only fifteen days coming twice in a month. As this method of counting days is not used in daily transactions, children are so convenient with it. On the contrary in all transactions and in school they use the date, the name of the month and year according to the Christian calendar. Hence they were found to be quite familiar with these.

It was found, however, that the day of the week and day of the month were known much earlier than the name of the month and the year. So if all the four are lumped together the test, as we have remarked elsewhere, becomes practically a test of giving the two letter only. It was thus though advisable to split it up in to two parts- the day of the week and day of the month going in year VII and the name of the month and the year and going in year X, the test of year VII and year VIII in point of hardness.

Requiring all the four items Binet and Burt put the test in year VIII and Terman with the same item in year IX. Burt gives the order of difficulties (So that the latter once are more and more difficult) as (1) Day of the week, (2) month, (3) year and (4) day of the month. Our statistics put it as (1) day of week, (2) day of month, (3) year and (4) month.

CHAPTER FIFTEEN

TESTS FOR YEAR VIII

VIII.1. Finding Value of Coins. (15 second)

TEXT AND PROCEDURE

Arrange three one- paisa pieces and three two-paisa pieces on the table before the child thus: 1,1,1, 2,2,2. Then say,

'You know this? (Pointing to the one paisa piece)

'You know this? (Pointing to the two-paisa pieces)

'Now tell me how much is the value of all these coins?'

Remarks. This test is exactly parallel to Binet's test. The coins are similar to the French coins of *sous* and double- *sous*. American investigators have used stamps, as they have no similar coins in America. The test with stamps is harder than the one with coins. Burt uses pennies and halfpennies, but this makes the calculation fractional and may be a little harder. He assigns the test, however, to year VIII. Our statistics show that it is a little too hard for year VII, but easy enough for year VIII, and we have, therefore, assigned it to this year. This is one of the best tests. It is so easy to give and to score. It does not depend to any great extent on school instructions, nor do children of superior homes have any advantage over those of inferior homes, since all children have almost equal opportunities of handling these coins.

VIII. 2 Repeating Five Digits. (1 out of 3. Read 1 per second)

TEXT AND PROCEDURE

Secure the child's attention and read the following series to the child distinctly and at a uniform speed of about one per second. Avoid sign-song ways of reading. Re-reading any of the series is not allowed above the level of seven years of age. Below the level of seven years, children are fidgety and the greatest problem is to secure their attention. Hence re-reading was allowed in their case only with the series; but such success was not counted.

3-1-7-5-9; 4-2-8-3-5; 9-8-1-7-6.

Remarks. In giving these 'digit' test the examiner should begin with the series one step below the one suited to the mental age of the child and then give the higher series. This provides just a little practice but remove the effects of fatigue. Later on as the examination continues the higher test of digits should be given as they come in among age series of tests. Burt differs in this and gives the 'digit' tests all together; and this gives the good deal of practice, but at the same time produces fatigue very quickly. It is difficult to say which of these causes influences the results most. Further, presumably if a child fails with five digits and by chance succeeds with one of the series of six digits, he would be taken as having succeeded with five. On the whole, on looking to the location of these 'digit' test by Burt and comparing it with the location of others, it appears that by his procedure the tests are rather too easy for children. Burt also gives the digits at the rate of one every second above this level. If they are given at the rate of one every second below the level of seven years the children begin to repeat the digits before the examiner finishes giving them.

Psychologically the test of giving digits lies between that of giving nonsense syllables and that of giving sense syllables. Sense syllables in sentences having meaning have a good many association and are held together in a *gestalt* by the meaning of the whole sentence. Digits have some meaning and as such can be retained better than absolute nonsense syllables. At any age roughly four times the number of sense syllables

can be given as the number of digits. In this connection it should be borne in mind that some of the digits are made of more than one syllable. This is particularly so in the Kannada language as contrasted with the Marathi. In Kannada, out of the digits from 1 to 10, two are of three syllables each and eight of two syllables each. In Marathi on the contrary, owing to the fact that the last syllables in this language is not usually emphasized, two of these become practically of two syllables each and eight of one syllables each. Thus as regard the number of syllables the Marathi digits resemble the English digits, while in Kannada the difficulty is much greater. But statistics show that the number of syllables does not much affect the number of digits given. This is one more argument in favor of our assertion that meaning of language is more important in psychological associations than the empty shell of sound syllables.

VIII. 3 Comprehension. Third Degree. (2 out of 3. Read once. After 5 to 10 seconds read a second time and then allow 30 seconds more.)

TEXT AND PROCEDURE

- (a) 'What's the thing for you to do when you have broken something which belongs to someone else?'
- (b) 'What's the thing for you to do when you are on your way to school and notice that you are in danger of being late?'
- (c) 'What's the thing for you to do if a playmate hit you without meaning to do so?'

When repeating the question its form must not on any account be changed.

Remarks. The 'comprehension; as well as some other tests have not been properly timed, nor is the rule as to the limitation of repetition of the question scrupulously observed in some previous revisions. Terman says that question may be repeated once or twice. Burt says repeat once if necessary, and twenty second are usually allowed to elapse although the directions seem to be that one should not be too pedantic over the time limit. It is a matter of experience that a good deal of difference is made by repeating a question once or twice or three times, more particularly a comprehension question. Backward children who fail to comprehend question a read once or even twice, may begin to comprehend it when read a third time. To avoid all such vagueness the procedure is made more exact in the present revision. Only one repeating is allowed and that from five to ten second after the first reading. After the second reading a maximum of thirty seconds is allowed for the reply. A great many children, of course, answer the comprehension questions promptly, but a few dull children take too long and so the time limit helps the examiner to determine when to proceed to the next question without loss of time.

Questions (a) and (c) were included by Binet in the easy series located in year IX and question (b) is from his difficult series located by him in year X. Terman rightly grouped these three together and located them in year VIII. For reason already sufficiently explained, it is no use grouping together very easy and very difficult questions, because the score is then determined by the hardest questions, the easiest questions serving no purpose in the group, and so it is essential that the easier and the more difficult sub-test should be separated, grouped severally, and re-assigned to the proper ages.

VIII.4 Definition. Superior to Use. (2 out of 4. 1 minute each)

TEXT AND PROCEDURE

- (a) What is a balloon?"
- (b) What is a tiger?"
- (c) What is a football (or a ball)?"
- (d) Who is a soldier?"

Use the word 'ball' instead of 'football' more generally with girls, since some of them are unfamiliar with the latter.

Remarks. This is the hardest test to score in all Binet's series of tests. After studying the definitions of objects given by children, Binet came to the conclusion that children of the mental level of about six years define objects in terms of their use and those of about nine years define them in terms of description or genius. This classification of responses is quite good so far as broad principles go, but when we come to actual details a number of difficulties arise and different investigators have classified the different categories. We shall first of all point out the nature of the two classifications and then discuss a few examples.

A. Definitions by use.

Use includes (1) action or functional use; and (2) use for us.

- *Cyril Curt, Mental and Scholastic Tests. P. 42.*

Examples of (1) it goes up (for balloon); it eats us (for tiger); it bounces (for football); he carries a gun (for soldier).

Examples of (2) to sit in (for balloon); it is in a circus (for tiger); to kick (for football); he keeps watch (for soldier or policeman).

B. Definitions, superior to use.

These include (1) description of shape, size, colour, mental of which it is composed, action, etc; and (2) higher class or genius, with some kind of differentia.

Examples of (1): it is like a motor-car (for balloon); it has stripes on it (for tiger); it is of leather (for football); he carries gun, he wear a kind of cap (for soldier).

Or again: it goes high up, men sit in it (for balloon); it is the forest, its eats people (for tiger; boy kick it, it goes up (for football); he carries gun, he catches thieves (for policeman).

Example (2): it is a kind of motor-car, or, it is a car that goes up in the air (balloon), (both are correct definitions; but 'a car', particularly as it is given in the Indian language without any qualifying word, is correct): it is an animal that its people, or, it is a kind of animal (for Tiger): it is a big ball, or it is a kind of ball (for football) (simply a ball' is in correct); he is a man who catches thieves (for policeman).

A number of observations required to be made on this test. In the first place Binet uses the same words for both the ages -VI and IX. Perhaps this was a better method from point of view of pure psychological investigation. Terman however, departs from this procedure and uses words which more generally evoke use definitions in year V and other words which more generally evoke description of classificatory definition in year VIII. We have generally followed Terman. Whatever the words selected when once they have been thoroughly standardized, they can be safely used. Beside, Terman's method would seem to avoid the some ambiguous responses.

Secondly, other investigators have classified definitions by actions as belonging to the lower mental level. On carefully studying the responses in the present investigation it was found that quite a larger number of intelligent children, even much above the mental level of eight years to which the higher type of responses belong, define objects in terms of more actions than just one. Hence it was thought that when an object was defined in terms of only one action, it belong to the mental level of five years, but when it was defined in terms of more than one action, it was regarded as a descriptive definition and assigned to year VIII. This later type of definition proceeds from the desire of the mind which, not being satisfied with only one action, proceeds to give more actions than one, which greatly restrict the application (the denotation) of the definition. To restrict the application of the definition to a smaller class of objects is analogous to giving the genus and then the differentia.

Thirdly, Definitions by genus without any kind of differentia or a qualifying word have been scored minus. Children even of low intelligence give definitions in terms of genus without a qualifying word. For example, they say, 'a football is a ball'. This is minus for the higher age. On the contrary if a child says 'a

football is 'a kind of ball', he passes in the higher test. It is almost as good a definition as 'a football is a big ball'. It proceeds from the desire of the mind to restrict the application of the word 'ball' to a smaller class of objects by adding 'a kind of'.

VIII. 5. Naming six Coins. (No error. Give in order indicated five paisas. One Paise, quarter rupee, half rupee, ten paise, two paise).

TEXT AND PROCEDURE

Procedure same as in V. 5.

Remarks: Binet and Burt (following Binet closely) use nine coins and locate the test in year IX. A good many other investigators do the same and locate the test in IX or X. Terman, however, uses six coins and locates it in year VIII. Our statistics with the six Indian coins go to show that the test is suitable for year VIII. The quarter-rupee and the half-rupee may be named as such or in terms paise.

VIII.6. Reading and report. (2 facts: 10 errors: 2 minutes.)

TEXT AND PROCEDURE

The following is Binet's selection as adapted by Terman. With necessary changes as required to suit local conditions:

Bombay/September 5th/A fire/last night/burned/three houses/near the center/of the city/. It took some time/to put it out/ The loss was fifty thousand rupees,/and seventeen families/ lost their homes./ In saving /a girl/who was asleep/in bed/ a fireman/was burned/on the hands.

Put the above passage, printed without bars on the card contained in the packet of the test material, before the child and say:

'I want you to read this for me as nicely as you can'.

Ask the child to read aloud and help him in case he cannot read any individual words, counting these as errors.

Carefully record the number of errors made and the time taken. It is convenient to use stop-watch in all cases where time is to be carefully recorded. If one is not available have at least a watch with a seconds hand. When the child has finished say,

'Very well done. Now, I want you to tell me what you read. Being at the beginning and tell me every thing you can remember'.

If the child stops after saying a few words. Say further,

'And what else? Can you remember any more of it?'

If the child misunderstands and thinks that the passage is to be given word for word, say,

'Tell me in your own words all you can remember of it'.

Do not ask any supplementary questions. Only urge him on encouraging so that he tries his utmost.

Remarks: It is essential that the child's response should be taken down verbatim. This is a different task but it can be done with a little practice if only the first letter of every word is written down as the child talks and then completed after the child has finished. In scoring for memories it is enough if the ideas are correctly reproduced. The exact words are not required. The passage as given above divided by means of bars is a great help in scoring. After the child's response is taken down underline the memories and then count. It will be noticed that this test occurs in year VIII, IX, and X with different scores. So accordingly to our general rule it should be scored in the highest year, the conditions of which are satisfied by the child's responses, and should therefore not be made use of for other ages. The same thing applies to the 'vocabulary' test or the 'free

association' test. In the eighth year the child has no great mastery over reading and so he should be allowed two minutes for reading with not more than ten errors. This standard is determined from the actual responses of children. It should be observed that as the child advances mentally he makes fewer mistakes, takes less time to read and remembers more facts. It may then be asked why it is necessary to lay down all the three conditions. It is always safer to impose all these conditions, because a child of low intelligence by a good deal of coaching may learn to read well, but it is less likely that he will remember the required number of facts. If, on the contrary, the limit is not set, some children may read very slowly or even twice over and so be able to remember more facts. In fact the valuable factors both of speed and accuracy are beautifully employed in this test. This test, however, should be omitted in the case children who have not been taught reading or who have not attended school for at least two years.

Burt says that Binet purposely sets no time limit as speed of reading depends on school practice. He concurs with Binet in giving the following as the average time of reading:

- At 8 years, 45 seconds
- At 9 year, 40 seconds
- At 10 years, 30 seconds
- At 11 years, 25 seconds

It should be remembered that these are not the *average* times and time *limit* should therefore be much higher than this. Terman allows 35 seconds in year X. The passage in English contains only 68 syllables, while that in Kannada contains 134 syllables and the one Marathi 116 syllables. The number of facts is kept the same as in the original, namely 20. We have, therefore, set down the following as the time limits:

- In the 8th year, 2 minutes
- In the 9th year, 1 minute
- In the 10th year, 40 seconds

The actual average times of reading were:

- In the 8th year (i.e. at average age 7 ½ years) 55 seconds
- In the 9th year (i.e. at 8 ½ years) 53 seconds.
- In the 10th year (i.e. at 9 ½ years), 41 seconds
- In the 11th year (i.e. at 10 ½ years), 40 seconds

A few words about the psychological importance of reading are necessary here. Many critics say that children should not be unnecessarily penalized if for want of opportunities they are unable to read. There is certainly a good deal of truth in this but only up to a certain limit. Even when children have sufficient opportunities, as in the case of all children regularly attending school, it has been found by actual experiment that there is a good deal of difference between different children in their ability to read. Intelligent children certainly read much better and with better understanding.

In fact good reading does not depend merely on ability to recognize the letters, as they are inconspicuous in such continuous reading. The same is true to a lesser extent even with words. It is the ideas conveyed by the words, the meaning of the passage that is grasped by the mind. It is doubtful whether the complete images evoked by the words stand forth in every intelligence reading or hearing. But it is possible that these images remain in the background as in a subconscious world and are associated with the meaning as a whole, which runs smoothly and swiftly in the conscious word of our mind. The reading of a backward child is therefore lacking in spirit and meaningless, because the child's mental imagery is so poor that the letters or words that child reads cannot evoke any meaning in his mind. The letters and words are to the child as so many meaningless syllable. His reading would resemble the reading of nonsense syllables by an intelligent child or adult. Further Languages development means ability to think in the abstract with the aid of language signs as vehicles of thought. And this ability to think in the abstract is one of the best signs of superior intelligence. These considerations fully justify our retention of this test as a test of intelligence, with the safe guard we have provided.

VIII Alternative 1. Tying a slip-knot. (model kept before the child. A spare thong. 2 minutes)

TEXT AND PROCEDURE

Use round leather thongs such as those used for football covers. Make the ready in advance of the experiment and show only the completed knot. The loop should be sufficiently large and the knot loose so that the turns of the thong may be easily seen. Say,

'You know what kind of knot this is, don't you? It is a slip-knot. I want you now to take this other thong and tie the same kind of knot around my wrist'.

At the same time give the other into the hands of the child and present your wrists before him. The completed slipknot should be kept on the table before the child with the loop turned away from him. The child might inspect the knot by taking it in his hands. The standing part should be sufficiently long. Let the child actually try the knot. In the knot made by child, the standing part is sometimes smaller than the other end. Score this plus if the knot is otherwise all right. After the child finishes, whether the knot is correct or not, ask him,

'Is it exactly like this one?'

If it is incorrect he may correct it if the time limit is not exceeded.

Remarks: As Indian children were quite unfamiliar with the bow-knot, it was thought that the test would be found unsuitable in their case and an attempt was made to substitute a similar test familiar to them and equally difficult. The slip-knot was ultimately selected. This knot is generally used by Indians to tie copper *ghadas* to the ends of ropes used in drawing water from wells. But children of seven and eight years were actually found to be unacquainted with this knot. In doing this they sometimes took much more than two minutes, which is too long, as the maximum time allowed is only two minutes. This seemed therefore to test the real intelligence of children more than it would have done, had the children had previous acquaintance with the knot. The test showed a steady increase in the percentage of passes as the age increased, but the correlation of the test with mental age as the scale as a whole was rather low, being 0.55. This test is put down as an alternative test in year VIII.

VIII, Alternative 2. Ball and Field (Inferior Plan)

TEXT AND PROCEDURE

Draw a circle about 6 centimeters in diameter, leaving a small gap in the side next to the child. This can best be done by cutting a circle of the required size out of thick card, and passing a red or blue pencil round the margin leaving a gap of required width. The circular line drawn should be of sufficient thickness.

Then say

'Let us suppose that your playing ball has lost in this round field. You have no idea what part of the field it is in. You do not know what direction it came from, how it got there, or with what force it came. All you know is that the ball is lost somewhere in the field. Now, take this pencil and mark out a path to show me how you would hunt for the ball so as to be sure not to miss it. Begin at the gate and shown me what path you would take'.

Adhere strictly to this wording. In particular avoid using a word like 'around' which might suggest a circular path to be traced.

If the child simply points the path with his finger, say,

'No: you must mark out your path with the pencil so that I can see it plainly'

Sometimes the child stops after tracing a bit of path and might say or think the ball is there. Then say,

'But suppose you have found it yet, which direction would you go next? This last formula will have to be repeated several times.

After the child has finished see if any plan governs his drawing. The drawing is scored for two ages the eight and the fourteenth.

Requirement of Year VIII. Some sort of plan should be evidenced in the drawing though perhaps not fully suited to serve logical purposes. If there be no sort of plan at all the performance is a failure. For example, mere random lines which may cross or re-cross one another or a single straight line or curved line, are failures. On the contrary If there be some attempt to make a search by some sort of zig-zag lines, or lines on the 'wheel plan', that is radiating from the center of the circle towards the circumference, on the 'fan plan', that is spreading out like a fan usually from the gate and so on, the attempt count plus. In all these plans, which are plus for year VIII but not for year XIV the path traced is usually not continuous and the lines are not parallel.

Requirement for Year XIV. If the line is continuous and parallel so that no big gaps are anywhere left unsearched the plan is plus for year XIV. The following are such typical plans as given by Terman:

1. A very nearly perfect spiral usually beginning at the gate or the center of the field.
2. Concentric circles. (In this case the circles need not be joined together).
3. Transverse parallel lines going from side to side joined at the ends.

The printed scoring card will be an aid in assessing the plans.

Remarks; This test is the one added by Terman in the Stanford revision. One does not feel quite at ease in administering the test. In the first place the children cannot realize that the map drawn is that of a very being field. They do not see the necessity of going round. They seem to think that they can stand in one place and look all round for the ball. Secondly, often it is very difficult to see if there is any plan in the drawing of the child. Sometimes even an intelligent child takes a wrong direction and fails in his plan. The first impulse of a child on seeing the map of the field is to close the gap with pencil. This is a common characteristic of the mind, which likes to complete what is left unfinished. Every gap even in acquired knowledge has to be filled in. We have placed the test as an alternative one in year VIII, and as one of the regular ones in year XIV. The correlation of the test as used for year VIII with mental age is 0.71 and as used for year XIV, it is 0.66.

CHAPTER SIXTEEN

TESTS FOR YEAR IX

IX. 1. Repeating Four Digits Reversed. (one out of 3. Read one per second)

TEXT AND PROCEDURE

The series are; 6-5-2-8: 4-9-3-7: 3-6-2-9:
Procedure same as in VII.3

IX. 2. Making change. (2 out of 3 correct. 15 second each. Read only once again if required).

TEXT AND PROCEDURE

Give the problems orally in the following order:

- (a) If I were to buy 4 rupees worth of cloth, should give the shop keeper 10 rupees. How much money would I get back?"
 - (b) If I bought 12 rupees worth and gave the shop-keeper 15 rupees, how much should I get back?
 - (c) If I bought 11 rupees worth and gave the shop-keeper 20 rupees, how much would I get back?
- The problems are to be solved orally. If two answers are given the second is to be scored according to the general rule.

Remarks: Binet uses actual coins and makes the problem one of shop-keeping. This procedure makes it a little more concrete, and perhaps slightly easier. The child is made the shop-keeper and he has some boxes to sell and some cash. The cash placed before him is in definite coins. The experimenter wants to purchase one of the boxes and offers a higher coin and asks for change. The child is actually required to make up the change from amongst the coins and hand it over to the experimenter. Binet places this test in year IX. Burt, following the same procedure with English coins, places it in year VIII. We have followed Terman's method of verbal arithmetic with Indian coins and have found it easy enough for year IX and slightly too difficult for year VIII.

The test requires on the part of the child not only a knowledge of the processes of addition and subtraction, but the ability to visualize the transaction and find out which of the two operations, addition or subtraction, is to be employed in this transaction. The process of imagining a transaction from auditory sensations is really a hard one and cannot be successfully tackled by feeble-minded children of very much higher age. They begin to add or sometimes even to multiply instead of subtraction.

IX. 3. Giving Similarities—two things. (2 out of 4 any real likeness is plus. 1 minute each)

TEXT AND PROCEDURE

Say to the child

'I am going to name two things which are alike in some way, and I want you to tell me how they are alike'.

- (a) Mango and banana—in what way they are alike. Similarity.
- (b) Iron and Silver
- (c) Steam-ship and Tonga
- (d) Wood and charcoal

After the first pair it is not necessary to repeat the whole formula. Simply say.
'in what way are-and-alike?

The child very often, probably not liking to lift the heavy burden of finding a solution to the problem. Say he does not know, or they are not alike. Then it necessary to say to him in a persuasive tone,

'they *are alike* in some way; you must tell me in what way they are alike'?

If a difference is given instead of similarity, say

'No I want you to tell me how they are alike. In what way are-and-alike?

Remarks: The test of giving similarities is psychologically harder than that of giving dissimilarities. While the letter is easy enough for year VII, the former even with the lenient way of scoring it, namely making two out of four enough for a plus, is hard enough for year IX. The test of giving similarities was used by Binet in 1905, but was not standardized for any age. It was standardized by Terman and located in year VIII. Feeble-minded, as well as younger, children more often give dissimilarities. Which is shown that the mind first tries to distinguish between objects by noting differences before noting similarities. This test is timed by Terman. But for reasons already explained it has been found better to time it.

XI, 4. Using three words in a sentence. (2 out of 3. Oral. 1 Minute each. Repeat formula once again, if necessary)

TEXT AND PROCEDURE

The three sets of words are:

- (a) Boy, ball, river.
- (b) Work, money, men.
- (c) Trees, rivers, lakes.

'You know what a sentence is, of course. A sentence is made up of some words, which say something. Now, I am going to give you three words, and you must make up a sentence that has all three words in it'.

The answer must be given orally. Similarly with the other sets of words, but with these sets use the shorter formula: 'Now make up a sentence having the words...in it'.

If the instructions are not comprehended repeat them once again, but never change the formula in any way.

If it is found that the child is laboring under the misapprehension that the sentence is to be made only with these three words and that no other words are to be used, say,

'The three words must be put with some other words, so that all of them together will make up a sentence'.

If the boy thinks that the three words are to be used in three sentences, say,

'No, make only one sentence using all the three words'.

To be regarded as plus, (i) the sentence must be a simple sentence, and (ii) it must make good sense and should contain no absurdity.

Remarks: Binet's procedure is to give only one trial with the words- pairs, rivers, and fortune,. He also required a written response, and this makes the test a little harder, locating it in year X. We have followed Terman's procedure of giving three trials with three different sets of words and requiring an oral response. This

procedure is certainly better than Binet's and leaves to mere chance. With this procedure the test is well suited to year IX as was found by Terman also. In the third set of words we have replaced 'desert' by 'trees' as it was found that Indian children have generally no knowledge of deserts.

In scoring the responses, we have not met the difficulties mentioned by Terman or Burt in English construction of sentence. Terman has allowed a compound sentence with two distinct ideas as plus for year IX. Burt has allowed two distinct ideas as plus for year X. Our responses show that only one simple sentence with one idea is easy enough for year IX. With English sentences such responses would be hard enough for year X or XI. As regards the second requirement of the sentence, that is containing no absurdity, we also met with no serious difficulty in scoring. The children who gave correct responses from the point of view of sentence structure generally gave sensible sentences.

Burt says' * A set of sentences in which the thought is well coordinated into a unitary story of description, passes (that is for Year IX, which requires one idea or sentence). "London is a big place. It has river in it. And many people come there to make money". We have distinctly disallowed this. Nor have we found necessity of scoring this plus with the formula we have used. Binet allows an absurdity in the sentences, but not Terman. We have found that it makes no difference in score since, as said above. When the first condition is satisfied the second is invariably so.

This test is a very valuable one. It is of the general type of 'completion' tests. For success in it, the association formed by children of each of these words with other ideas must be very rich. So that when three words are uttered, children can join them into one coherent idea with the help of these performed association. A more intelligent child has always-keener observation and forms a greater number of such associations.

*. Cyril Burt, *Mental and Scholastic Tests*. P.52.

IX, 5. Reading and Report: (6 facts: 5 errors: 1 minute)

TEXT AND PROCEDURE

See VIII.6

IX. 6. Free Association, 35 Words in 3 minutes. (Record every half-minutes score separately)

TEXT AND PROCEDURE

Say,

'Now, I want to see how many different words you can name in 3 minutes. When I say 'ready' you must begin and name the words as fast as you can, and I will count them. Do you understand? Be sure to do your best, and remember that just any words will do, like "clouds", "dogs", "Chair", "happy", "come", "go", Ready, go ahead'

The instruction may be repeated once again if they are not understood. But generally there is no necessity to it.

Do not disconcert the child staring at him. Keep your eye away from him or on your record sheet. Record the words verbatim if possible, but if there is no assistant, this is generally not possible. It is interesting to record at least the class of words given, such as concrete objects abstracts ideas, verbs, adjectives, and so on. Repetitions should not be counted.

If the child stops altogether, wait for 15 seconds and then say, 'Go ahead as fast as you can. Any words will do'.

Keep on urging like this if the child stops for than 15 seconds.

Some children begin to give sentences or to count: then say,
'Counting (or sentences) is not allowed. You must name separate words. Go ahead'.
Proper names are not allowed; but mythological names may be allowed.

Remarks: This is a test which was found to be most deceptive in its working, particularly in the form in which Binet used it, namely, giving sixty words in three minutes. While even many normal adults could not pass in this test, with a score of sixty words, there were a few children even of nine and ten years of mental age who could give more than sixty words. From the percentage of passes, the test in Binet's form was found difficult even for the adult stage. Then the difficulty sometimes arose, whether a child who passes in this test but fails in all the tests of years XII and XIV, should be scored plus only in this test in year XVI. For these reasons the test is relegated to the class of alternative tests in year XVI. For interesting comparison, actual number of words given by all children above the level of seven years of age was recorded, and the mean and median scores of children of the several chronological ages were worked out. The table below gives the mean and median scores at the several ages:

<i>Chronological Age (in years)</i>	<i>Number of Children</i>	<i>Mean Score Score in 3m</i>	<i>Median Score in 3m.</i>	<i>Remarks</i>
7 to 8	22	35	35	Low score in Tenth year probably due to detained children in Primary IV
8 to 9	50	38	37.5	
9 to 10	61	34	30	
10 to 11	86	42	40	Note that there is practically no rise above the 14-year level
11 to 12	75	43	40	
12 to 13	70	48	47	
13 to 14	59	55	56	
14 to 15	36	52	48	
15 to 16	26	56	53	
16 to 17	28	52	48	
17 to 18	13	54	48	
18 to 20	20	58	60	

It was thus found that the normal score for age IX was 35, and that for 'very superior adults' was 80. At these two extremes, the test is more reliable than at year XVI. Now and then a child is found who sits silently throughout the three minutes although encouraged several times to give words. Sometimes a child gives three or four words and then goes on repeating these same words several times, at the same time appearing to make an effort to find new ones. This is an interesting form of stereotype and is a symptom of weak minds.

When this widely divergent speed of association revealed by Indian children is compared with that of western children, one is inclined to believe that the very slow environment of eastern children is responsible for this. The western child is surrounded by men and objects that are moving fast and is forced constantly to think and react to these very speedily to that quick thinking, and the speedy movement of associated ideas in the minds, grow into a habit with him. On the contrary, the environment of an eastern child, particularly rural child, being very slow, there is not necessity for him to think speedily at all. The same thing is also to be seen in Indian schools. The teachers never pay regard to the cultivation of speed in work and the children go on at their own snail's pace. In the street the schoolboy lazily plods along in tune with the surrounding moving objects, the slow moving card, the leisurely cowboy and so on. Of course, in some of the larger Indian cities life is very much quicker, but the population of our cities, as compared to the general population, is less than one-fifth. So the speed of work of our general population is very much slower, and speedy work means a speedily thinking mind. The lesson we have to learn from this, for our educational system, is to be devise ways and means to cultivate speed in our school work. This need is all the more urgent on account of our slow environment. Our arithmetical problems and other pieces of schoolwork may be timed and the normal speed of accomplishing these pieces may be determined and children trained to attain this speed. Only by such methods can we hope to remedy a fundamental defect in our educational and social systems. Writer for some time tries a method of

classroom teaching which greatly enlivened the children and was instrumental in cultivation speed in their work. When an example or piece of work was set, the children were asked to work it out and return their answer books, which were arranged on the master's table in order of precedence. The books were then marked in such a way that the earlier children got more marks. For example, the first boy if his work was right got 10 marks, the second and the third got 9 marks, the fourth and the fifth got 8 marks and so on. This enlivened the children very much and after a short time the children understood the value of quick working. This is mentioned only as an example of how teachers may devise simple classroom methods to attain the objects in view.

The habit of rapid thinking is certainly an important factor for success in this test. Some psychologists have thought that the surrounding objects afford clues to words and required children to close their eyes during the test. Far from helping the children, the presence of a number of objects is hindrance in the way of selecting words quickly. It was observed again and again that though a child began naming one of two surrounding objects he could not proceed. When he named a table or a cupboard he seemed to exhaust the objects around him. Nor could he see that the 'drawers' or 'handles' or 'books' (in the cupboard) were new words. After giving only one or two names of surrounding objects he usually fell back on his mental resources and began to give words from his imagination. What Binet has said about *absent* objects applies equally well to *present* objects. He says: * 'Little children exhaust an idea in naming it. They say for example, "hat" and then pass on to another word without noticing that hats differ in colour, in form, have various parts, different uses and accessories, and that in enumerating all these they could find a large number of words'. There is therefore no point in making children close their eyes which, in the case of some children, being unusual, even disconcerts them and obstructs their thought processes. The room in which children are tested may, if so desired, be as bare as possible, as Terman advises; but, even is found to be unnecessary. The fact that, though children are surrounded by a number of identical objects, some of them can name a good many and others hardly any, is an excellent illustration of the fact the same physical environment can assume various different forms according to the mental ability and equipment of the observers. Thus the sight of the same flower may excite only a vague colour sensation in a child, may present a beautiful and aesthetic shape to an artist or may appear to be made up of several important parts such as stamens or pistil to a botanist.

*Quoted from Terman, *The management of Intelligence*. P. 273

The most intelligent way of tackling the problem is to give the names of objects are, parts of the objects, connected with the object that is named. For example, a child gives the word 'tree' and then names its parts such as 'branches', 'leaves', 'flowers', or gives names of objects associated with the given word, such as 'garden', 'creeper', and so on. Such an intelligent method is followed by very few children. Only about two or three children in the whole of our testing did this. The majority of children named an object and then gave one or two objects associated with the given word and passed on to think of objects more distinctly associated with it.

Only a very few of the children go on finding abstract words or words having abstract relationship with the given word. No doubt, only advanced children are capable of doing this; but merely from the point of view of giving the maximum number of words, such a method is not very successful one.

It is interesting to note some of the causes of failure in this test, first of all, it is the poverty of mental content and previously formed mental association that is the main cause of failure. As Terman says: 'the main factors in success are two, (1) richness and variety of previously made association with common words; and (2) the readiness of these associations to reinstate themselves'. But apart from this poverty of mental associations, there are certain other factors, which cause failure in this test even with some intelligence subjects. Some children fail owing to their very emotional nature; they are too shy or nervous. In fact, shyness plays greater role in this test than in any other. Some intelligence children get the impression that they are to give very difficult or out of the way words. Hence they waste in finding them. A few are led away by verbal associations of similarity of sound, and begin words rhyming with the given words, although they are warned against it. This occurs more particularly when the test of finding rhymes is given before this test.

Children who are talkative succeed with this test better than thoughtful but more reticent children, although the former may be far less intelligent than the latter. Some children are given to hasty work and hasty replies. Such children fail in some other tests merely on account of their haste but they get a counterbalancing advantage in this test. From this point of view, the test deserves a place in this scale since some children who undeservedly lose in some of the other tests merely from their over-haste, justly get some advantage in this test.

The rate at which words are given does not remain constant. Usually the number of words given in the first half minute is larger and they fall off gradually towards the end of the period. The average number of words given by 10-year-old children in the successive half-minute periods is as follows:

	Half-minute periods					
	1 st	2 nd	3 rd	4 th	5 th	6 th
Bombay-Karnatka	10	6.2	5	4.8	4	3.8
Stanford Revision	18	12.5	10.5	9	8.5	7
London Revision	19.3	13.4	10.3	8.5	7.3	6.6

Burt's figures however are obtained in a slightly different way. They are the average number of words in the successive half-minutes given by children who pass in the test. Further, the average age of Terman's group seems to be 10 years, that of Burt's group 9.5 years, while in the present study it is 9.5 years.

The corresponding average scores in the successive half-minute periods in the present revision for adolescents of fifteen and sixteen years of age are as follows:

	Half-minute periods					
	1 st	2 nd	3 rd	4 th	5 th	6 th
Average age of 14 ½ years	16	9	8	7	6	6
Average age of 15 ½ years	17.7	10.6	8.5	7	5.8	6.4

For comparing simply the speed of association in different half-minutes it is better to word out the percentages of words given in these half-minute periods. They are:

	1 st	2 nd	3 rd	4 th	5 th	6 th
Bombay-Karnataka	29.6	18.3	14.8	14.2	11.8	11.2
Stanford revision(Terman)	27.5	19.1	16	13.7	13	10.7
London Revision (Burt)	29.5	20.5	15.7	13	11.2	10.1

It will be seen that the biggest drop in all the revision is after the first half-minute. There is no warming-up period.

Lastly, some of the very backward children give sentences instead of words, even though they have been told to give words and not sentences. This seems to suggest that the units of ideas are sentences and not words. When a child thinks of a horse, can he get an image of the horse simply, or does he imagine things in some such form as 'this horse is black'. 'a horse is yoked to a Tonga', 'I saw a horse in the bazaar'? There are very strong reasons to suppose that the latter form is the unitary form of our ideas though we may not be sufficiently trained in the use of language to express these ideas in the form of sentences. The subject is one which requires exhaustive study.

On the whole for various reasons detailed above the test of Free Association (IX, 6; XVI, Alt. 2.; XXII, 6) had better be dropped as a test of intelligence as recommended at the end of Chapter Eight. Its results are not in harmony with the results of other test in the same age-groups.

IX, Alternative. Test of Vocabulary, twenty words. (Equivalent to 7200 out of 36,000 words of Marathi language; or to 5600, out of 28,000 words of Kannada Language)

TEXT AND PROCEDURE

Place the card of printed words before the child, and say,

'I want to find out how many words you know. Listen and when I say a word, you tell me what it means'.

'What is an orange?

'What is a bonfire'?

'Roar, what does it mean'?

If the child hesitated being different about giving a proper definition, say a little encouragingly,

'You know what a bonfire is. You have seen a bonfire, Now what is bonfire?'

If the child is still silent, say further,

'Just tell me in your own words: say it any way you please.

'All I want to find out is whether you know what a bonfire is'.

If the child answers is not quite satisfactory. Or is imperfect, say,

'Explain. I don't understand. Explain what you mean'.

Adhere strictly to the above formulae. Never illustrate the use of a word in any way, such as by using it in a sentence. Rigid definitions are not expected from children. Sometimes in giving meanings even a change of a part of speech is allowed.

*1. These words are given only for the sake of illustration. The actual words of the vocabularies in the two Indian languages are given on the vocabulary cards of these versions.

Remarks: In assessing the definitions given by children great leniency needs to be shown, because the power of expression of children, particularly below the age of twelve years, is very poor. In the twelfth year they just begin to define abstract words correctly (see XII,3). Hence below this level child shows that in some way he knows the meaning of a word the definition should be scored plus. Nouns may be rendered into adjectives or verbs, or again in terms of long explanatory sentences but, so long as the correct idea is expressed, the response should be scored plus.

For scoring purpose, it is enough if the child defines the requisite number of words, for example 20 words from both cards together in this year; each card consists of 50 words. But it is interesting to find the total vocabulary of the child in every case. It is done as follows. This list consists of 100 words selected from a small Kannada dictionary containing about 28,000 words. These 100 words are representative of the 28,000 words in dictionary. Thus if a child can define only 20 words out of 100 he would be able to define 20 per cent of the whole number of words in the dictionary; that is, his total vocabulary according to the dictionary; that is, his total vocabulary according to the law of chance is 5,600 words. The Marathi dictionary from which the list is made up consists of 36,000 words, so the average Marathi vocabulary of a child of 9 years is 7,200 words. Strangely, the average number of words defined by Kannada and Marathi children in the several ages came very nearly to the same figure. So we have retained the same figure for both Kannada and Marathi children for all the ages. The following table gives the same figure for both Kannada and Marathi children for all ages. The following table gives the means and median scores of Kannada and Marathi children separately in the several

years:

<i>Age next birthday</i>		8	9	10	12	14	16	<i>Above</i> 16
<i>Kannada</i> {	<i>Mean</i>	20	23	25	31	37	45	52
	<i>Median</i>	20	23	25	30	38	46	52
<i>Marathi</i> {	<i>Mean</i>	14	20	25	31	49	49	50
	<i>Median</i>	13	19	25	27	48	46	51

The vocabulary test should not be given unless the mother tongue and the language learnt by children in the school are the same. In Kannada districts quite a number of children have Marathi as their mother-tongue and learn Kannada in schools, and vice versa in Marathi districts bordering on Karnataka. In such cases, it has been standardized which of the two languages a child knows better.

Hence this test has been standardized only on children whose mother-tongue and school language coincide. Contrary to expectation the test has been found to be a very good test of intelligence. One might have thought that the test depended a good deal on schooling, but it is not so. The correlation of this test with mental age as determined by the scale as a whole is 0.89. Just as an intelligent child picks up from his environment more knowledge than a duller one of the same age, so also he picks up more words from the language. Thus wider vocabulary means wider knowledge and higher intelligence.

CHAPTER SEVENTEEN

TESTS FOR YEAR 'X'

X, 1. Arranging Five Weights. (2 trials of 3 correct)

TEXT AND PROCEDURE

Use five pill-boxes 3 centimeters or a little less in diameter, all identical in appearance and load them with lead foil or lead shot until they weigh 3, 6, 9, 12, and 15 grams respectively. The contents should be properly packed in cotton wool and should not rattle. The lead shot or foil should be as far as possible in the very center. Mark these boxes(weights) on the under surface by letters like P, S, Y, C, H, so that after the child has arranged the boxes(weights,) the examiner can silently turn them over and check the arrangement. Place these boxes(weights) on the table in an irregular fashion before the child and say.

'See these boxes, They all look alike, don't they? But they are not alike. Some them are heavy, some are not quite so heavy, and some are still lighter. No two weigh the same, Now, I want you to find the heaviest one and place it here. Then find the one that is just a little lighter and put it here. Then put the next lighter one here, and the next lighter one here, and the lightest of all at this end (pointing each time at the appropriate spot).

Do you understand?

'Remember now, that no two weights are the same. Find, the heaviest one and put it here, the next heaviest here, and lighter, lighter until you have the very lightest here. Ready, go ahead

Note that the instructions are repeated with a shorter formula. This is always necessary.

If the child still does not understand, repeat the whole of the instructions once again. No further help should be given. The child should not be shown how to weigh the boxes in his hand; finding out the correct procedure is part of the test.

Some children get the impression that the performance should be done as quickly as possible; other take longer time and check their first arrangement. For uniformity after the first arrangement, it is better to say, 'See again if you have placed them properly'.

Remarks: Always record the order in which the child places the boxes. According to Weber's law, the case with which the differences between successive weight boxes are distinguished should be in ascending order beginning from the end of greatest weight, the hardest difference to distinguish being that between 15 and 12 grams and the easiest one that between 6 and 3 grams. On refereeing to the actual statistics we find that the percentages of passes in estimating the difference between the successive pairs are:

Difference between 15 and 12 grams	77%
Difference between 12 and 9 grams	91%
Difference between 9 and 6 grams	94%
Difference between 6 and 3 grams	86%

This is a corroboration of the truth of Weber's law. The slight fall in the last pair appears to be due to the fact that the lightest weights are so light that it is difficult even to feel their weight on the hands.

This is not merely a test of 'sensory discrimination'. If it were it would, probably, be less value as a test of intelligence; for, as has been observed already, even lower animals sometimes have greater sensory discrimination than man. Binet calls it a test of 'sensorial intelligence'.

The test requires in the first place proper comprehension of the instructions; in fact, this is an important factor in the proper solution of the problem. We have therefore always insisted on laying down accurate procedure in giving the instructions. The wording of the test should not be changed even in the slightest degree. The speed with which the instructions are given, the proper enunciation of the phonetic sound, the emphasis on keywords, the number of repetitions of the instructions, should all be carefully laid down. In comprehending the instructions, the children are required to visualize the several times of performance contained in the next instructions, keep them in mind till all the items are given, weave them together so as to form one whole, and decide upon the goal to be reached. After the goal to be reached is discovered a great effort of attention is required to keep it in mind until the several adjustment and adaptations, that are required to reach the goal, are made.

* After the formula there is sometimes the chance of a correct arrangement being spoiled, but we are unable to help this. The formula is to be given in all cases. Terman expressly disallows this step, though some other workers have allowed it.

This test is one most interesting tests from the children's points of view. It is a test, which requires some *doing* in the part of children. Children are fond of doing concrete things than of listening to things or thinking in the abstract. The test, therefore, serves, as a great relief when given after a good many other tests that require abstract thinking. If it is given at the beginning it arouses the child's interest and he takes to testing more kindly afterwards. The correlation of the test with mental age however is not very high. It is 0.57.

Note down any peculiarities in the performance. Intelligent children lift the weights and balance them by slight movement up and down in the hand, sometimes successively in the same hand, sometimes simultaneously in the two hands. The former method is the better of the two. Backward or younger children often fail in grasping the idea of arranging the weights in descending order of weight.

X, 2. Repeating Twenty to Twenty-two Syllables. (1 out of 3: or 2 with 1 Error each).

TEXT AND PROCEDURE

- "The apple tree makes a cool pleasant shade on the ground where the children are playing".*
- "It is nearly half-past one o'clock; the house is very quite and the cat has gone to sleep".
- "In summer the days are very warm and fine; in winter it snows and I am cold".

Procedure same as in VII.1

* The Kannada and Marathi versions are mere translations, but the number of syllables is kept the same

Remarks: It is interesting to note how the memory for syllables increases with age. The following tables gives comparison:

Age in years	No. of Syllables given	Increase in age in years given	Increase Syllables	Increase Per year
III	7			
IV	13	1	6	6
VII	18	3	5	1.7
X	22	3	4	1.3
XIX	30	9	8	0.9

The increase per year in the number of syllables given, steadily decreases as the child advances in years. Compare the increase in the number of digits given as age increases (see p.161)

X, 3. Naming the Months (20 seconds: 1 error in naming. 2 checks of 3 correct.)

TEXT AND PROCEDURE

'Name all the months of the year'

'What month comes before April?'

'What month comes before July?'

'What month comes before November?'

Ask these questions straightaway and give no other help, even to start him off with any one month provided he completes the cycle.

Keep a record of the time from the moment the child names the first month. Sometimes intelligence children may not recollect the name of the first month at once and may waste a good deal of time.

Remarks: See remarks under VII, 4, on time orientation.

This test is criticized on the score that it depends a good deal on schooling or training. But it is one thing to know the months by rote and quite another thing to have time orientation of this order. The checks in particular and test X, Alternative 2, go to show whether this much time orientation has appeared. On the whole the test has served its purpose well and there is not much weight in the criticism.

X, 4. Drawing Designs from Memory (1 correct, 1 half-correct, Expose 10 seconds)

TEXT AND PROCEDURE

Before exposing the card say to the child.

'This card has two drawings on it. I am going to show them to you ten seconds, then I will take the card away and let you draw from memory what you have seen. Examine both drawings carefully and remember that you have only ten seconds'.

Provide the child with paper and pencil and then expose the card for ten seconds, taking care that the card is presented to him in the right position, with the Greek key pattern to his left and the truncated pyramid to his right. After 10 seconds quickly remove the card out of sight and cover it up or insert it in the packet, so that there is no temptation on the part of the child to cast his glance in that direction. Let the child begin to draw as soon as possible after the card is removed. The drawings on the scoring card will help the examiner to mark each diagram correct, half-correct or wrong. The following hints will be found useful in scoring.

Correct: The peculiarities of the drawings as regards right angles, the different individual parts and eccentricity of the inner rectangle are all correctly reproduced. The drawing need not be neatly drawn from the artistic point of view nor the straight line perfectly drawn. One of the rectangles in the truncated pyramid diagrams may be drawn square or one of the top squares in the Greek key pattern may be turned outside. The eccentricity in the truncated pyramid may be shown in any direction. The whole diagram may be inverted if otherwise perfectly correct.

Half-correct: The drawing are generally correct, but some (only one) important peculiarity may be wrongly drawn or not drawn at all: for example eccentricity is not shown at all, both rectangle are drawn as square or vertical rectangles in the truncated pyramid; angles are rounded off instead of being drawn square; the turns in the Greek key patterns are drawn sometimes less sometimes more in number.

Failure: When the figure is incorrectly drawn as regards more than one of the essential peculiarities.

Remarks: It is difficult to lay down exactly in words what constitute a success, a half-success, or a failure. The sample drawings show on the scoring cards will be a great help. Burt has attempted to analyse the errors and lay down the method of scoring, but his scoring seems to be little too lenient. * The first two diagrams of the truncated pyramid which he shows as successful will have to be taken as half-correct according to the above instructions.

According to Binet, the chief factors required for success in this test are 'attention', 'visual imagery', and 'little analysis'. The child has to concentrate his attention on the drawings, fix the visual imagery in his mind, with some analysis as to how the line is placed and their relative sizes. There is another important factor which intelligent children make use of and which is not mentioned by Binet. This is the motor memory. The child as he sees the diagrams traces the lines in his mind imaging his fingers going round the diagrams. When he does this, the chances of his success are greater as his motor memory comes to his help where his visual imagery fails. According to Terman, analysis is an important factor. He points out that one of the diagrams consists of thirteen lines and other of twelve and this number is much beyond the memory span for line, unless these lines are grouped and the grouping analysed and remembered as a lesser number of units. For example he says: 'The design to the right, which is composed of twelve lines, may be reduced to four elements: 1) the outer rectangle; 2) the inner rectangle; 3) the off-centre position of the inner rectangle; and 4) the joining of the angles. Of course the child does not ordinarily make an analysis as explicit as this; but analysis of some kind, even though it be unconscious, is necessary to success. * It should be remembered also that the time of exposure in this experiment is much more than time usually allowed in tachistoscopic studies by which memory span is determined.

*Cyril Burt, *Mental and Scholastic Test*. Pp. 53.-5

*Terman, *The Measurement of Intelligence*, p. 261

Binet and Burt placed the Greek key pattern to the right and found that failures on this were much more frequent than failures on the truncated pyramid pattern. This is probably due to the fact that the figure on the left hand receives our greater attention, because as a result of our reading habits it is examined first. Terman reversed the position and found that the failures were almost equally divided between the two. The more difficult figure was placed to the left and secured greater attention. Our statistics agree with Terman's and show that the failures on the Greek key pattern are almost exactly equal to the failures on the truncated pyramid pattern. A good many children, however, drew the second figure first though it was placed to the right in the original. This is probably because they thought it easier and they could attempt it first.

Binet (1911), Terman and Burt all agree in placing this test in year X. Our statistics also produce the same conclusion.

X, 5. Finding Rhymes. (3 rhymes for each word. 1 minute for each part. 2 out of 3 correct).

TEXT AND PROCEDURE

Say,

'you know what a rhyme is, of course, A rhyme is a word that sounds like another word. Two words rhyme if they end in the same sound, understand? Take the two words "hat" and "cat". They sound alike and so they make a rhyme. "Hat", "rat", "cat", "bat" all rhyme with one another. Now, I am going to give you a word, and you will have one minute to find as many words as you can that rhyme with it. The word is "day".

Name all the words you can think of that rhyme with "day".

If the child begins to give words without meaning, say,

'The words you give must have meaning'.

If the child fails with this, give three words that rhyme with 'day' and say,

'All these sound alike with "day", Do you understand?

Proceed further after this, or directly after the child has given correct rhymes with the first, as follows.

'Now you have another minute to name all the words you can think of that rhyme with "mill".

Similarly with the third word 'spring'.

If the child gives words that have no nasal sound, say,

'I want words with a nasal sound'.

Scoring would be done liberally taking as correct proper names and negative terms formed by adding a prefix to the given word, provided they rhyme properly.

Remarks: Binet's procedure of giving this tests, as also that of Burt, seems to be much harder than this one, which is Terman's. Binet gives only one word and requires three rhymes. In the present procedure three trials are given requiring three rhymes for each of which two have to be correct. Again Binet's word is tri-syllabic, and Burt's disyllabic, which are both consequently harder than Terman's monosyllabic words. Our words are disyllabic, but they are easy enough, perhaps easier than monosyllabic words, as monosyllabic words would be harder to give in the Indian languages than disyllabic ones. Besides, the illustration given in Terman's version are more elaborate than Binet's. This seems to be responsible for the wide divergence in the location of the test. Binet (1911) locates it in year XV, Burt in year XII, Terman in year IX, while the present revision places it in year X.

The test is found to be very good test of intelligence and it satisfies very well all the requirement of such. Its correlation with mental age as determined by the scale as a whole is 0.78, and the percentage of passes in the test from year to year show fine rising curve. Psychologically the test requires the child to fix the goal permanently in mind and under the guiding force of this goal to find other words of similar sound by a trial and error method. After a word is found, it must be compared with the given word the memory of which is also firmly retained and the child sees whether the new word satisfies the condition laid down. This is no mean task for the associative powers of the mind. A little remissness in any of these essential requirements results in failure. Backward children either fail to keep the goal fixed in mind, or to remember the given word, or to awaken their powers of memory by which the whole store of vocabulary is searched for similar words by the method of trial and error.

Thus when the word "hill" is given the child makes words like "fill", "rill", "bill", and sees which of these words are in his vocabulary. The mind of a backward child works like that of a normal child in a sleepy condition. The mind is dissociated and cannot hold together all the conditions laid in the problem. One or more of these slip out and hence the solution is incorrect.

X, 6, Reading and Report (8 facts. 2 errors: 40 seconds)

TEXT AND PROCEDURE

See VIII, 6

Remarks: Terman allows 35 seconds for reading, while we have allowed 40 seconds. But Terman's English passage contains 68 syllables, while our Kannada passage contains 134 syllables and the Marathi passage 116 syllables.

X, alternative 1. Vocabulary, Twenty-five words.

TEXT AND PROCEDURE

See IX, alternative test

X, Alternative 2. Giving Month and Year (Both Correct)

TEXT AND PROCEDURE

'What month is it?'

'What year is it?'

The name of the month must be given, and not simply the number, and the year-both according to the Christian calendar.

See VII. Alternative test.

CHAPTER EIGHTEEN

TESTS FOR YEAR XII

XII, 1. Detecting Absurdities (3 out of 5. ½ minute each. After 5 seconds read time.)

TEXT AND PROCEDURE

Secure the child's attention and say

'I am going to read a sentence which has something foolish in it. Some nonsense. I want you to listen carefully and tell me what is foolish about it'.

Then read each of the following twice, as stated above, emphasizing the key words:

(a) A man said "I know a road from my house to the town, which is downhill all the way to the town and downhill all the way back home".

What is foolish about that?

If the child answer is not clear say again,

'I am not sure that I know what you mean. Explain what you mean. Tell me what is foolish in the sentence I read?' Similarly the rest.

b) An engine driver said that the more carriages he had on his train the faster he could go'.

c) Yesterday the police found the body of a girl cut into eighteen pieces. They believed that she killed herself.

Sometimes children say this is foolish because nobody kills herself. Then say, 'she might have been tried on life and might have thought of killing herself'.

d) There was a railway accident yesterday, but it was not very serious. Only 48 people were killed'.

e) A bicycle rider being thrown from his bicycle in an accident, struck his head against a stone and was instantly killed. They picked him up and carried him to hospital, and they do not think he will get will again'.

Any sensible answers are correct.

Remarks: Of the five sub-tests in this three are; Binet's, namely, c), (d), and (e). The first two are substituted by Terman for Binet's two sub-tests, which are as follows:

(1) 'I have three brothers-Jack, Tom, and myself'

'What is silly in that?'

(2) 'A man once said: "If I should ever grow desperate and kill myself, I shall not choose a Friday to do it on; for Friday is an unlucky day, and would bring me bad luck'.

'What is foolish in what the man said?'

These two appear to be slightly harder than Terman's. The difficulties in these tests pointed out by Terman do not exist in Indian languages, but we have preferred to retain Terman's selections. Terman, however, requires four to be correct out of five, while Binet would require only three out of five. The latter procedure is better, and with this procedure we found that the test is suitable for year XII; that is, about 40 percent of children between

10 and 12 years of age and 74 percent of those between 12 and 14 years pass it. With Terman's procedure, that is on the basis of four out of five, the test is found to be hard enough for year XIV in which 53 percent pass the test.

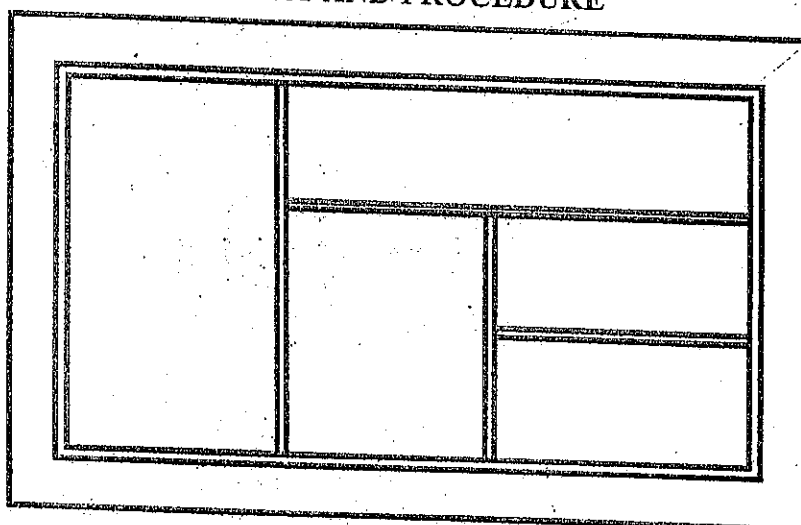
The Test is one of the series, in that it hardly requires any schooling or training and children of this age with common sense or 'mother wit' can easily pass it. It correlates very highly with mental age determined by the test as a whole, the correlation being nearly 0.9. It requires on the part of the child the ability to *visualize* the items given and find out the logical incongruity between the several items. For this of course the child must have acquired a previous knowledge of certain facts, such as the slope of a road and the appreciative mass connected with it obtained by actually walking over such roads; or the experience that a heavier vehicle is harder to pull and moves with less velocity, and so on. The greater the number of items to be reconciled and the less organized the acquisition of previous experience, the more difficult the test. The logical incongruity assumes various forms in the different sub-test. In the first the child is required to know all the properties of a sloping road, such as that walking down it is easier than walking up, water flow down such a road and not up and hence to reason that the same road cannot be downhill both ways.

*Cyril Burt, *Mental and Scholastic Tests* Pp. 56 and 57, which give Binet's tests adapted for London children.

For the second problem, the child is required to have possessed kinesthetic memories of heavy and light carriages. The child in various ways in his childhood gets experience in pulling toy carts of other loads and preserves these memories in his nervous system connected with the muscles that are used in dragging such load. Such kinesthetic memories now help the child to reason that a heavier train can not move faster than a lighter one. In the third problem, the child has to visualize the girl's body lying cut into eighteen pieces and to bring into proximity with this his knowledge that the girl would die immediately she cut herself into two pieces and therefore could not go on cutting herself into pieces. In the fourth problem, the visualization is also very important but more difficult. The child must create a mental picture of a train that is derailed or some such thing and of forty-eight people those are dead. Then only does the child see the gravity of the situation. In the fifth, also very much same kind of visualization is necessary. The child's attention must be focused on the picture of the cyclist dying on the spot. Then only can he find the incongruity of this statement with those that follow.

XII, 2. Construction Puzzle. (Healy and Fernald. 3 times in 5 minutes)

TEXT AND PROCEDURE



The above is a drawing of the dimensions of the form board with the several pieces fitted into it. It can be locally made by a careful carpenter. It is an open rectangular wooden block whose external dimension are 9.5cm X 12 cm X 1.2 cm deep, while inside the excavated hollow measures 7.5 X 9.75 X 0.6 cms deep. The wood pieces that fit into this hollow are 0.6cm in thickness and are five in number. Their dimensions are 7.5 X

3 cms, 6.75 X 2.5 Cm, 2.5 X 3.75cms.

Place the outer frame of the form-board on the table in front of the child with longer side nearest to him*. Put all the smaller pieces in a heap before him in a little further away than the frame. The smaller blocks fitted in their proper position should not be seen by the child. Then say to the child:

'I want you to fit these blocks in this frame as quickly as possible. If you do it rightly they will all fit in and there will be no space left over. Go ahead'.
For a plus he must do it three times in a total time of five minutes.

Remarks; It is very interesting to observe children performing this feat. Intelligent children do it with proper understanding of space relation. Backward children cannot project their imagination even slightly, and proceed with the performance till the last move even though, obviously, success by these moves is impossible; or they put the first piece in such a way that a small gap is left at its side, where obviously no other piece could go; or again they try to insert a broad piece in a much narrower space and go on trying to force it in though it is quite clear that it cannot go in. There is always an element of trial and error in the performance of both intelligent and backward children, but in the case of intelligent children there is more *insight* and less *trial and error* while in that of the backward there is more of trial and error and less of insights.

*Terman places it with the shorter side nearest to him. Apparently both are alike in the results.

This test is quite apart from other tests in that it requires no linguistic ability and depends only on the ability of manipulation. It is therefore a very good test as a change. Terman says: 'The test has a lower correlation with intelligence than most of the other tests of the scale'. Our finding however is quite the reverse. It has a very high correlation with mental age as determined by the scale as a whole, namely 0.85 and it shows a steady rise in the percentage of passes from year to year. We have never found a very backward or a very young child succeeding in it by virtue of mere trial and error in the specified time, nor have we found any very intelligent child failing in it. On the contrary it is a test in which children take very great interest, as intelligent *doing* is always better liked by children than mere verbal reproduction or abstract thinking. Anyone who has studied Kohler's experiments on apes will easily see how even simple acts, which appear very simple to us, require very great insight and cannot possibly be performed merely by the method of trial and error. *Terman says nothing of doing the performance as quickly as possible but it appears to be better to tell the child what exactly is expected of him, without making him nervous by unnecessary hurrying during the performance itself.

XII, 3. Defining Abstract Words. (3 out of 5. 1 Minute each to start defining)

TEXT AND PROCEDURE

The words are

(1) Pity, (2) Revenge, (3) Charity, (4) Envy, (5) Injustice

Say,
'What is pity?'
'What do we mean by pity?'
If the child answers in terms of the given word itself as 'pity means to pity someone', say,
'Yes, but what does it mean to pity someone?'

Logical definitions are not required. It is enough if the meaning of the word is brought out by any sort of explanation or by examples. It is good to analyse what elements are required in the definition in order to score them plus.*

*See Kohler, *Mentality of Apes*, pp. 130-33.

In one instance the chimpanzee, when he was given two sticks which could be telescoped at the ends by fitting the end of one into the hollow of the other tried to reach the bananas by pushing one of these sticks outside the cage with another that he held in his hands.

Pity has the idea of (i) tender feeling or helping attitude, (ii) when another person is in difficulty or is suffering.

Revenge has the idea of (i) another person doing some kind of injury to us and (ii) our answering it with a similar or a greater injury, either in *thought* or *execution*.

Charity has the idea (i) giving something to other or showing kindness to others, (ii) when they are in need, or for *religious* purpose.

Envy has the idea of (i) feeling uneasiness or dissatisfaction, (ii) when others are in an advantageous position, have obtained some sort of materials gain or superior position.

Injustice has the idea (i) one's having done something and (ii) another person returning some sort of punishment or act for it that the first act did not deserve. Sometimes the same person (i) may do or think something and (ii) may deny having done it or thought so.

Remarks: Binet's words are 'kindness', 'injustice', and 'charity' and requires two correct responses out of three. Terman has five words, 'pity', 'revenge', 'charity', 'envy', 'justice' and requires three correct responses out of five. We have followed Terman except that we have substituted 'injustice' for 'justice'. Terman vernacular word for 'justice' is 'Nyaya'. This word unfortunately is colloquially used in sense almost the opposite of justice. It is used in the sense of 'Quarrel'. Hence the word had to be discarded and the opposite word 'injustice, or 'Anyaya', had to be used. This word has no ambiguity about it. Binet word 'kindness' has been discarded by many subsequent investigators. The definitions of this word given by children cannot easily be scored, perhaps on account of the word being too easy and incapable of being expressed in simpler terms. 'pity', 'revenge', and 'envy' have also been used by other investigations and admit of easy scoring.

The test was located by Binet (1911) in year XII, with a slightly different procedure as stated above, by Terman in year XII, by Burt in year XIV, by Kuhlmann in year XI, and by Bobertag in year XII. There is thus very great uniformity in the location of the test, which speaks well for its value.

A child's ability to define words can be analysed in the first place in to two parts; his conception of the exact connotation of the terms and his power of expression. The former depends on his logical acumen and the latter on the development of his linguistics ability. Both of these are found from various kinds of tests to grow by side with intelligence. When we examine the vocabulary of children we found that they pick up the names of the most common concrete objects first possibly those connected with their own bodies or the names of the most common objects in their environment. Thus they learn to name their mother and father first, then they name their hand, eyes, nose, etc. (III, 2); then other less familiar objects in their surrounding such as cat, dog, and so on. Here again there are two stages; the stage of recognition and the stage of expression. These two stages exist also in our acquisition of new words as we add to our vocabulary throughout life. For is it not within the experience of every person that when he meets new words in his reading he seeks to know their meaning? At this stage the person cannot use the word. Later on as he becomes more familiar with the word the associative bonds between the meaning and the system of labial and vocal muscles and the correlative system of nerves which are concerned with speech becomes so strong that as soon as the person intends to express a certain meaning those words automatically come out of his vocal organs. When a child learns the meaning of a word like 'cat' or 'tree' he does so after a long time. First he carefully studies the things which are called 'cats'. It is by slow observation of the characteristics that are common to all cats that he comes to recognize cats. He commits several mistakes in the early stages. Perhaps he calls a puppy by the name of a 'cat'. The he is corrected and unconsciously he tries to see what further characteristics he is to add to his idea of a cat in order to exclude dogs from that denomination. Thus slowly by isolating more and more of the abstract

characteristics which are common to larger groups of the same class of objects, the child is able to learn the correct application of terms or their use. Still greater difficulties experienced with abstract terms, such as 'charity' or 'revenge'. Even here the child learns most quickly the names of those abstract terms the ideas of which come earlier within his experience. Thus a child would know the meaning of 'truth' or 'falsehood' much earlier than 'revenge' or 'charity'. The child slowly studies the various acts, which explain the abstract terms and unconsciously beings to generalize the use of the terms. The standardization of such test has brought to light various facts in connection with the development of the abilities of children and as the results of further standardization is published; our knowledge of the growth of children's abilities will become fuller and fuller.

The correlation of the test with mental age as determined by the scale as a whole is 0.88.

XII. 4. Repeating Five digits Reversed (1 out of 3. Read 1 per second)

TEXT AND PROCEDURE

The series for this year is:

3-1-8-7-9; 6-9-4-8-2; 5-2-9-6-1

Procedure same as VII, 3 and IX, 1.

XII. 5. Interpretation of Fables (4 marks). (2 correct of the equivalent in half-credits, each fables correctly interpreted carrying 2 marks. Time limit 2 minutes after final query is put).

TEXT AND PROCEDURE

Say to the child

'you know what a fable is? You have heard fables? A fable as you know is a little story, and is meant to teach us a lesson. Now I am going to read a fable to you. Listen carefully and when I am through I will ask you to tell me what lesson the fable teaches us. Ready, listen;

(a) Hercules and the Wagoner'

'A man was driving along a country road, when the wheels suddenly sank in a deep rut, the man did nothing but look at the wagon and called loudly Hercules to come and help him. Hercules came up, looked at the man, and said, "put your shoulder to the wheel, my man, and whip up your oxen". Then he went away and left the driver'.

'What lesson does that teach us?

If the child simply repeats the story as a great many do, say only again,

'What do we learn from this?'

If the child's answer is not clear, say,

'What do you mean? Or "Explain"; I don't quite understand what you mean.

Record the answer of the child verbatim and proceed to the next fable.

'Here is another fable. Listen again and tell what lesson this fable teaches us?'

b) The Milkmaid and her pails'

'A milkmaid was carrying her pail of milk on her head, and was thinking to herself; The money for this will

buy four hens; the hens will lay least 100 eggs; the eggs will produce at least 75 chicks; and with the money which the chicks will bring I can buy a new dress to wear instead of the ragged one I have on. At this movement she looked down at herself, trying to think how she would look in her new dress, but as she did so, the pail of milk slipped from her head and dashed upon the ground. Thus all her imaginary schemes perished in a moment'.

Similarly with the rest.

c) The Fox and the Crow

'A crow, having stolen a bit of meat, perched in a tree and held it in her beak. A fox, seeing her, wished to secure the meat, and spoke to the crow: "How handsome you are and I have heard that the beauty of your voice is equal to that of your form and features. Will you not sing for me, So that I may judge whether this true?" The crow was so pleased that she opened her mouth to sing and dropped the meat which the fox immediately ate'.

d) "The Farmer and the Stork"

'A farmer set some traps to catch cranes which had been eating his seed. With them he caught a stork. The stork, which had not really stealing, begged the farmer to spare his life, saying that he was a bird of excellent character, that he was not at all like the cranes and that the farmer should have pity on him. But farmer said, 'I have caught you with these robbers the cranes, and you have got to die with them'.

e) "The Miller, his Son, and the Donkey"

'A Miller and his son were driving their donkey to a neighboring town to sell him. They had not gone far when the child saw them and cried out. "What fools those fellows are to be trudging along on foot when one of them might be riding!" The old man, hearing this, made his son get on the donkey, while he himself walked. Soon they came upon some men. "Look" said one to walk". On hearing this, the Miller made his son get off, and he climbed upon the donkey, himself". Further on they met a company of women, who shouted out. "Why, you lazy old fellow, to ride along so comfortably, while your poor boy thee can hardly keep pace by the side of you!". And so the good-natured Miller took his boy up behind him and both of them rode. As they came to the town a citizen said to them, "why, you cruel fellows! you two are better able to carry the poor little donkey than he is to carry you". "Very well", said the Miller, "we will try", so both of them jumped to the ground. Got some ropes, tied the donkey's legs to a pole, and tried to carry him. But as they crossed bridge the donkey become frightened, kicked loose and fell into the stream'.

For a correct response, which gives the correct moral of the fable in general terms give 2-marks. For responses, which give plausible but not exact answers give half credit. As Terman says, these latter responses are of two kinds: '(1) Interpretations, which are stated in general terms and are fairly plausible, but are not exactly correct, and (2) those which are perfectly correct as to substance, but are not generalized'.

An example of the first kind is, 'if we pray to God in the time of difficulty, he suggests a remedy to us'.

An example of the second kind is, 'He ought not sit silent calling out loudly for help. He ought to try himself'.

Remarks: This test was first standardized by Terman in 1911. It is one of the best tests of intelligence as it requires on the part of the child a thorough comprehension of the story and *generalization* of the moral contained in it. The correlation of the test with mental age as determined by the scale as whole is 0.94. Perhaps a little schooling might be a help in showing the child what a moral is and how to draw it. Beyond that, the children are thrown entirely on their own resources. Again and again it was observed that though the children knew the story and perhaps had also read the moral, it did not help them at all. They had to put forth their best efforts anew to draw the moral and several times failed to do so.

The fable has often been used to test the delinquency of children. It is argued that a child who cannot draw the moral from a concrete situation as in the fable, would not be able to draw it from a social situation and hence would not be able to guide his conduct. Thus a test that sifts out dull people also sifts out delinquents. This proposition however is challenged by a good many investigators, who say that dullness and delinquency are by

no means concomitant. However, it seems possible that in a good many cases delinquency is due to the person's not being able to comprehend the gravity of the situation not to imagine the results of his actions, which lack of comprehension is certainly due to his mental retardation. To this extent however any test of intelligence would also be a test of delinquency. But here are also cases where delinquency is due to the environment in which the person grows up and the delinquent is helped by his intelligence to be expert delinquent rather than to cure himself of his delinquency by the width of vision which he gets on account of his intelligence

XII, 6. Interpretation of Pictures. (3 out of 4. Time limit 2 minutes for each picture)

TEXT AND PROCEDURE

The formula in this year is the same as in VI, 5. Use the pictures in the following order: (1) Railway station, (2) Re-captation at home, (3) Motor accident, and (4) Domestic scene.

Say to the boy or girl,

'What is this picture about?'

'What is this picture off?'

Generally the same formula as in VI, 5 prompts a response of the interpretation type from children of this mental age. But, if in some cases the reply is doubtful, return a second time to such pictures with the new formula:

'Explain this picture'

If the reply of the child is not clear, say,

'Explain what you mean'.

Remarks: See remarks under VI, 5 as to what constitutes of a picture. The interpretation of the picture is the meaning of the picture as a whole, generally the one the artist intends. But the interpretation given by the children need not be the same, any plausible and not inconsistent interpretation should be scored plus.

The picture we have selected agree remarkably well with Terman's pictures as to difficulty of descriptions or interpretation. Terman places the description of his pictures in year VII while we put it (for our pictures) in year VI. Terman places interpretation of his pictures in year XII, and we also placed ours in the same year. The selection of pictures is therefore a very important affair; some pictures may admit of interpretation at as high as stage as the adult one, while others may admit of the same as early as the tenth year. Hence results the great divergence in the location of the test. The form of the questions is no less important. Spontaneous interpretations such as those given by our first formula would place the test a little higher than those obtained after specific questions such as 'Explain this picture' 'What is the meaning of this picture?' Binet locates the test in Year XV and Terman and Burt both in year XII. The correlation of the test with mental age is 0.84.

XII, Alternative 1. Vocabulary, thirty words (both Kannada & Marathi)

TEXT AND PROCEDURE

Same as in IX. Alternative test

XII. Alternative 2. Repeating Six digits. (1 out of 3. Read one per second)

TEXT AND PROCEDURE

The series for this year is:

3-7-4-8-5-9; 5-2-1-7-4-6; 4-7-1-5-8-2.

Procedure same as in the 'digit' tests of previous years.

XII. Alternative. 3. Comprehension. Fourth degree (2 out of 3 repeat questions after 5 seconds. Time ½ minutes)

TEXT AND PROCEDURE

- a) What ought you to say when someone asks your opinion about a person you don't know very well?
- b) What ought you to do before beginning something very important?"
- c) Why should we judge a person more by his actions than by his words?"

Procedure as in VIII, 3.

Remarks: These are the three most difficult of the Binet comprehension questions. They were separated by Terman out of the whole lot and located in year X. We found them difficult enough for year XII, even a little too difficult here. Burt places them in year XI. The correlation of the test with mental age as determined by the scale as a whole is 0.85.

CHAPTER NINETEEN

TESTS FOR YEAR XIV

XIV, 1. Induction test: Finding a rule. (Gets rule by sixth folding. Unfold and show the paper to the child after he gives his answer to each cutting)

TEXT AND PROCEDURE

Cut out small pieces from old news paper or blank sheets about 15 cm. X 20 cm. in size. Take one of these and say to the child.

'now watch what I do'

Then fold it once and in the middle of the folded edge cut out a small triangular notch with a pair of scissors. Then say,

'How many holes will there be in the paper when it is unfolded?' After the answer, whatever it may be, unfold the paper and hold it up for the subject's inspection.

Then take another piece, fold it once as before and say,

Now, when we folded in this way and tore out a piece, you remember it made one hole in the paper. This time we will give the paper another fold. Now many holes will there be? With the third piece of paper say,

'When we folded it this way there was one hole and when we folded it this way there were two holes. Now, I am folding it again. How many holes will be it this time when I unfold it?'

Recapitulate every time, as for example with the sixth fold as follows.

'When we folded it this way there was one hole. When we folded it again there were two, when folded it again there were four, when we folded it again there were eight, When we folded it once again there were sixteen: Now tell me how many holes there will be if we fold it once more?

After the answer unfold the paper in the child's presence as usual. Take care that in the formula the words 'once', 'twice', etc., as 'when we folded it once', when we folded it twice, etc., are not used as they may help and sometimes misguide the children. Leave them free to get their own rule. After the child's final correct answer to the sixth fold say,

'How did you get it?' This question should not be asked before the sixth fold, though the previous answer may be correct.

For a pass the rule must be correctly stated.

Remarks: This test was suggested by Binet's 'paper-cutting' test and was first standardized by Terman in 1914. It is fairly good test of intelligence with a correlation of 0.61 with mental age as determined by the scale as a whole. School instruction has no effect on it. It is also a test that arouses to a high degree interest and curiosity of children. The only disadvantage in it is that it is likely to be communicated to others. It is more easily remembered than many of the other tests by the subjects that have undergone it, no doubt on account of its concrete nature and the interest that it arouses in them.

This test forces the child to use the trial and error method in an intelligent manner. The trial and error is not simply a blind trial with any numbers whatsoever but it seeks to discover a rule. The most common type of reasoning used by children in this test is: 'one'; 'two'; 'three'. He finds this incorrect and probably reasons out that the correct number four is obtained by omitting one number in the middle: thus he is led to say 'six' next and

again finds this incorrect. Intelligence children generally find the rule here: namely the next number is obtained by doubling the previous one: but some children still go on and say 'twelve' next. After this, normal children of this age usually get the rule but there are still some who give 'twenty four' as the next number.

Very few children try the method of visualizing the folds and creases and reasoning out the number of holes by the deductive method. This deductive method is therefore not the natural method with children.

Terman places the test in year XIV. Our statistics find it a little too difficult for year XII, and easy enough for year XIV, in which year we have placed it.

XIV. 2. Dissected sentences. (2 out of 3. 1 minute each)

TEXT AND PROCEDURE

Use the cards with the following disarranged sentences on it:

- (a) To asked paper my teacher correct I MY
- (b) A defends dog good his bravely master
- (c) For the started an we country early at hour.

Place the printed cards before the child, leaving the first sentence open and covering the remaining with a cardboard or a sheet of paper. Say to him,

'Here is a sentence that has the words all mixed up so that they don't make any sense. If the words were changed around in the right order they would make a good sentence. Look carefully and see if you can tell me how the sentence ought to read'.

If within one minute the child cannot solve the problem, simply read out the correct order to him pointing to each word as you read.

After the child gives the rearranged sentence always read out to him the sentences as he has given it and ask him finally,

'Is that right?'

If the child misunderstands and adds new words, which is, however, very rare, say to him, 'No outside words are to be added', and give him a fresh trial.

Otherwise no supplementary question of any kind or explanations are allowed. Then give the second and third sentence, each time covering the rest with a sheet of paper.

Remarks: The sentences are arranged in order of difficulty as we found them. In translating the sentences into the Indian languages we have retained the original sense as far as possible and have obtained the same number of words in each sentence as in Terman's version. Burt's translation however, differs from Terman's in the arrangement of words and in one case in the number of words also, for example, the first sentence contains eight words according to Terman's, Burt and our versions.

Our arrangements of words is very nearly the same as Terman's, but Burt's arrangements differs. The second sentence contains seven words according to Terman's, Burt's and our version, and our arrangement of words is very nearly the same as Terman's, but Burt's arrangement differs. The third sentence contains nine words according to Terman's version and ours, but only eight words according to Burt. In this sentence, however, in our version the sense had to be changed a little thus adding some new ideas in order to keep the number of words the same. Consequently the arrangement of words also is a little different from Terman's. Burt uses the word 'morning' for Terman's 'early hour'. Burt's arrangements of words is also different as in the

other sentences. The test is located in year XII by Binet(1911). Terman, Burt and Bobertag. Our statistics find it a little too hard for year XII, and so we have placed it in year XIV, as does also Saffiotti.

In assessing the answers any arrangement of words that makes good sense and is in common use is correct. In Indian languages the cases of nouns are made by inflecting the words themselves. Thus by merely looking at the nouns it is possible to pick out the subject. But children generally do not look to the grammatical forms in rearranging the words. The *meaning* is the governing principal. They concentrate on making good sense; for example, in the Marathi version of the first sentence the words '*mazyā*' and '*maza*' are used indiscriminately with '*shikshakana*' and '*pareekshecha*'. We score both ways plus.

All workers agree in saying that this is one of the best tests of the series. Its correlation with mental age as determined by the scale as a whole is 0.88. The percentage of passes steadily increases with age. It is easy to give and easy to score: it does not take much time; it interests the child. It is of the type of 'completion' tests and it requires the child to rearrange the words so as to complete the sense. Certain key words serve as a clue and lay down the skeleton of the sentence. This skeleton is to be given flesh and blood n\by rearranging the words properly.

In the procedure we have laid down that the sentences as given by the child should invariably be read out to him. This is quite necessary as the exercise is oral and even intelligent children sometimes may forget what they have said. When we read it a second time we are sure that the child has done his best and not made a mistake merely out of chance.

XIV, 3. Arithmetical Reasoning (2 out of 3. 1 minute each)

TEXT AND PROCEDURE

Use the card with the following arithmetical problems printed on it.

- a) If two pebbles cost 5 paise, how many pebbles can you buy for 50 paise?
- b) If a man's salary is Rs. 20 a week and he spends Rs, 14 a week, how long will it take him to save Rs. 300?
- c) At 15 rupees a yard, how much will 7 feet of cloth cost?

Place the card before the child and hide from view all except the first problem. Point to the first problem and say,

'Read this aloud for me'

Help him if he cannot read some of the words. After he finishes say,

'Find me an answer to this'

The subject must not use paper and pencil and the answer must be given in one minute.

For a plus, two of the three answers must be correct. Correct procedure without a correct answer is of no use. No second attempt is allowed. However, if the child corrects himself spontaneously he should be allowed to do so, provided the time limit is not exceeded.

Remarks: This is the second test in which simple arithmetical problems are used as test of intelligence, the first being IX, 2. Making change. At first sight these problems appear to be too simple for the age and many teachers whose opinions were sought agreed but when they are actually standardized this is the result. In IX, 2 the processes involved are simply addition and subtraction. In this year they are mostly multiplication and division. The figures to be manipulated are small and the problems are simple and straightforward. Such processes are often met with daily life by children and by adults who have not had much schooling. In school, children of these ages learn much more complicated processes. It is not, therefore, the processes themselves

but their application that tax the intelligence. Backward children use the process of multiplication where simple addition ought to be used and so on. Hence these tests serve as very good tests of intelligence. The present test shows a steady increase in the percentage of passes as age increases and has a correlation of 0.77 with mental age as determined by the scale as a whole.

The test was standardized by Terman. He selected the problems from Bonser's *Study of the Reasoning Ability of Children in the Fourth, Fifth, and Sixth School Grades*. Our location of the test agrees with Terman's.

XIV, 4.3 Problems of Enclosed Boxes (3 out of 4. $\frac{1}{2}$ minute for each problem)

TEXT AND PROCEDURE

The problems are,

One large box containing

- a) 2 smaller, 1 inside of each
- b) 2 smaller, 2 inside of each
- c) 3 smaller, 3 inside of each, and
- d) 4 smaller, 4 inside of each.

Show the child a small card board box about 8cm, by 5 cm, in height without opening the lid and say,

'You see this box; it has two smaller boxes inside of it, and each one of the smaller boxes contains a little tiny box. How many boxes are there altogether, counting the big one? First the large box, then two smaller ones, and each of the smaller ones contains a little tiny box'. The second time, say,

'Suppose now this box has two smaller boxes inside and each of the smaller boxes contains two tiny boxes, how many altogether? Remember, first the large box, then two smaller boxes, and each smaller one contains two tiny boxes'.

Similarly, the third time say,

'Three smaller boxes, each of which contains three tiny boxes'.

The fourth time say,

'Four smaller boxes, each containing four tiny boxes'.

The problem is given orally and solved by the child also orally, that is, without the aid of paper and pencil.

Remarks: This test was devised and standardized by Terman. He places it in year XVI, but our statistics show that it is easy enough for year XIV; even a good many fairly intelligent children between 10 and 12 years of age pass the test. It correlates fairly well with mental age as determined by the scale as a whole, the correlation being 0.61.

For success in it the test requires either visual or tactual imagery, or both, of a high order. It is doubtful how far verbal imagery alone would solve the problems. Mentally backward children are unable to build mental pictures of a very complicated nature; the child has to comprehend the instructions and as they are being given the mental pictures is being added to and held firm until it is completed. In this respect the test differs a good deal from the other test in the scale and deserves a place in it.

XIV. 5. Giving Similarities-three things(3 out of 5. 1 minute each)

TEXT AND PROCEDURE

The following are the sets of words for this age:

- a) Wool, Cotton, Silk
- b) Snake, Cow, Sparrow
- c) Book, Teacher, News paper.
- d) Scissors, Paisa, Piece of wire
- e) Rose, Potato, tree

See IX, 3.

Give full formula:

'I am going to name three things, etc'. in the beginning

With the rest say,

'In what way are.....alike?'

Sometimes the child says they are not alike and does not want to make an attempt. They say,

'These three are alike. Tell me in what way they are alike'. Many children express similarity by saying.

'They are all useful'. Then say,

'In what way are they all useful?'

Any kind of real likeness is enough.

Remarks: This test is similar to IX, 3. (Which See) except that similarity between three things required instead of two. The best way of giving the similarity is to put them in the higher class by logically. But such a direct and rigid classification need not be expected from children of this age. Indirect way of classification or any method of showing resemblance is enough.

Terman sets no time limit to this test. But we are of the opinion that a time limit is a great help to this test because if for nothing else it helps the examiner to know when to proceed to the next item, and useless to wait for more than one minute for the answer. Instead of 'knife-blade, penny, piece of wire', we have used 'scissors, paisa, piece of wire'. The rendering of 'knife-blade' into the Indian language is clumsy; the use of 'knife-blade', being only a part of an object, is undesirable. Again we have 'wool, cotton, silk' instead of Terman's 'wool, cotton, leather' as leather is not used for as many purpose in India as in west even including its use for shoes.

The test shows a steady rise in the percentage of passes from year to year and it has a high correlation with mental age. The correlation being 0.88.

XIV, 6. Ball and Field, Superior Plan.

TEXT AND PROCEDURE

Same as in VIII, Alternative 2, Which may please be seen for discussion. For help in scoring sees sample word of answers.

XIV. Alternative. Vocabulary. Forty words from vernacular lists.

TEXT AND PROCEDURE

Same as in IX. Alternative Test.

CHAPTER TWENTY

TESTS FOR YEAR XVI (AVERAGE ADULT)

XVI.1 Interpretation of fables (8 marks)

TEXT AND PROCEDURE

Same as in XII.5

XVI.2 Reversing hands of clock (2 out of 3 Error not must exceed 3 minutes. Time limit 1 minute)

TEXT AND PROCEDURE

Before giving the test see that there is no watch or clock with in the child's sight. Say 'suppose it is 6.22 o'clock, i.e. 22 minutes after 6: can you see in your mind where the large hand would be and where the small hand would be'.

'Now, suppose that the two hands were to change places so that the large hand takes the place of the small hand and the small hand takes the place of the large hand. What time would it then be?

Similarly with 11-10 and 2-46.

The range of answers for the first problem is 4-30 to 4-35. For the second 1-53 to 1-58 for the third 9-10 to 9-15.

Generally the child of this age knows how to read clocks and watches. If there by doubt, test actually by making read your watch accurately. If he happens not to know, drop the test and substitute one of the alternative test.

Remarks: The answer is generally given by children in half a minute. Terman allowed 3-2 minutes, we have allowed 1 minute and this is enough. The position 8-10 is very clumsy. If it were made 8-13 the answer would be 2-41 and not 1-42 as with 8-10. The difficulty and nature of the test would, therefore, be changed a little, and so we have changed it to 11-10, the answer to which is 1-56 and nature of the test is retained as it was, at the same time removing the clumsiness of 8-10.

The test is given by Binet in his 1905 series though omitted in those of 1908 and 1911 as the series was not an age scale, the age where this test fits in not indicated. Terman located it in year XIV. Godard and Kuhmann used only 2 of the sub tests, the (a) and (c) omitting (b) the probably on accounts of its clumsiness as we have shown. They want both problems to correct and located it in year XV. Our statistics show that the test is a little too difficult for year XIV, and so we have placed it in year XVI.

This test is very valuable problem in constructive imagination of the visual type. It requires the child, first of all, to create a mental picture of the given position of time and then with the help of it to construct another picture with the hands changed. Both pictures do not appear to remain in attentive consciousness at one and the same time. After the first is found and the second is being constructed, the is momentarily wiped of but again reconstructed several times for comparison, attention oscillating from one to the other as many times as its is necessary to solve the problem. This is no mean feat of mental gymnastics. The test correlates high with mental age as determined by the whole scale, the correlation being 0.82. The percentages of passes rises gradually from year to year but is not very high even in the highest age. A little less than 40 percent of the children between 12-14 years of age pass this test, about 51 percent of those between 14-16 and 59 percent of those above 16 years of age.

XVI.3 Giving differences between Patil and Kulkarni. (2 out of 4 in any for 1 writing work, .2 Settlement of disputes or supervision of villages, .3 Power and .4 Appointment and lemuris)
(In large cities and else where if children are found to be unfamiliar with Patil and Kulkarni, this test may be omitted and one of the alternative ones substituted)

TEXT AND PROCEDURE

'You know a village Patil and you know a village Kulkarni. Now give any real main difference between them that you know of.'

If the child gives only one or two differences or some trivial differences and stops, say, 'any differences will do. Give me any differences that you know of. For success two of the following four in any form and even though mixed together are sufficient:

Village Patil

- 1 Has no writing or account work
- 2 Has settlement of disputes or supervision of village
- 3 Is a superior officer(power)
- 4 Does not require much literacy and the post is generally Hereditary.

Village Kulkarni

1. Has writing and account work
- 2 Has no such work
3. Is subordinate to the Patil
4. Has, as a rule, to pass a literacy test and the post not generally hereditary

There may be small differences in these items on account of local circumstances and these should be taken into account in scoring.

Remarks: The Binet test of 'giving three differences between a president and a king' is unsuitable in India. Indian children had no knowledge of a president and only indirectly of a king. We had, therefore, to substitute for it a similar test but suitable for Indian conditions. Two such were tried, : (1) 'the differences between a collector and the President of a municipality, and (2) the differences between the village Patil and village Kulkarni'. Of these, the later seemed to better as most children come into contact with the village Patil and village Kulkarni and a much smaller number with the collector and the president of the municipality. The first would give a slight advantage to town children, while the second would give the advantage to village children. As most of test in the scales give a slight advantage to town children, it was thought a slight advantage to village children in this test was desirable. The test does not depend much on schooling. There is lesson on the Patil and the Kulkarni in one of the primary reading books, but that lesson is read by children at an age when they are not expected to know the full implications of the statements made in it. The children were, therefore, found to be thrown entirely on their own resources and began to find the differences from what they knew of Patil and Kulkarni. The test is found to be fairly good test of intelligence, the correlation between it and mental age as determined by the entire scale being 0.58. There are very few children of this age in large cities who do not know about a Patil and a Kulakarni. If one is found the test may dropped and one of the alternative tests substituted in its place.

Psychologically the test is similar to other tests for finding differences and similarities, because finding similarities presupposes knowledge of dissimilarities. Compare, for example, VII.6: IX,3: XIV, 5: XIX,3.

Since the test was standardized some years ago conditions in India have changed immensely. India has become a democratic country and the post of President of the country has been instituted. It would be interesting to standardize the original test of Binet 'giving three differences between a President and a King' now. The posts of village Patil and Village Kulkarni have gone out of vogue in the changed political situation and the test would not be suitable in the changed circumstances.

XVI. 4. Repeating Six digits Reversed. (1 out of 3. Read 1 per second)

TEXT AND PROCEDURE

Use the series:

4-7-1-9-52: 5-8-3-2-9-4: 7-5-2-6-3-8:

Procedure same as in similar previous tests. See VII, 3; IX, 1; XII, 4.

Remarks : In 'digit' tests a lower series should always be tried before a higher series is given.

XVI, 5. Problem. Questions. (2 out of 3. Query on (a) and (b). Read a second time after 5 seconds. Time Limit 1 minute to begin reply.)

TEXT AND PROCEDURE

Say,

'Listen carefully, and see if you can understand what I read'. Read the following passages twice over as indicated above one after the other slowly and emphasizing the key-words so as to make the meaning clear.

- (a) A man who was walking in the woods outside his town stopped suddenly, very much frightened, and then ran to the nearest policeman, saying that he had just seen hanging from the limb of a tree a.....a what?
- (b) 'My neighbor has been having strange visitor. First a doctor came to his house, then a lawyer, then a clergyman(or priest). What do you think happened there?'
- (c) A villager who had come to town for the first time in his life saw a man riding along a street. As the man rode by, the villager said: "This man is lazy, he walks sitting down". What was that man riding on that caused the villager to say "He walks sitting down"?

Examples of satisfactory and Unsatisfactory answers:

(a) Satisfactory: A man who hanged himself: suicide.
Unsatisfactory. A snake: A ghost: a nest of Birds: a thief was running away after committing a theft, hanged himself on the branch of tree: a monkey: adventitious roots of a banyan tree: a bat: the tail of a monkey: a cat: a bag: a honeycomb: a tiger: a bundle of cloths hung in the dark: some wonderful thing: a fallen trunk of a tree: a thief about to jump down: a bird: a fruit: Mangoes.

(b) Satisfactory: Illness resulting in death- doctor to give medicine, lawyer to settle money-matters, priest to perform religious rites. Somebody is dead - doctor to see if he is dead. Lawyer to settle disputes about division of property, and priest to perform funeral rites. Somebody is ill and the doctor comes: then he is about to die and lawyer comes to make a will: finally he dies and the priest comes to perform the funeral rites. Murder- doctor for post-mortem, lawyer to note down facts of the case, and priest for funeral rites. Some accident - doctor to examine the dead man, lawyer in order to conduct case, and the priest for funeral rites.

Somebody is poisoned - doctor for medical examination, lawyer to note down the law-points and priest for the funeral rites. Somebody recovers after serious illness - doctor to give medicine, lawyer to take the patient's signature and the priest for test after recovery. Somebody is ill and then dies- doctor for treatment, lawyer for management of property, and priest for purification of the household (praya-schitta) and for receiving charity.

Unsatisfactory: Doctor for illness, pleader for some business, priest for some ceremony. Doctor for illness- money might have come for a feast. Murder- no explanation. Doctor for illness, lawyer for quarrel, and priest for marriage.

© Satisfactory/Bicycle.

Unsatisfactory: A horse, A cart, a motor car; a palanquin: a tonga: on foot: carrying another person. On the ground: a bullock: a small animal; a lame man was seated on the ground and propelling himself along; a monkey; a calf.

Remarks: The first two problems are Binet's the third is added by Terman. In the first problem Binet's uses the name of a forecast near the town where the subject is being examined. Terman omits this. Binet's form is better as it helps the child to visualize the events and makes the problem more concrete and definite. Binet required the answers to both problems correct: while Terman requires two out of three to be correct. Thus Binet's form of test is harder than Terman's. Binet located this test in year XV of his 1911 scale. But with the identical form of test in year XIII, and Terman with the amended procedure in year XIV. Our statistics requiring two out of three problems correct would place the test in year XVI. The test is very fairly good test though many children hazard an answer by mere guessing. The correlation of the test with mental ages as determined by the entire scale is 0.80.

The test requires the child to visualize elements and discover a situation where all these elements will fit in. It is thus psychologically a form of the completion test of the abstract type. It is interesting how many children answers reveal the strong power of perceptual associations, which children are powerless to inhibit, and are unable to find out a proper solution by reasoning. Thus in the first problem a great many answers give 'a bat'. Children have seen these bats hanging from the branches of trees and as soon as they try to visualize anything hanging from the branch of a tree, the previous association formed brings up the image of a bat. Similarly in the third problem when we translated 'a white man' by 'a European' in the Indian language the answer invariably was 'a motor car' because Indian children have usually seen Europeans 'riding' in a motor-car, and when the words 'European' and 'riding' are given, the association brings in 'a motor-car'. After a little trial we were obliged on this account to change the phrase 'a European' into simple 'a man'. These instances exemplify the great power of perceptual images to occupy the mind and inhibit thinking, which is necessary in finding out a situation in which the given elements will fit. The fixity of the imagery is thus sign of low intelligence. Higher intelligence can construct and destroy imagery very rapidly. In good many cases a plausible answer is first given and then some sort of explanation is manufactured, particularly in (b).

Binet and Terman did not time the problem, nor did they lay down many times the problems should be re-read. The greater the number of times the passages are read, the greater the chances of backward subjects finding an answer. Similarly we cannot by indefinitely waiting for a reply, and so we have for the sake of uniformity thought fit to time the test and lay down the procedure accurately.

If the test is scored on the basis of two out of two (both Binet's i.e. the first two) after Binet it is harder still and would be suitable for the Superior Adult Group according to our statistics.

XVI.6. Repeating Seven Digits (1 out of 2. Read 1 per second)

TEXT AND PROCEDURE

This series is : 2-1-8-3-4-7-9: 9-7-2-8-4-6-5:

Procedure same as in previous 'digit' tests.

Remarks: If the test is scored on the basis of 1 out of 3, it is slightly easier and is suitable for year XIV. Terman has the same digit, namely 3, twice in the first series and 7 twice in the second series. This sometimes disconcerts the child and diverts his attention. He sometimes interrupts and says the digits occurs a second time, and so the test is spoiled. We have changed this.

XVI. *Alternatives 1. Vocabulary. Forty Five Words.*

TEXT AND PROCEDURE

Same as IX, Alternative test.

XVI. *Alternative 2. Free association. Sixty words in three minutes*

TEXT AND PROCEDURE

Same as in IX, 6.

CHAPTER TWENTY ONE

TESTS FOR YEAR XIX (SUPERIOR ADULT)

XIX. 1. Using a code: (Vernacular code to be used. 2. Errors -6 minutes. Inverting numbers or writing one of the numbers or writing one of the numbers of a letter count half error. But if all the numbers are inverted and the message is otherwise correct no errors are to be counted).

TEXT AND PROCEDURE

Place before the subject the card on which the code is printed and say: 'say this secret code. You will see from this that you can write any syllable in the same way. Now examine the method of writing syllable carefully. To illustrate, first the word "go to school" (in Kannada) are written herein ordinary script and same again in the secret code. The first syllable (in the Kannada script) is the seventh of the class of consonants beginning with Therefore, we write the first consonant.....and the number seven beneath it. Thus.....means.....Further, the first syllable.....is the second in order in the series of the syllables..... therefore we write the number 2 above. Thus.....means the first syllablesimilarly, the last syllable, etc,

"Now I will give half a minute. In that time you must study this carefully without speaking a word. Then, I will take away the card and ask you to write something for me in this secret code. Now look at this"

After the half minute say:

Now you must write something for me. Remember how you are to write it. When you have to write a syllable look at the class of consonants to which it belongs. Then write down the first of these and *below* that put the serial (belonging to the series) number of the required consonant in that class; then write the number of required syllables in the series of the syllable *above* the letter". Then remove the card from the sight of the subject and say,

"now write the word.....in this code".

Remarks. The English code test was devised by Healy and Fernald and appears in their *Test for practical mental classification*. Published in 'Psychological Review monograph' (1911). It was standardized and located in year XV by Dr. Godard and then by Terman in his Stanford revision and located in year XVI (Average adult). As the Indian script are quite different a new code test had to be devised for this script.

The result of this test, therefore, cannot be compared with those of the English code since there is very little in common between them except the name. The code appears to be simple at first sight but when actually put to the test it is found to be too difficult for the adult group but is suitable for the Superior adult group. Psychologically it requires the subject to put into use the highest conceptual processes of analysis and synthesis. The power of complementing verbal statement is also a great factor. The directions are rather long, but the instinct of curiosity keeps up the interest of the subject.

In the direction we have stated that if the subject inverts all the digits 'No' error is to be counted. The reason is that it is natural for you not to remember without much drilling which of two alternative is correct, just as children find it very hard to remember 'right' and 'left' without drilling. Hence if the subject follows on of the alternative to the logical end, no error is to be counted.

The consonant in the Indian languages are classified as 'gutturals', 'dental', 'labials', etc. These are called the *Vargas* (classes). There are certain other which are grouped together in a heterogeneous mass and are called the *Avargiyas*, the un-classed group, but for simplicity in giving direction we have called this group the average which is really an inaccuracy, but excusable for the sake of convenience.

The test shows a high correlation with mental age as determined by the scale as a whole, namely 0.89. More

than 36% of children between 14-16 year of age pass it and nearly 40% of adults i.e. subjects above 16 years of age.

XIX. 2. Ingenuity Test (2 out of 3.5 minutes each. Directions may be repeated. If the subjects fails on the, experimenter explains it).

TEXT AND PROCEDURE

Say to the subject,

(a) A mother sent her boy to the river to get seven pints of water. She gave him a 3-pint vessel and a 5-pint vessel. Show me how the boy can measure out exactly 7 pints without guessing at the amount. Begin by filling the 5-pint vessel'.

The answer must be found orally. The subject must explain the complete solution. Tell him no marking on the vessel with chalk or by any other method is allowed. If the subject fails to give the solution within 5 minutes explain this problem and proceed to the next.

(b) Same as above, except that 5 and 7 are given to get 8.

Begin by filling the 5-pint vessel.'

(c) Same as (a), except that 4 and 9 are given to get 7.

Begin by filling the 4-pint vessel

Remarks: The test was devised and standardized by Terman. It is rather a hard test though at first sight it does not look so. Terman says: 'Only an insignificant number pass the test below the mental age of 14 years and about two-thirds of Average Adult fail. Of our superior adults somewhat more than 75% succeed.' Our superior show that it is passed by 14% of children between 12 and 14 years of age. 32.5% of children between 14 and 16 years and 36% of those above 16 years of age. The correlation of the test with mental age is 0.65.

Psychologically the test requires a very high power of visual imagery. The vessels must be imagined and the operations of pouring water, adding, subtracting, etc. must be done with these imaginary vessels. There is also an element of trial and error process. The subject tries various processes and finds out which process leads to the required result.

The weakness of the test is that it is most likely to be communicated to other children. The problems can be easily remembered and being of a novel nature or something of the type of 'catch' questions that children take pleasure in asking others. When this is found to be the case the examiner should not use the test.

XIX. 3. Differences between en Abstract Terms. (3 out of 4. Wait 1 minute for reply)

TEXT AND PROCEDURE

What is the difference between

(a) Laziness and idleness?

(b) Pride and Vanity?

(c) Poverty and misery?

(d) Dishonour and disrepute?

If the subject simply gives the definition of the two words separately without pointing out the essential difference between the two terms, say, 'Yes but I want you to tell me the difference betweenandtell me only the difference'.

Remarks: The selection of the proper pairs of words is very important here. There must be similarity of meaning and yet an essential difference between the two terms. Binet's pairs in his 1908 scale were:

- (i) Pleasure and happiness
- (ii) Evolution and revolution
- (iii) Event and advent
- (iv) Poverty and misery
- (v) Pride and Pretension.

He cuts these down to three pairs in his 1911 scale:

- (i) Idleness and laziness
- (ii) Event and advent
- (iii) Evolution and revolution

Burt, who otherwise follows Binet very closely, uses the first, second and fourth of the Binet 1908 pairs.

In selecting the pairs of words mere translation is of no use, because in the different language the subtle differences in such pairs of words vary to a very large extent. Sometimes it is very difficult to find exactly parallel pairs. Further the difficulty of the pairs may vary much in different languages. Terman uses the following four pairs:

- (i) Laziness and idleness
- (ii) Evolution and Revolution
- (iii) Poverty and Misery
- (iv) Character and reputation

We have used four pairs in the Indian Languages which are translated into English as nearly as possible as indicated above in the English version of our tests but not exactly. The fourth pair we tried at first was 'honor and reputation'. This raised some difficulties and we changed it later on into the contrary terms 'dishonor and disrepute'. In giving differences any really clear contrast should be accepted. Giving the meaning of the terms only without bringing out the contrast is of no use. The test appears in year XIII of Binet's 1908 scale and in the adult group of his 1911 scale, in the forms we have indicated above. Terman has it in year XVI (Average Adult group). Burt locates it in year XV. We have found it difficult enough for the Superior Adult group.

This test taken along with XII, 3 'defining abstract words', and demonstrates clearly how our languages ability evolves from year to year. While 'defining abstract words' is found easy enough for year XII, finding differences between abstract terms' is difficult even for the Adult group. It is suitable only for the Superior Adult group. It is passed by 26% of children between 14 and 16 years of age and by 41% of the Adult group, i.e. those above 16 years of age. It correlates well with mental age, the correlation being 0.78.

XIX.4. Binet's Paper-cutting test. (If given must come before XIV. 1. The creases must be shown in pencil and the diamond shaped holes in the middle of the two halves of one of these creases 2 minutes to complete drawing.

TEXT AND PROCEDURE

Take a piece of paper as stated in XIV.

Say,

'Watch carefully what I do. See, I fold the paper this way (folding it once over in the middle)' then I fold it this way' (folding it again in the middle, but at right angles to the first fold) Now, I will cut out a notch right here'. So saying cut out a triangle notch in the middle of the single edge where there are no separate leaves.

Then leave this folded paper with the notch exposed to view, but pressed flat against the table. Give the

subject a pencil and another piece of paper similar to the one used and say:

'Take this piece of paper and make a drawing to show how the other sheet of paper would look if it were unfolded. Draw lines to show the creases in the paper and show what results from the cutting'.

If the paper is not square but little longer one way, the holes may be either on the longer creases or on the shorter. But their number and position in the middle of the two halves of the creases must be correct. The exact shape and size of the holes are not taken into account. The child must not fold the paper given to him before drawing.

Remarks: As Terman remarks, this test requires the power of constructive visual imagination. The child has successively to construct visual images of the piece of paper, the first folding and the creases so formed, then the second fold, the creases so formed and the holes made by the cutting. This is a highly valuable test for higher ages, for which it is very difficult to get suitable tests. The correlation of the test with mental age is 0.63. Nearly 28 percent of children between 14 and 16 pass it and only 32 percent of those above 16.

Binet located this test in year XIII in his 1908 scale and in the adult group in his 1911 series. Terman locates it in the Superior Adult group and Burt in year XV.

XIX. 5. Repeating Thirty Syllables. (1 out of 2 Absolutely correct)

TEXT AND PROCEDURE

The following are the renderings of the Marathi Passage to be given for repetition in this year

- (a) Rama likes very much to go to the grandmother, because she tells him funny stories.
- (b) Yesterday I saw a pretty little dog in the street. It had curly brown hair.

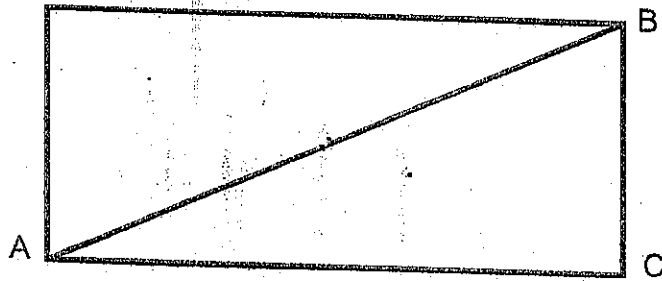
Procedure, as in previous repetition of syllables tests. Note that in this year one of the two passages must be absolutely correct.

Remarks: Binet's corresponding test is of 26 syllables and he located it in year XV in his 1911 scale. Burt located it with the same number of syllables in year XIV. Terman uses 28 syllables and places it in Year XIV or Adult group. Binet and Burt allow only one trial. While Terman gives two trials. We have used 30 syllables with two trials and find it difficult enough for the Superior Adult group. The test is passed by 25.5 percent of children between 14 and 16 years of age and by 30.5 percent of those above 16 years, that is, the Adult group.

The correlation of the test with mental ages is 0.68. But when all the tests of repeating syllables are correlated together with mental age the correlation is 0.89.

XIX. 6. Reversing Triangle in Imagination. Kärnataka form. (Time to complete diagram 2 minutes)

TEXT AND PROCEDURE



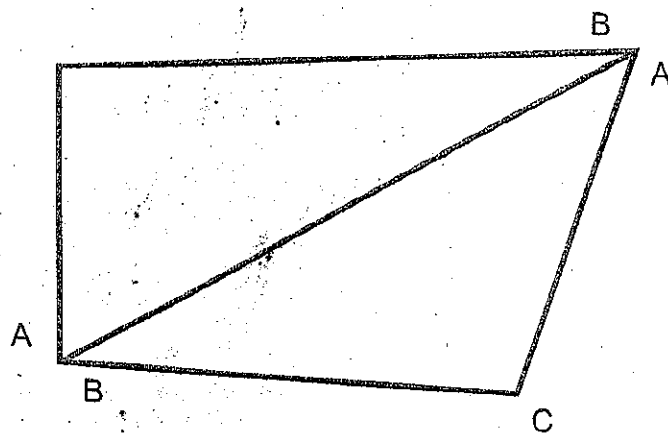
Place a stiff white card 15cm by 10cm, along the diagonal AB as in the 'divided card test' (VI, 3), before the child, with that cut edges joined together as shown in the diagram above, the side AC being towards the child (the letters A, B, C, are not to be shown). Then say,

'look carefully at the lower piece of this card (pointing to it). Suppose I now turn it over (showing the turning movement with the hand without removing the piece) and place this corner of lower piece (pointing to B of the lower piece) touching this corner of the upper piece (pointing to A of upper piece); and this corner of lower piece (pointing to A of lower piece) touching this corner of upper piece (pointing to B of the upper piece). What would it all look like? Now I am going to take the piece away (remove at this point the lower triangle only from View). Imagine it placed as I told you: and draw its shape in the proper position. Begin by drawing the shape of a top triangle'.

The finished diagram as drawn by the subject should be as shown below.

The essential points to be remembered in scoring the diagram are – (1) the figure should be roughly symmetrical about the diagonal; (2) the ends A, B, and B, A should be congruent; (3) the angle 'C' should be roughly a right angle.

Note that (2) and (3) are included in a way in (1).



Remarks: This test was suggested while giving Binet's test of reversing triangle in imagination, which was found to be too hard even for the Adult group. Even this test was found to be too difficult for the Adult group. Statistics showed that the present test was suitable for the superior adult group and Binet's original test was hard enough for the very superior Adult group. The present test is passed by 28 percent of adults and Binet's test by 4 percent of them. The correlation of the test with mental age is 0.55.

XII Alternative. Vocabulary, 55 words.

TEXT AND PROCEDURE

Same as in previous Vocabulary test

Remarks: With increase in mental age not only does the quantity of the vocabulary, i.e. number of words known, increase, but also its quality. The definitions of words given by this class are generally accurate, even abstract words being defined accurately.

CHAPTER TWENTY TWO

TESTS FOR YEAR XXII (VERY SUPERIOR ADULT)

XXII, 1. Comprehension of physical relations. (2 out of 3. May read a second time if necessary)

TEXT AND PROCEDURE

(a) Problem regarding the path of cannon ball.

Draw on a piece of white paper ; long horizontal line from end to end , towards the bottom of the paper. Above it a distance of about six centimeters and at the left end of the paper from the subject's side draw a short line about three centimeters long parallel to the first line.

Say to the subject, 'suppose this line is a level piece of ground and this short line is a gun which is placed horizontally parallel to the ground and is fired across this perfectly level piece of ground. Now suppose that this canon is fired off and that the ball comes to the ground at this point here (making a small mark at the farther end of the line which represents the level ground). Take this pencil and draw a line which will show what path the cannon ball will take from the time it leaves the mouth of the cannon till it strikes the found.

As Terman says the answers may be classified as follows:

- 1) A straight line is drawn joining the mouth of the canon and the point where the ball strikes the ground.
- 2) A straight line is drawn from the mouth of the canon to a point almost directly above the point where the ball strikes and then line drops suddenly to the goal.
- 3) The line rises gradually from the mouth of the cannon and then descends down to the goal, thus making something like a parabolic curve.
- 4) The line goes horizontally from the mouth of the cannon for some distance and then descends gradually to the goal.

Of these the fourth type of answer is the only one that is satisfactory. It need not, however, be mathematically accurate.

(b) Problem about the weight of a fish in water

Say to the subject: 'You know, of course, that water holds up a fish that is placed in it. Well, here is a problem. Suppose we have a bucket which is partly full of water. I place the bucket on the scales and find that with the water in it weighs exactly 20 kilograms. Then we put a 2 kilogram fish into the bucket of water. Now, what will be the weight of the whole thing?

Even though the answer may be correct ask further.

The response is correct when the subject says (1) The weight is 22 kilograms: (2) persists with this answer after the second query, and (3) gives some sort of plausible explanation like the following: The weight is there any way; the weight of water and the weight of fish both press down the scale-pan.

(c) Difficult of hitting a distant mark. Say:

'You know, do you not, what it means when they say a gun "carries 100 meters"? It means that the bullet goes that far before it drops an appreciable amount.

'Now, suppose a man is shooting at a mark about the size of a petroleum tin. His rifle carries perfectly more than 100 meters. With such a gun is it any harder to hit the mark at 100 meters than it is at 50 meters

If the subject answers in the affirmative ask him to explain. The correct explanation is that a small deviation

from the correct direction at the start becomes larger and larger as the distance is increased. The mathematical relation of this increase of deviation to distance is not required. It is enough if the subject gives only the general principle. The subject will have to be questioned a little until he understands what exactly is required.

Remarks: One feels rather uneasy in giving this test. It is found too hard for the average adult group and even a little too hard for the superior adult group. We have, however retained it in the very Superior Adult as it is so difficult to get good tests for the highest intelligence. It is passed only by about 7% of the subjects above 14 year of age and by 10% of the adult level. Our statistics in this respects are very disappointing as compared with Terman's.

Terman says: 'At the 14 year level less than 50% pass. Of average adults from 60 to 75% are successful. Few Superior adults fail.' The correlation of the test with mental age is 0.75.

The test seems to depend a good deal on acquired information. Two of the three problems are concerned with shooting. Very free Indian Children are conversant with any kind of shooting apparatus, even toy apparatus, with the results that they have no idea of the mechanism involved or of the path of the projectile. Of course they do a good deal of stone throwing but their experiences with regard to the path of the stone are very vague, and the test is consequently found to be very difficult for them. Psychologically, no doubt, it does require a good power of imagery but for the proper form of imagery of be roused the children must have had previous experience of the situations.

XXII, 2. Repeating Eight Digits (1 out of 3. Read one per second)

TEXT AND PROCEDURE

Give the digits with a uniform speed and without any rhyme or sing-song manner.
The series is: 7-2-5-3-4-8-9-6; 4-9-8-5-3-7-6-2; 8-3-7-9-5-4-2-6

Remarks: The test is passed by 13.3 percent of subjects between 14 and 16 years of age and by 7.2% of those above 16 years, which shows that memory for digits falls off after the sixteenth year. The lower percentage of passes in the adult level of this as well as of the three following tests seems to support the view that immediate memory fails off after the sixteenth year. (see footnote of p. 97)

XXII, 3. Repeating thought of passage heard (Read the passage only once in about ½ minute)

TEXT AND PROCEDURE

Say to the subject,

'I am going to read once only a little selection of about 6 or 8 lines. When I am through I will ask you to repeat as much of it as you can. It does not make any difference whether you remember the exact words or not, but you must listen carefully so that you can tell me everything it says'.

The subject's report should be taken down verbatim.

After securing attention the following passage should be read slowly - in about half a minute- and with expression, laying stress on the key-words.

'Many opinions have been given on the value of life. Some call it good, others call it bad. It would be more nearly correct to say that it is mediocre, for on the one hand our happiness is never as great as we should like, and on the other hand our misfortunes are never as great as our enemies would wish for us. It is this mediocrity of life which prevents it from being radically unjust'.

The subjects generally try their best to give the substance of the passage but if they stop in the middle encourage them to say as much as they can remember, in whatever words they like.

The arguments in the passage should be carefully reproduced and should contain the following three ideas: (1) Life is neither good nor bad (but medium); for (2) It (or our happiness) is not so good (or so great) as we wish it, nor (3) so bad as our enemies (or others) wish.*

Remarks: This is one of the most valuable tests in the series. It requires the subjects to comprehend the logical argument of a difficult passage, as it is being read, and to reproduce it. The selection is Hervieu's reflection on life and is excellent one. This test differs from those of repeating syllables in that, in this test, the sense of the passage is more important than the words in which it is clothed; while in the other the actual syllables are all important. Of course even in the latter the sense or meaning is a great help in retaining the syllables, since the child would be able to repeat far fewer nonsense syllables than syllables having sense. It differs from the test of reading and report in that in the latter the child him-self reads and his ability to comprehend a passage that is received through visual impression of tests, while in the former his ability to comprehend a passage received through auditory impression is tested. But from this we cannot say that comprehension of a passage received through auditory impression is more difficult than comprehension of a passage received through visual impression. For such a comprehension the passages must be of the same difficulty, but it is far from so, in the case of the present passage and the passage in IX, 5. The passage in IX, 5. is descriptive and concrete, while the present passage is reflective and abstract.

Binet assigns the test to the Adult group; Burt assigns it to XVI. Terman gives one more passage and requires one of the responses to be correct. His second passage contains a good many scientific terms and cannot be easily translated into the Indian languages. We have retained only one passage, namely the original one used by Binet, as this was found to answer the purpose very well. Terman assigns it to Superior adult group, while we think it hard enough for the Very Superior Adult group.

Abstract thinking and the ability to form rapid imagery are no doubt signs of great intelligence. Lower intelligence requires concrete objects and concrete illustrations as an aid to thinking. This is a great lesson in education. In the earlier stages the teacher provided the child with models and diagrams, from which the child learns new things; but as the imaginative power of the child is developed, the use of such concrete objects is lessened. In later stages such model are used only when explaining the most intricate machinery.

The test is passed by 15.8% of subjects between 14 and 16 years of age and by 13.3% of those above 16. The correlation of the test with mental age is 0.82.

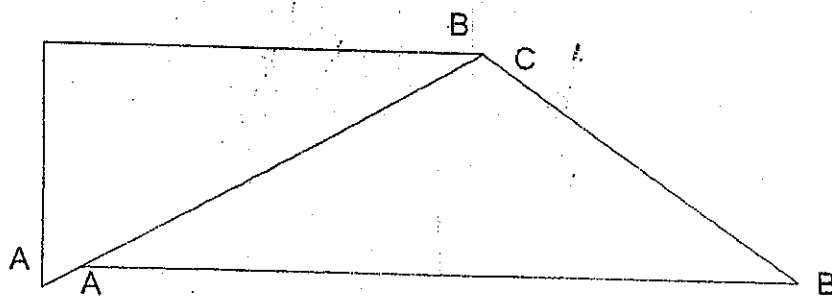
XXII, 4. Reversing Triangle is imagination. Binet's form (Time limit 2 minutes to complete the diagram).

TEXT AND PROCEDURE

Procedure similar to XIX. 6

After placing the divided card before the subject exactly as stated in XIX.6 say,
'Look carefully at the lower piece of this card. Suppose I turn it over and lay this edge (pointing to line A-C (see diagram given) without moving the card along this edge (pointing to A-B of the upper triangle) and suppose that this corner (pointing to 'C') is placed just at this point (pointing to B): What would it all look like? Now I am going to take the piece away. (Remove the lower triangle from view). Imagine it placed as I told you, and draw its shape in the proper position. Begin by drawing the shape of the top triangle'.

The finished diagram as drawn by the subject should be as follows (without the letter)



The essential points to be remembered in scoring the diagram are: 1). 'A' 'C' 'B' must be roughly a right angle. (2). 'A' 'C' must be shorter than 'A' 'B' and (3) 'B' 'C' must be the shortest of the three lines.*

Remarks: This test is one of the hardest tests in the whole series. It does require a little geometrical knowledge, the differentiation of a right angle from an acute angle, for example. The test is found surprisingly enough to be too difficult even for adults. The subject is required to form an image of the figure of the triangle, lift it in imagination, and apply it is directed. This is certainly helped by a proper knowledge of the size of the angles. It would be of interest to see what percentage of unschooled adults would pass the test as compared with adults who have had a course of schooling. Our present statistics show that the test is passed by 9.2 percent of children between 14 and 16 years of age and by 3.3 percent of adults, i.e. those above 16 years of age. The correlation of the test with mental age is 0.83. The test is located in the Ault group by Binet (1911) and in year XV by Burt. It is dropped by Terman as an unsuitable test.

*Cyril Burt, *Mental and Scholastic Tests*, P.67.